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- Model assessment and selection
  - Measures of predictive accuracy
  - Model selection
- Cross-validation
  - When is LOO applicable
  - PSIS LOO-CV
  - K-fold CV
  - Model Comparison
- Summary
- Additional material
  - Advanced PSIS
  - Information criteria


# Bayesian Statistics and Data Analysis
# Lecture 8b


Måns Magnusson
Department of Statistics, Uppsala University
Thanks to Aki Vehtari, Aalto University



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# Section 1

# Model assessment and selection



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# Model checking and predictive performance

- In L8a: What does the model miss?
  Posterior predictive checks compare features of observed and replicated data.



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# Model checking and predictive performance

- In L8a: What does the model miss?
  Posterior predictive checks compare features of observed and replicated data.
- In L8b: How well does the model predict?
  We assess predictions for the task of interest using held-out observations.



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# Model checking and predictive performance

- In L8a: What does the model miss?
  Posterior predictive checks compare features of observed and replicated data.
- In L8b: How well does the model predict?
  We assess predictions for the task of interest using held-out observations.
- The best candidate can still fit poorly.
  All candidates may miss important features. Predictive comparisons complement model checks; they do not replace them.



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# Predictive performance

- Modeling complex phenomena with models that are simplified
  All models are wrong... but some are useful.



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# Predictive performance

- Modeling complex phenomena with models that are simplified
  All models are wrong... but some are useful.
- True predictive performance is found out by using it to make predictions and comparing predictions to true observations
  - external validation



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# Predictive performance

- Modeling complex phenomena with models that are simplified
  All models are wrong... but some are useful.
- True predictive performance is found out by using it to make predictions and comparing predictions to true observations
  - external validation
- Expected predictive performance
  - approximates the external validation



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# Goal of model evaluation

- Model choice is a (model-)decision-theoretic problem



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# Goal of model evaluation

- Model choice is a (model-)decision-theoretic problem
- Evaluate the utility of a model $M$ for new unseen data $\tilde{y}$:

$$U_M = \int u_M(\tilde{y}; y) p_{\text{true}}(\tilde{y}) d\tilde{y} \,,$$

where $\tilde{y}$ is an unseen observation and $y$ are the training data. The utility $u_M(\tilde{y}; y)$ evaluates the prediction made by model $M$ after fitting it to $y$.



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- The expectation is with respect to $p_{\text{true}}$ ($f$ in BDA3)



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- The expectation is with respect to $p_{\text{true}}$ ($f$ in BDA3)
- Choose the model function to maximize our utility



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# Model choice utility

- Application specific utility/cost functions are important
  - eg. money, life years, quality adjusted life years, etc.



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# Model choice utility

- Application specific utility/cost functions are important
  - eg. money, life years, quality adjusted life years, etc.
- General utility: overall in the goodness of the predictive distribution
  - we don't know (yet) the application specific utility then good information theoretically justified choice is log-score for model $M$

$$\log p_M(y^{\text{rep}}|y)$$



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$$\log p_M(y^{\text{rep}}|y)$$

- We want the model that explain the data the "best"



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Subsection 1

# Measures of predictive accuracy



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# Bayesian point prediction accuracy

- Training residuals (the model is fitted to all of $y$)

$$e_i = y_i - E(\tilde{y}_i|y)\,,$$

where

$$E(\tilde{y}_i|y) = \int \tilde{y}_i p(\tilde{y}_i|y) d\tilde{y}_i\,,$$

i.e. the expected predicted value



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- Training mean squared error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_i^n [y_i - E(\tilde{y}_i|y)]^2\,.$$



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i.e. the expected predicted value

- Training mean squared error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_i^n [y_i - E(\tilde{y}_i|y)]^2\,.$$

- But how do we evaluate the posterior predictive distribution?



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# Probabilistic predictions: Scoring rules

- A scoring rule evaluates a predictive distribution $q$ at an outcome $z$:

$$S(q, z) \in \mathbb{R} \qquad \text{(higher is better)}.$$

Here $q$ is a probability density or probability mass function.



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  Here $q$ is a probability density or probability mass function.
- A proper rule has its highest *expected* score when the forecast matches the true distribution:

$$\mathbb{E}_{Z \sim p_{\text{true}}}[S(p_{\text{true}}, Z)] \geq \mathbb{E}_{Z \sim p_{\text{true}}}[S(q, Z)].$$



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- A proper rule has its highest *expected* score when the forecast matches the true distribution:

$$\mathbb{E}_{Z \sim p_{\text{true}}}[S(p_{\text{true}}, Z)] \geq \mathbb{E}_{Z \sim p_{\text{true}}}[S(q, Z)].$$

- A local rule depends on the forecast $q$ only through its value $q(z)$ at the observed outcome.



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# Densities and probabilities

- For a continuous outcome, $p(y_i \mid y)$ is a density. The probability of exactly one value is zero; probabilities come from integrating the density over intervals.



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# Densities and probabilities

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- A density can exceed 1, so its logarithm can be positive. For example, a $\mathrm{Uniform}(0, 0.5)$ density is 2 inside its support: $\log 2 \approx 0.69$.



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# Densities and probabilities

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- A density can exceed 1, so its logarithm can be positive. For example, a $\mathrm{Uniform}(0, 0.5)$ density is 2 inside its support: $\log 2 \approx 0.69$.
- For a discrete outcome, $p(y_i \mid y)$ is a probability mass, between 0 and 1. Compare scores on the same outcome scale and units.



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# Probabilistic predictions

- The log score is local and proper: higher is better.

$$S(p(\cdot \mid y), \tilde{y}_i) = \log p(\tilde{y}_i \mid y).$$



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# Probabilistic predictions

- The log score is local and proper: higher is better.

$$S(p(\cdot \mid y), \tilde{y}_i) = \log p(\tilde{y}_i \mid y).$$

- Evaluate the predictive density at an observed value $y_i$:

$$\log p(y_i \mid y) = \log \int p(y_i \mid \theta) p(\theta \mid y) d\theta.$$

This is a training score: $y_i$ was also used in fitting the model.



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# Pair discussion: log scores

**Discuss in pairs (1–2 minutes).**

For a new Bernoulli outcome $\tilde{y}$, model A predicts $\Pr(\tilde{y} = 1) = 0.8$ and model B predicts $\Pr(\tilde{y} = 1) = 0.4$.

**Which model has the higher log score if success occurs? What if failure occurs?**



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**Which model has the higher log score if success occurs? What if failure occurs?**

**Answer:** Using natural logarithms,

| | Model A | Model B |
|---|---|---|
| Success ($\tilde{y} = 1$) | $\log(0.8) \approx -0.22$ | $\log(0.4) \approx -0.92$ |
| Failure ($\tilde{y} = 0$) | $\log(0.2) \approx -1.61$ | $\log(0.6) \approx -0.51$ |

A scores higher for success, B for failure. A confident prediction earns a better score when right and a worse score when wrong.



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# Probabilistic predictions

- The log pointwise predictive density (lppd):

$$\text{lppd} = \sum_{i=1}^{n} \log p(y_i \mid y).$$

This is not generally the joint log predictive density $\log p(y \mid y)$: integrating out shared parameters can induce dependence.



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# Probabilistic predictions

- The log pointwise predictive density (lppd):

$$\text{lppd} = \sum_{i=1}^{n} \log p(y_i \mid y).$$

  This is not generally the joint log predictive density $\log p(y \mid y)$: integrating out shared parameters can induce dependence.
- Monte Carlo approximation with $\theta^{(s)} \sim p(\theta \mid y)$:

$$\text{lppd} \approx \sum_{i=1}^{n} \log \left( \frac{1}{S} \sum_{s=1}^{S} p(y_i \mid \theta^{(s)}) \right).$$



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Subsection 2

# Model selection



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# Model selection

- Evaluate model $M$ on unseen outcomes, one for each of $n$ prediction cases:

$$\text{elpd}_M = \sum_{i=1}^{n} \int \log p_M(\tilde{y}_i \mid y) p_{\text{true}}(\tilde{y}_i) d\tilde{y}_i.$$

  Here $\tilde{y}_i$ is a future outcome for case $i$ and $y$ are the training data. For regression, each prediction conditions on its covariates (not shown).
- This is the total expected log predictive density; divide by $n$ for an average per observation.



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- This is the total expected log predictive density; divide by $n$ for an average per observation.
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- $p_{\text{true}}$ is (almost always) unknown



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- This is the total expected log predictive density; divide by $n$ for an average per observation.
- The expectation is with respect to $p_{\text{true}}$
- $p_{\text{true}}$ is (almost always) unknown
- The utility for each case is $\log p_M(\tilde{y}_i \mid y)$: higher is better.



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# Section 2

# Cross-validation



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# Leave-one-out cross-validation (LOO-CV)

- Our target is the total expected log predictive density:

$$\text{elpd}_M = \sum_{i=1}^{n} \int \log p_M(\tilde{y}_i \mid y) p_{\text{true}}(\tilde{y}_i) d\tilde{y}_i.$$

The future outcomes $\tilde{y}_i$ come from the true data-generating process; $y$ are the observed training data.



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  The future outcomes $\tilde{y}_i$ come from the true data-generating process; $y$ are the observed training data.
- How can we estimate these expectations without collecting new data?



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# Leave-one-out cross-validation (LOO-CV)

- Use held-out observations to estimate expected log predictive performance.



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# Leave-one-out cross-validation (LOO-CV)

- Use held-out observations to estimate expected log predictive performance.
- Hold out observation $i$ and try to predict $y_i$ based on $\mathbf{y}_{-i}$
- Estimation of $\text{elpd}_M$ using leave-one-out cross-validation

$$\begin{aligned}\text{elpd}_{\text{loo}} &= \sum_{i=1}^{n} \log p_M(y_i|\mathbf{y}_{-i}) \\ &= \sum_{i=1}^{n} \log \int p_M(y_i|\theta)p(\theta|\mathbf{y}_{-i})d\theta\end{aligned}$$



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- Use held-out observations to estimate expected log predictive performance.
- Hold out observation $i$ and try to predict $y_i$ based on $\mathbf{y}_{-i}$
- Estimation of $\text{elpd}_M$ using leave-one-out cross-validation

$$\begin{aligned}\text{elpd}_{\text{loo}} &= \sum_{i=1}^{n} \log p_M(y_i|\mathbf{y}_{-i}) \\ &= \sum_{i=1}^{n} \log \int p_M(y_i|\theta) p(\theta|\mathbf{y}_{-i}) d\theta\end{aligned}$$

- Analogy: replace expectations over future outcomes with held-out observations. Each fitted model uses $n-1$ observations.



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# What does $p_{\text{loo}}$ measure?

$$p_{\text{loo}} = \underbrace{\sum_i \log p(y_i \mid y)}_{\text{lppd: training fit}} - \underbrace{\sum_i \log p(y_i \mid y_{-i})}_{\text{elpd}_{\text{loo}}\text{: held-out fit}} .$$

- The gap measures how much better the model predicts observations it has already seen than observations left out of fitting.

loo glossary: effective number of parameters.



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- The gap measures how much better the model predicts observations it has already seen than observations left out of fitting.
- Often interpreted as an effective number of parameters. It need not equal the number of coefficients, or even be an integer.

loo glossary: effective number of parameters.



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- The gap measures how much better the model predicts observations it has already seen than observations left out of fitting.
- Often interpreted as an effective number of parameters. It need not equal the number of coefficients, or even be an integer.
- In the coming example: $-26.8 - (-29.5) = 2.7$. Interpret this only after checking that the LOO computations are reliable.

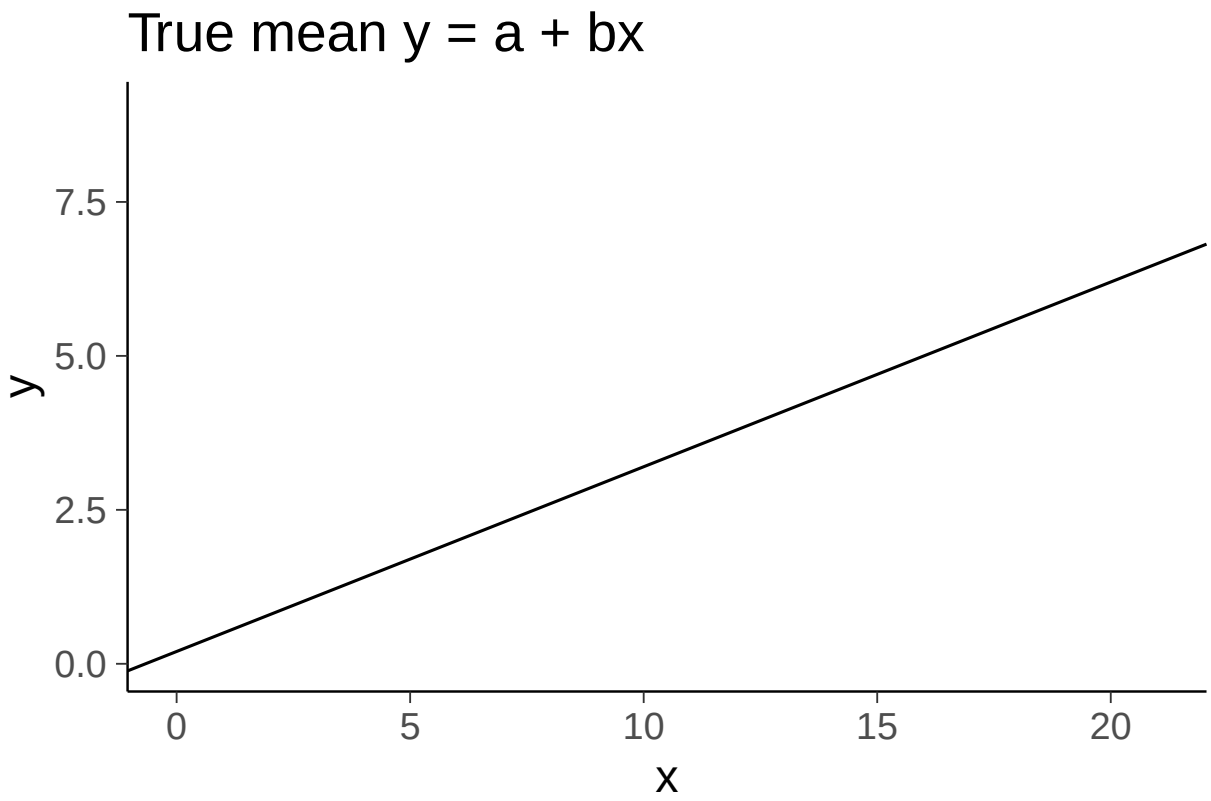
loo glossary: effective number of parameters.



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# LOO example: leaving out observation 18

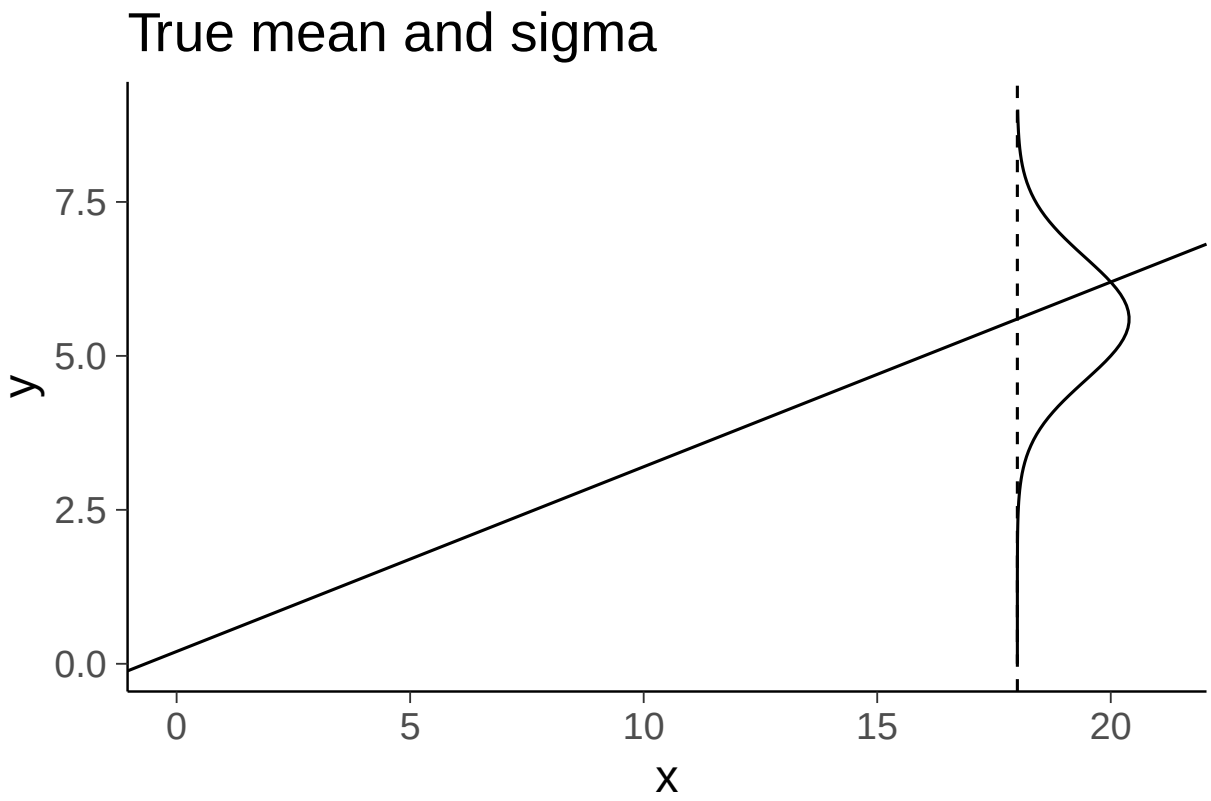




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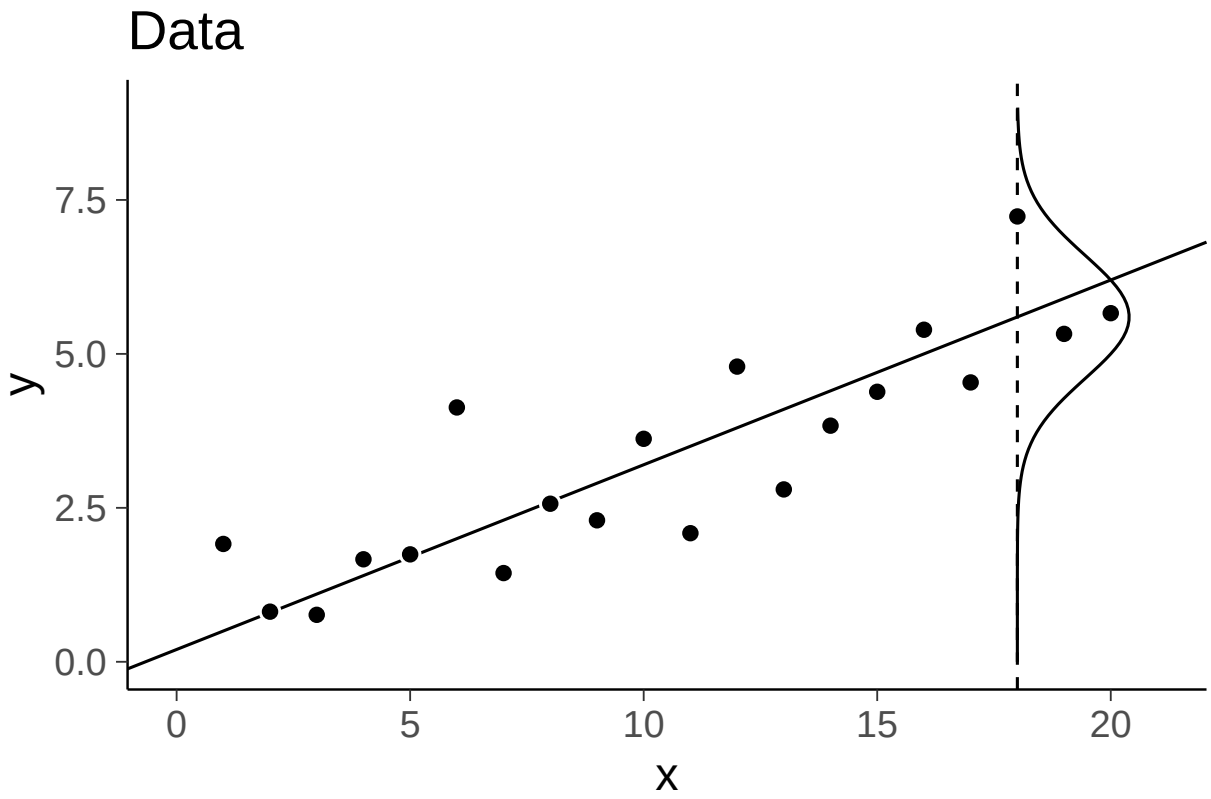




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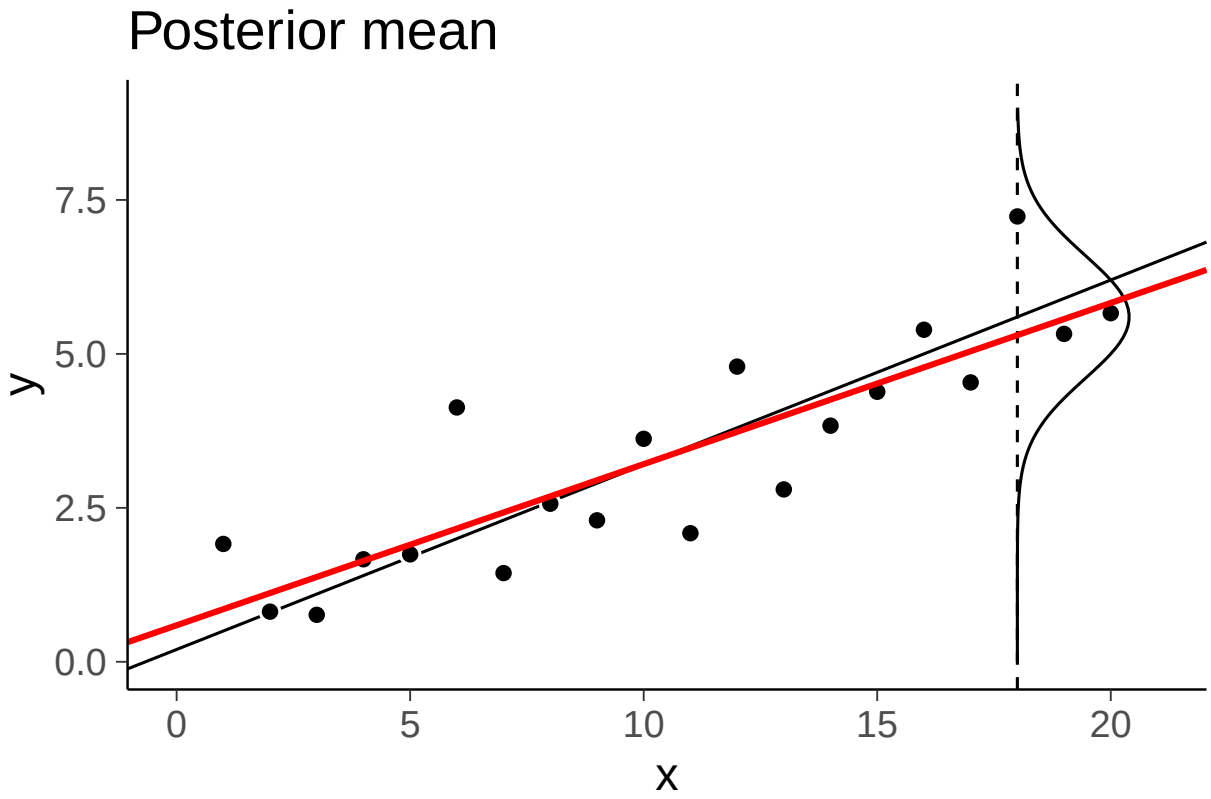




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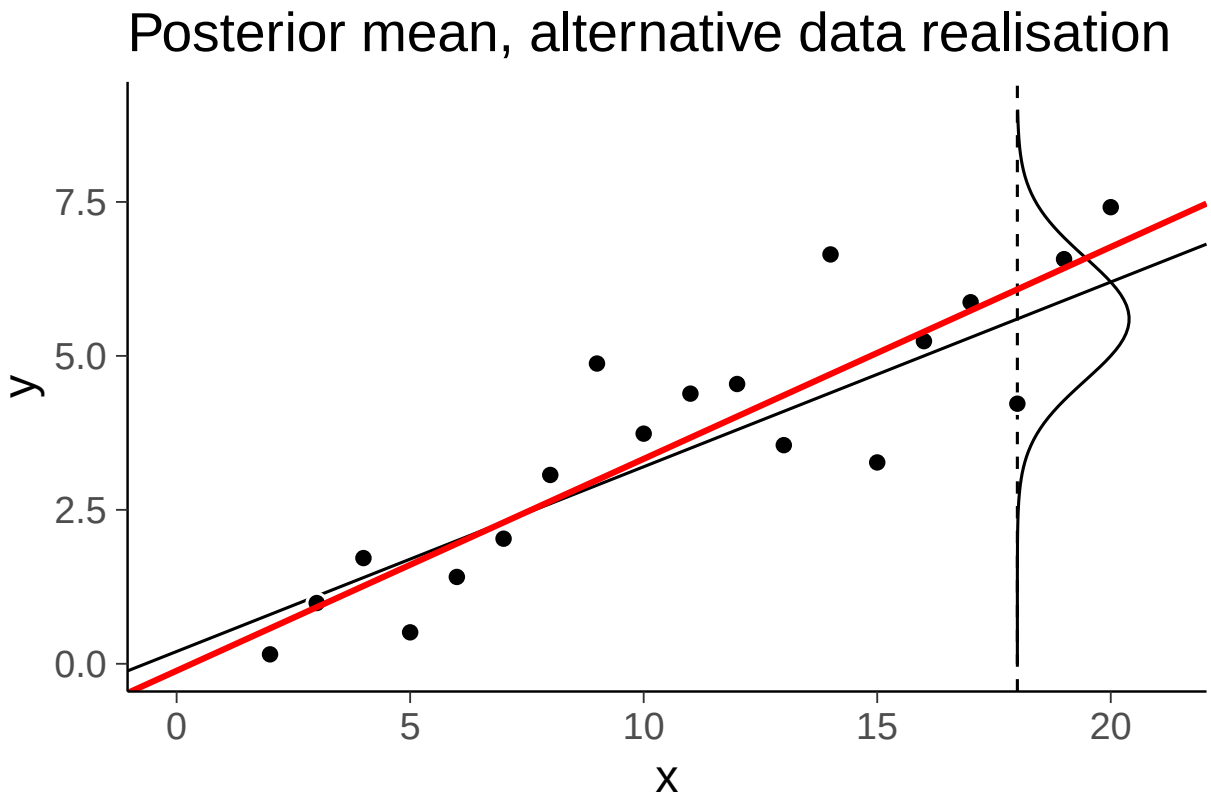




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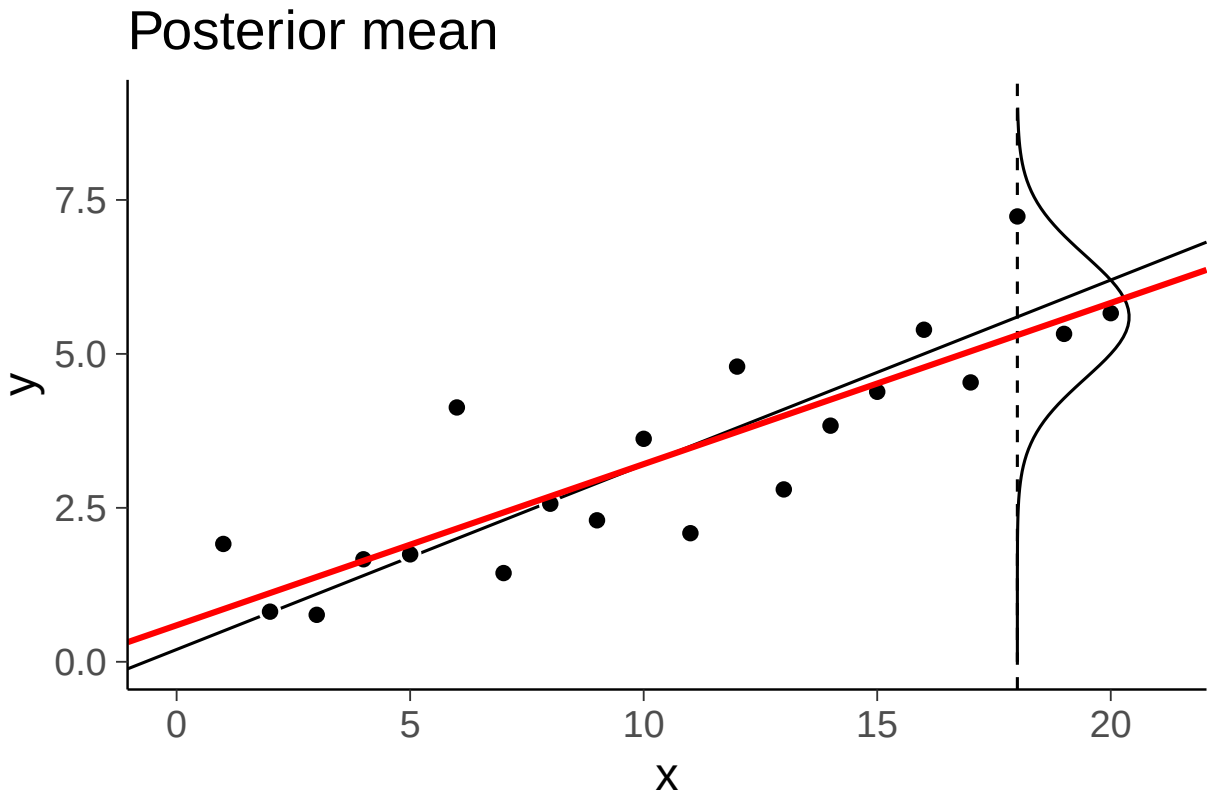




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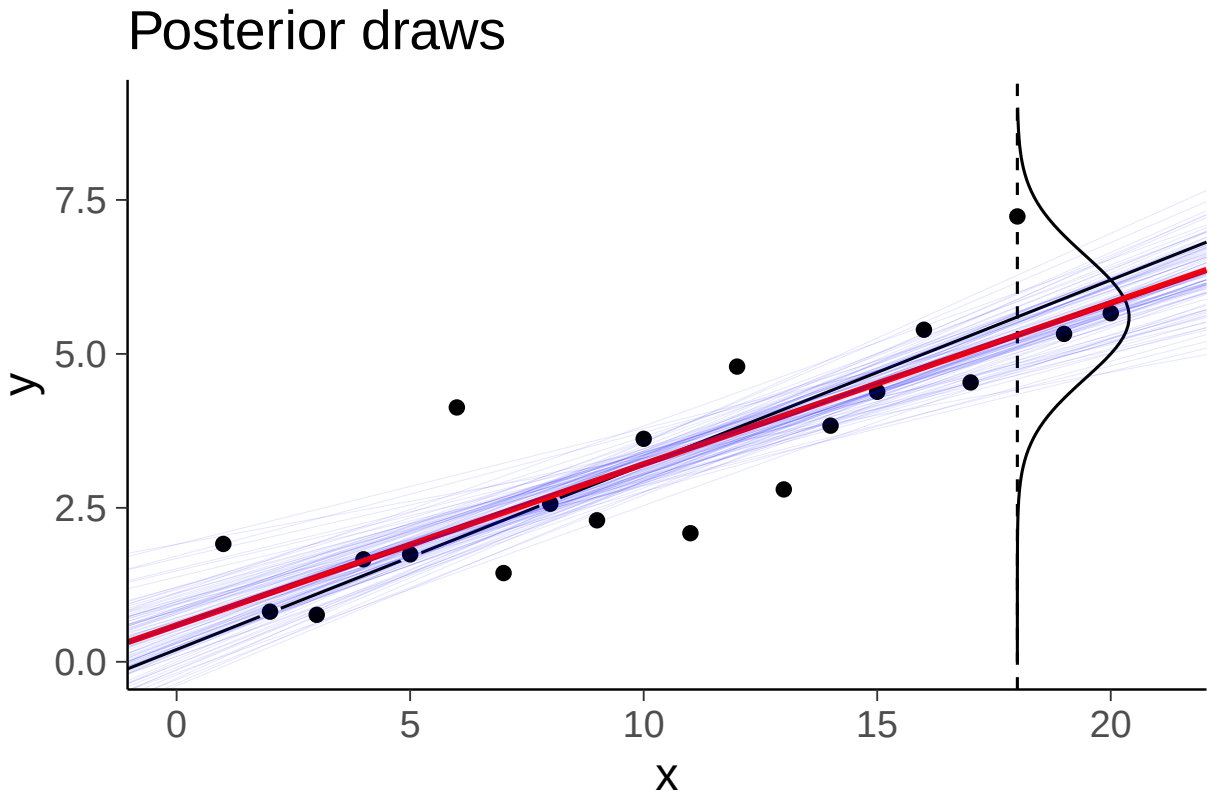




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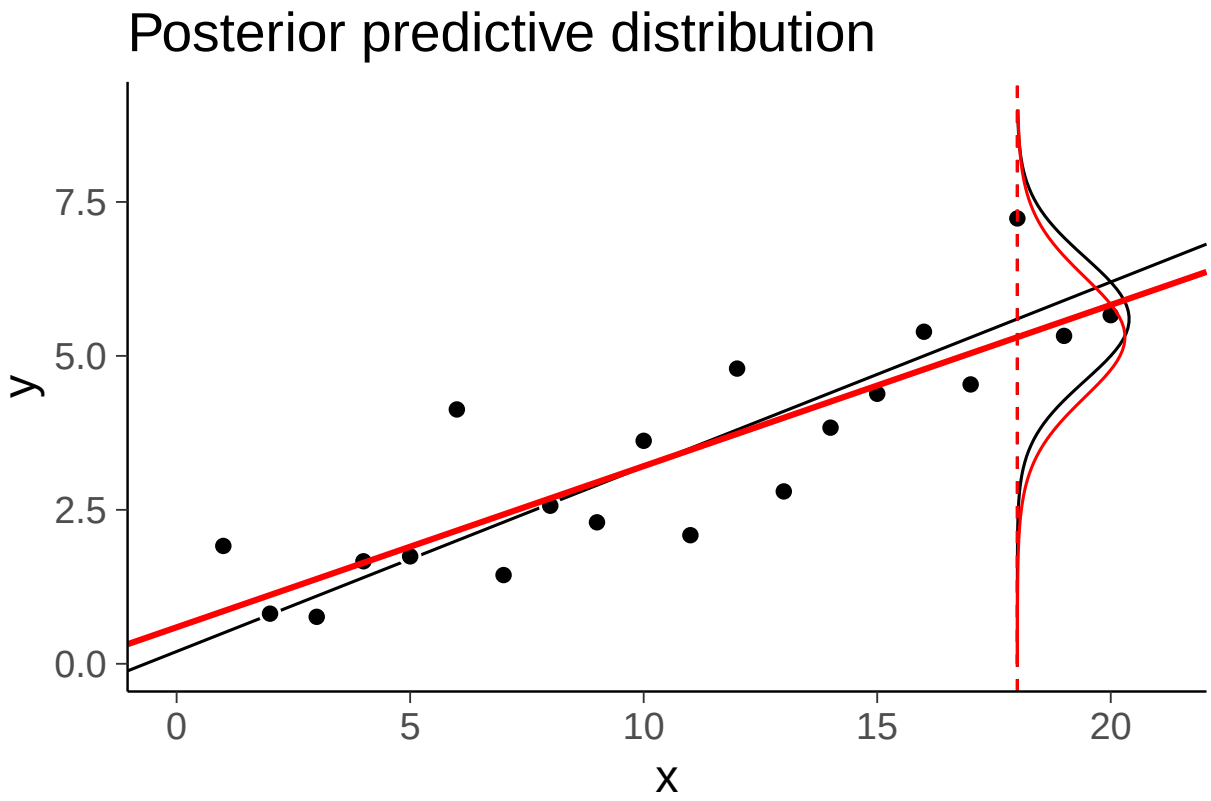




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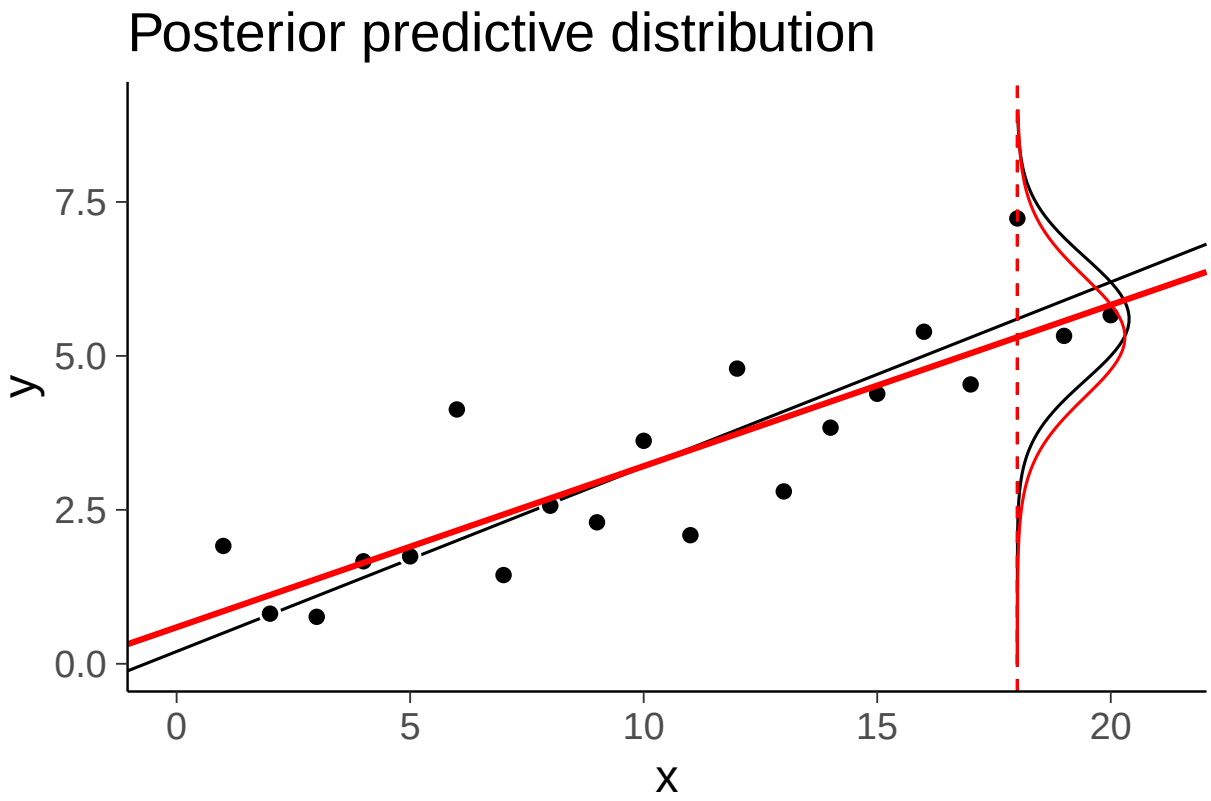




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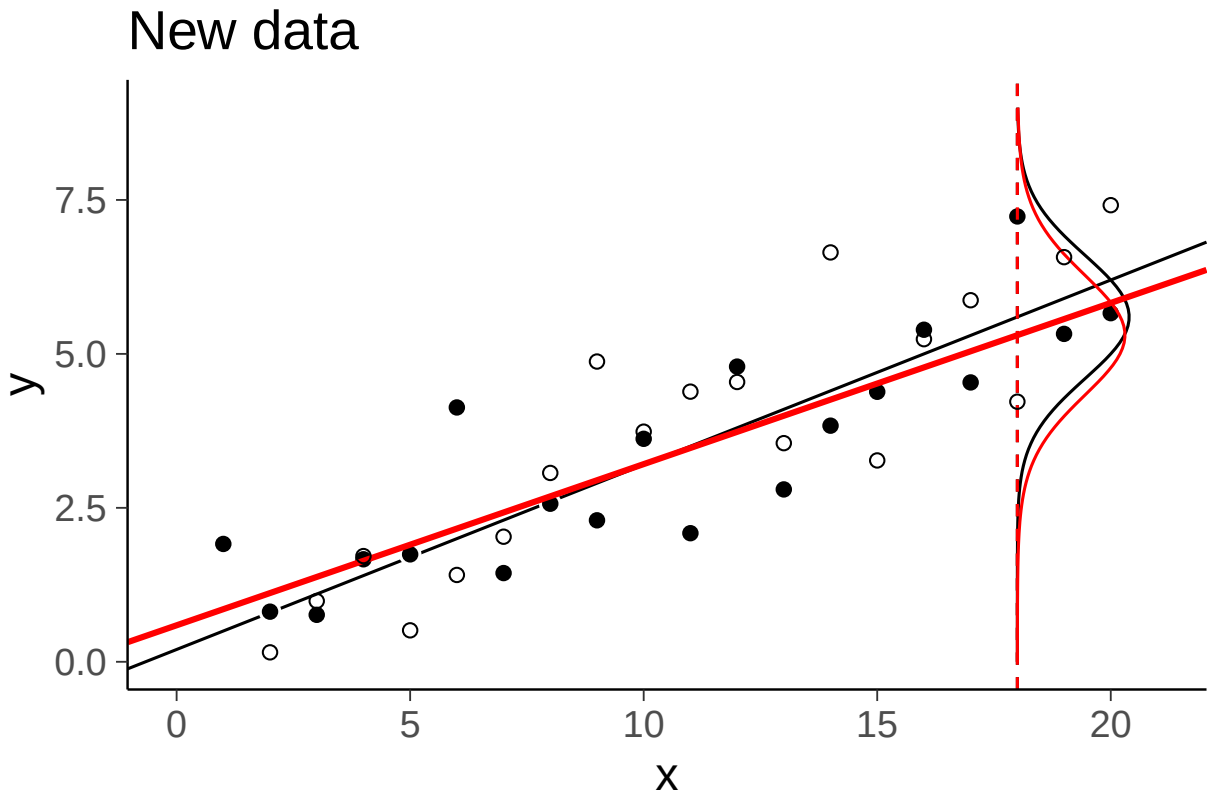

$$p(\tilde{y}|\tilde{x} = 18, x, y) = \int p(\tilde{y}|\tilde{x} = 18, \theta) p(\theta|x, y) d\theta$$



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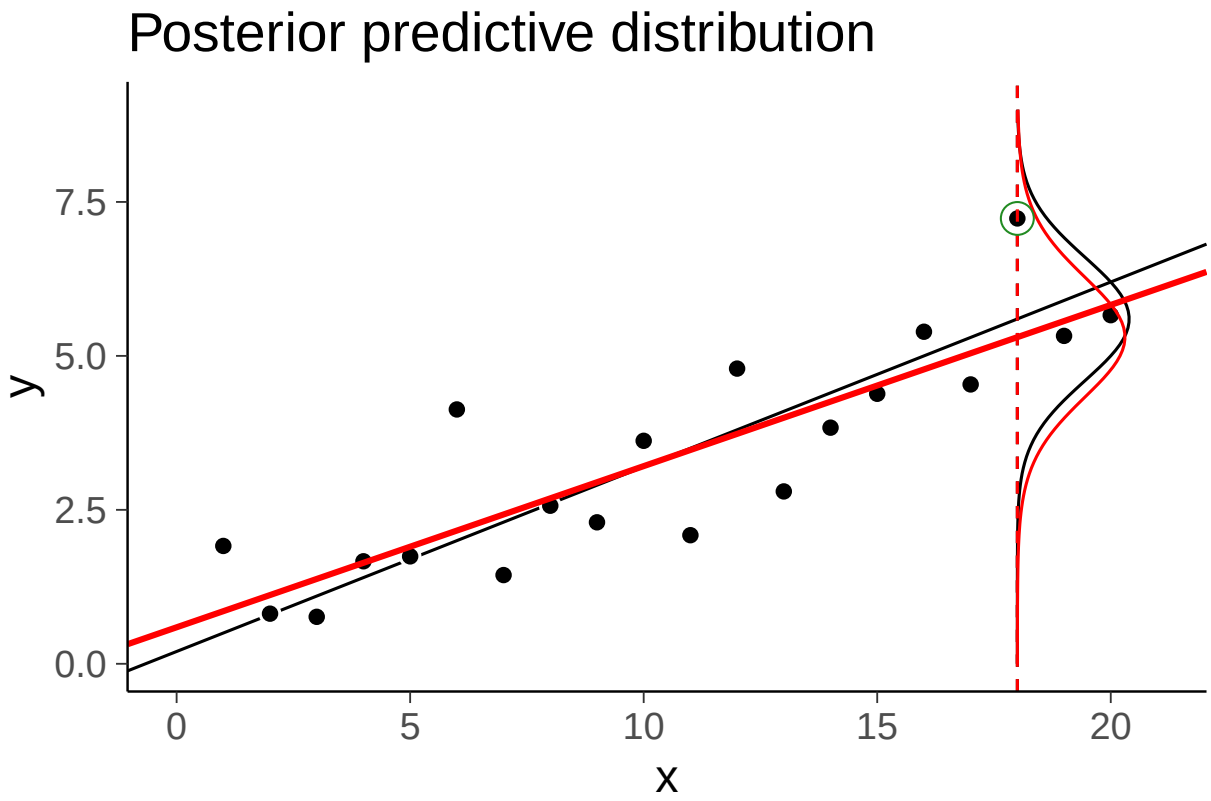




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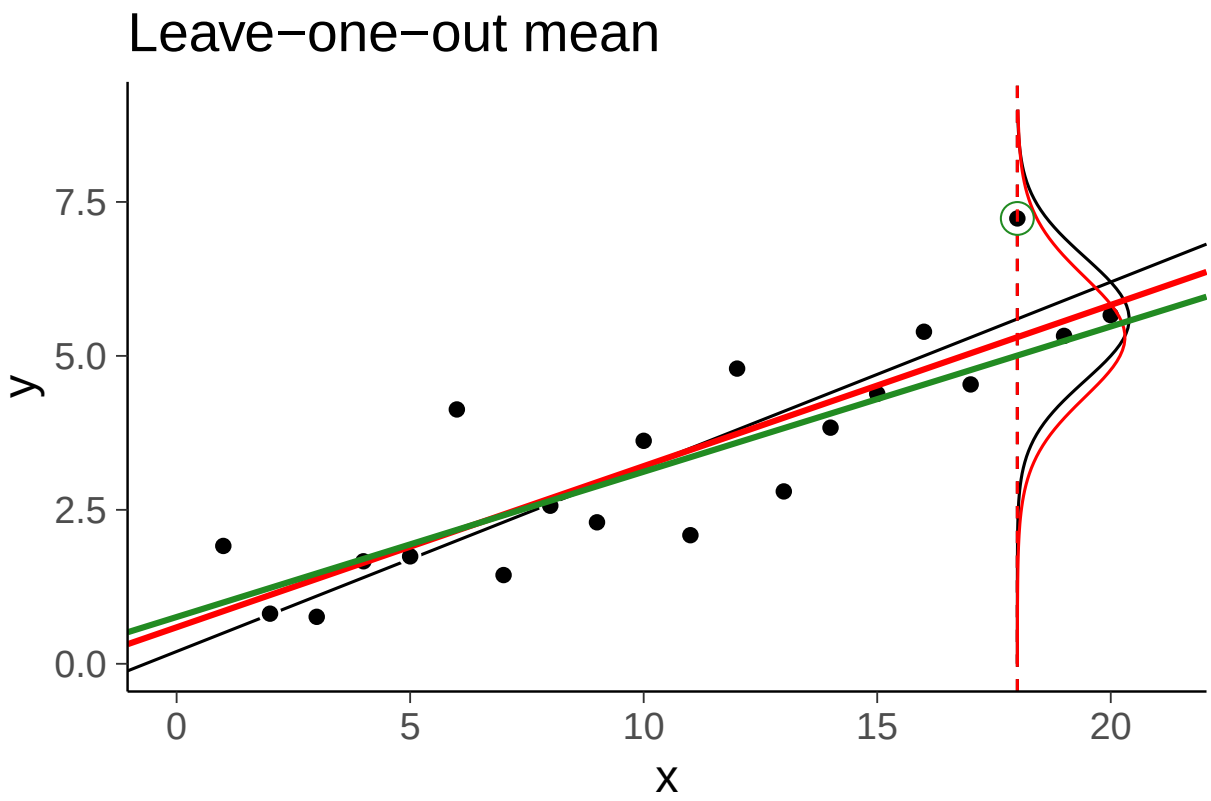




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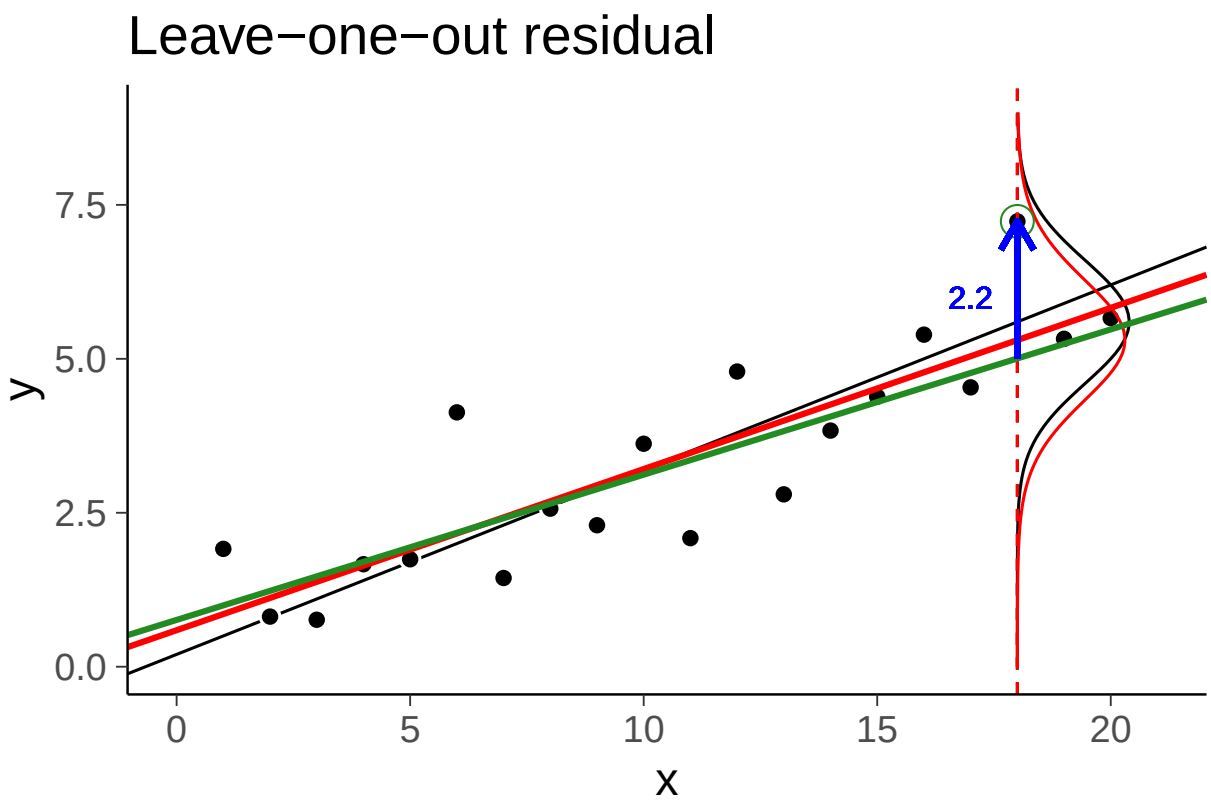




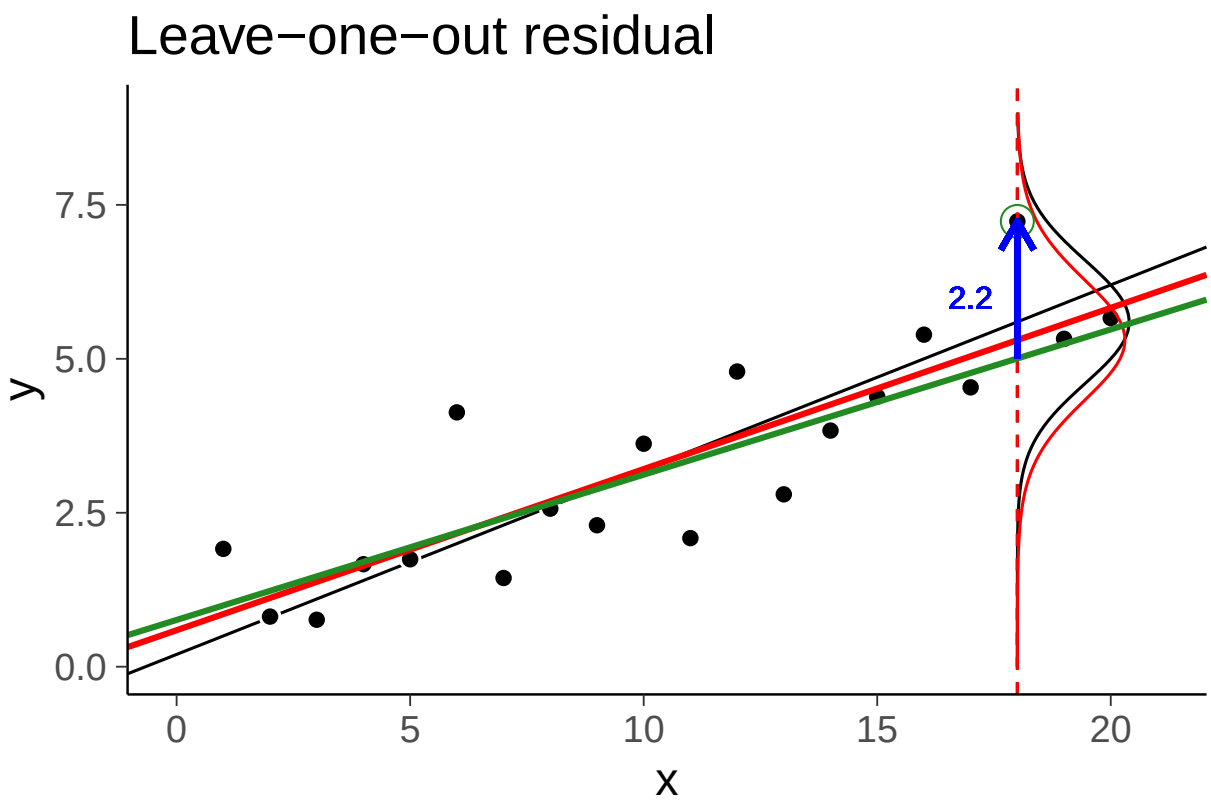
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# LOO example: a held-out prediction error



# LOO example: a held-out prediction error



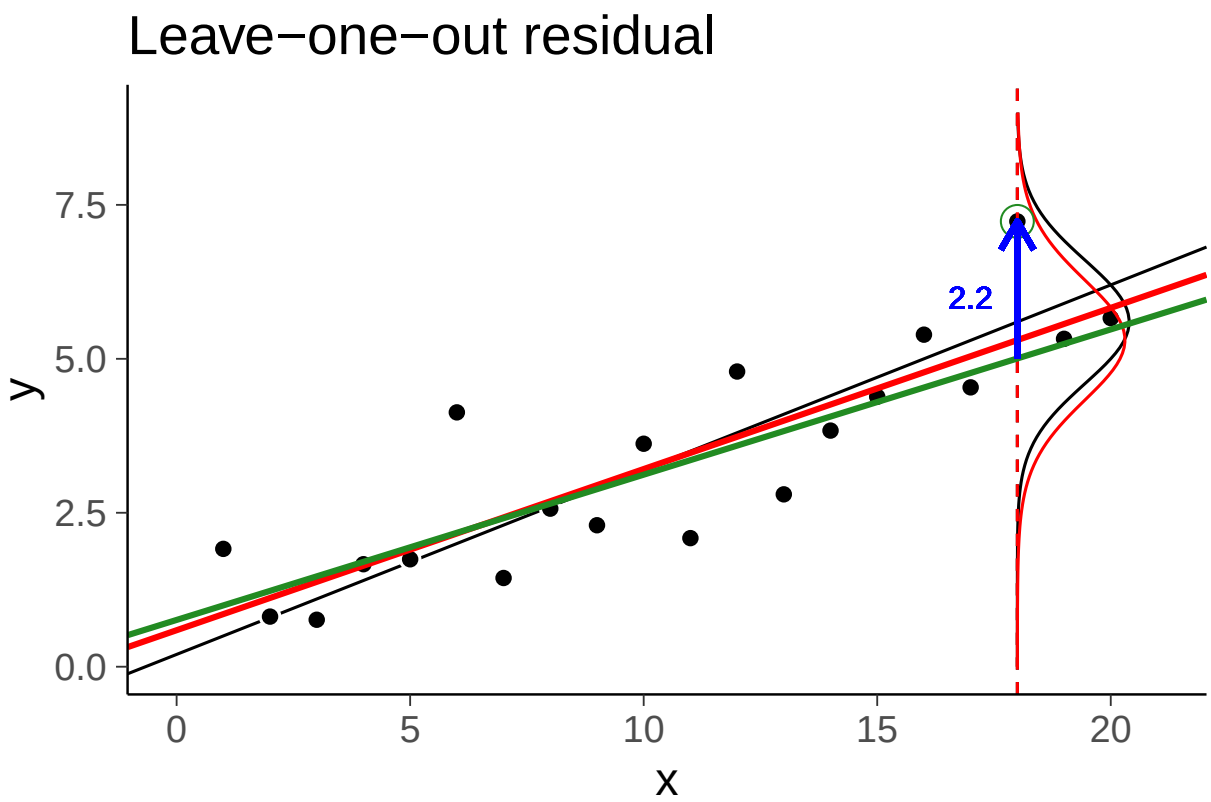

$$y_{18} - E[\tilde{y} \mid \tilde{x} = 18, x_{-18}, y_{-18}]$$



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# LOO example: a held-out prediction error




$y_{18} - E[\tilde{y} \mid \tilde{x} = 18, x_{-18}, y_{-18}]$

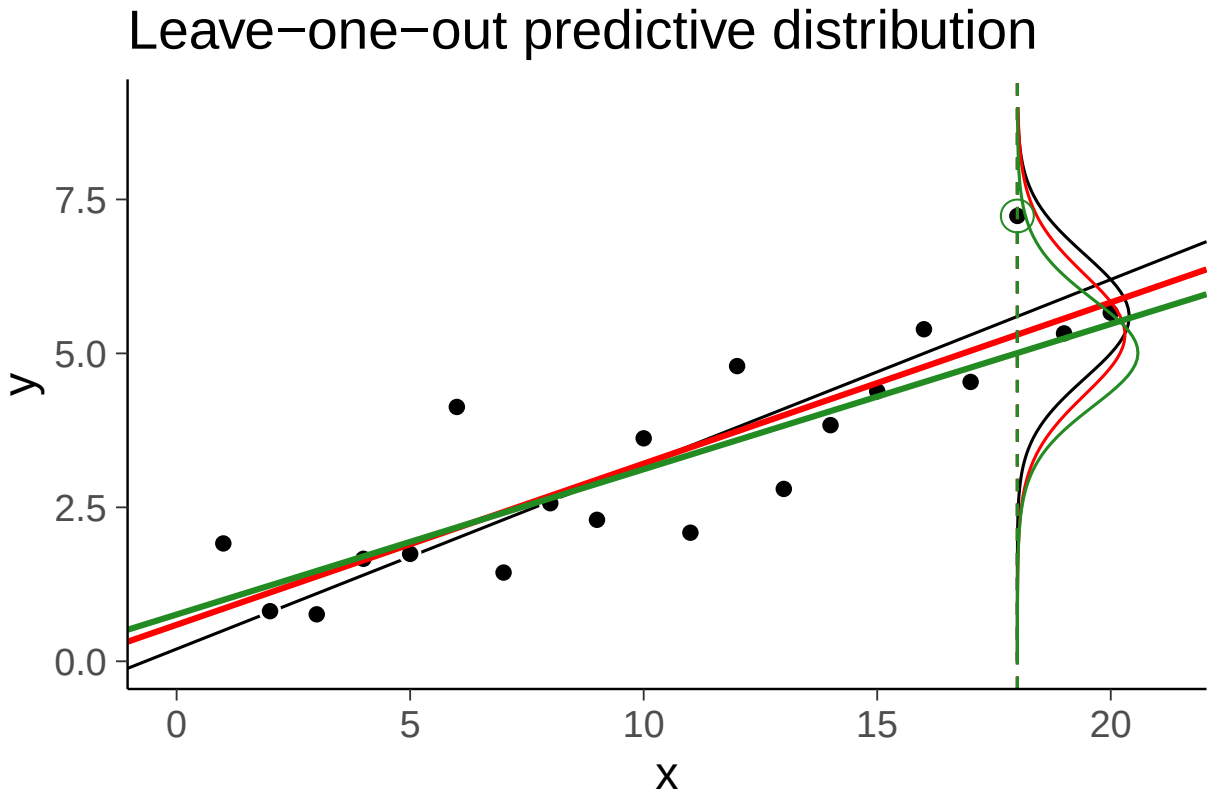
Held-out errors can be summarised using RMSE or an absolute-error quantile.



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# LOO example: the predictive distribution

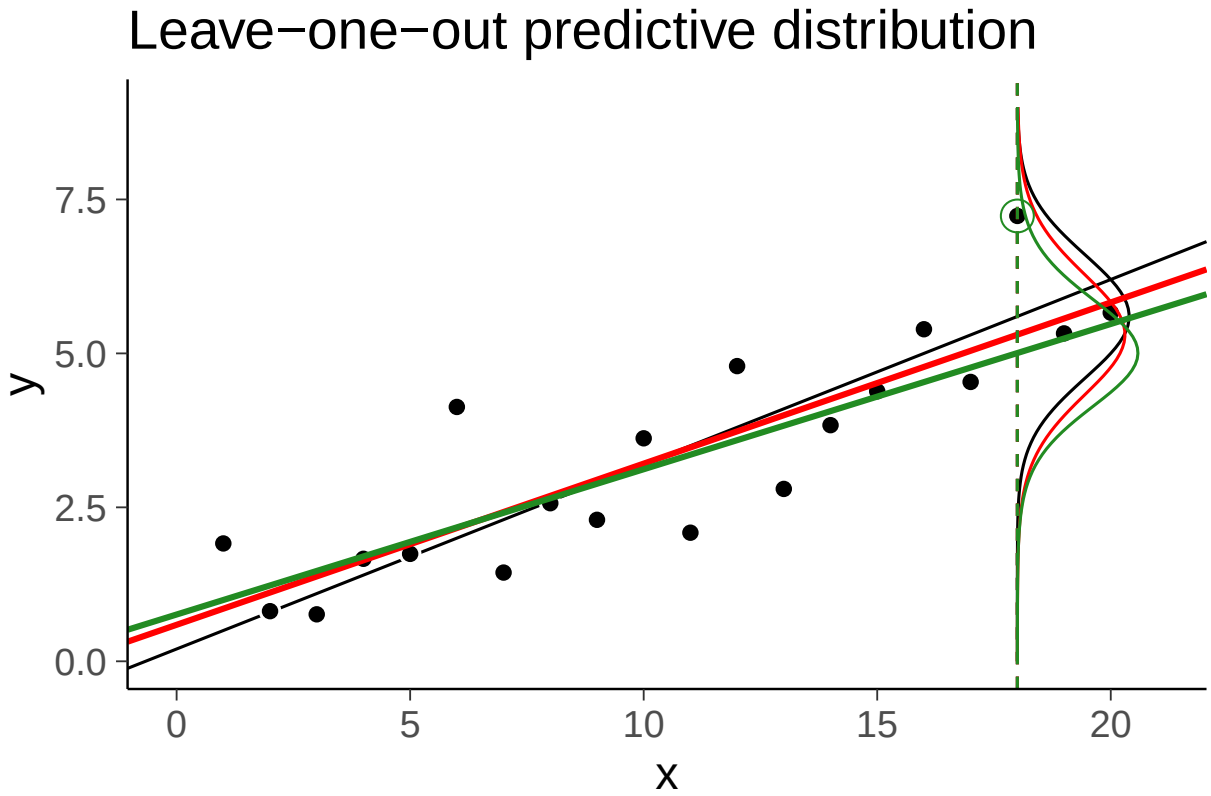




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# LOO example: the predictive distribution



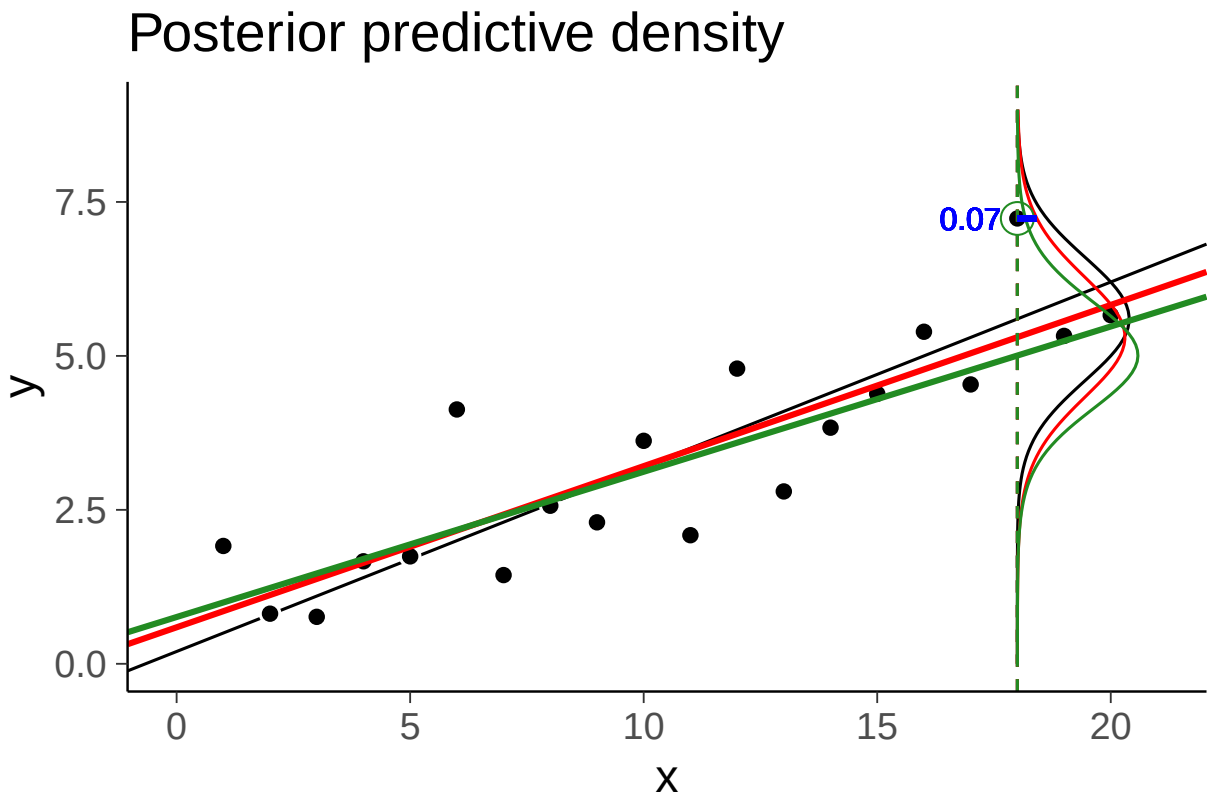

$$p(\tilde{y} \mid \tilde{x} = 18, x_{-18}, y_{-18}) = \int p(\tilde{y} \mid \tilde{x} = 18, \theta) \cdot p(\theta \mid x_{-18}, y_{-18}) d\theta.$$



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# Training and LOO predictive densities

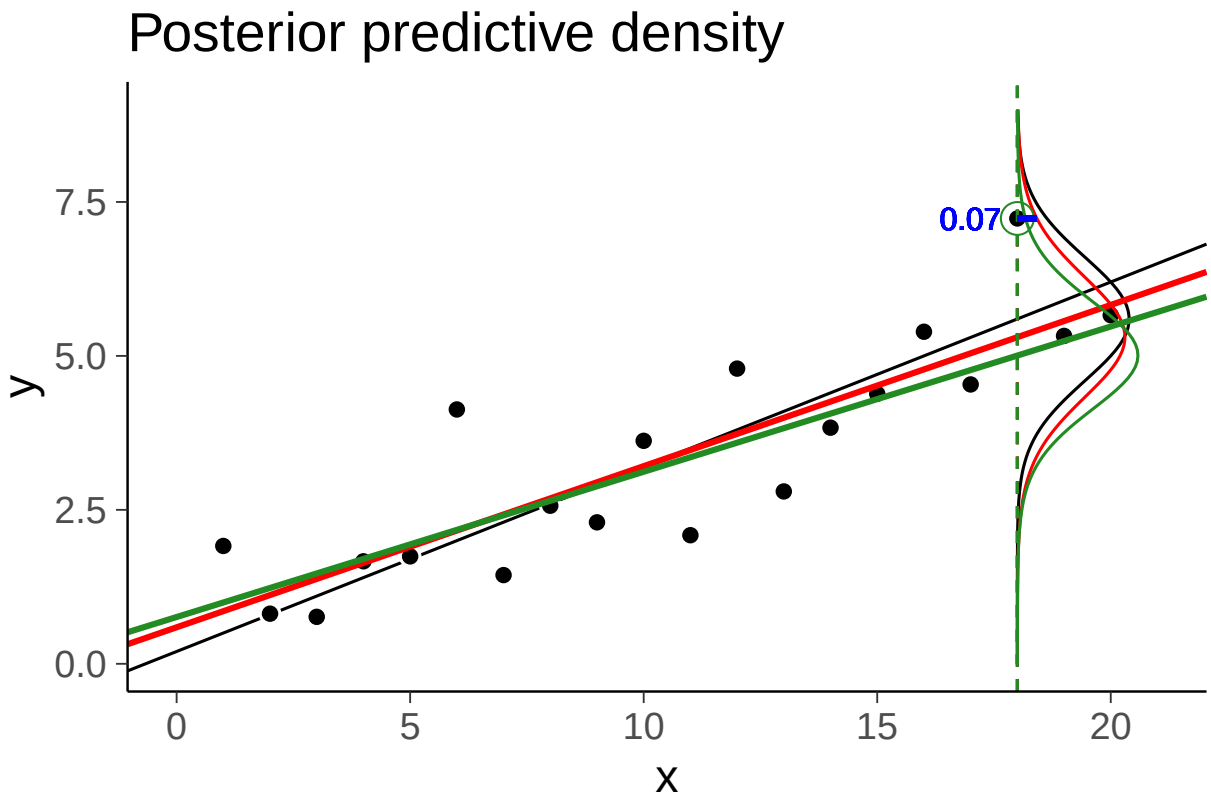




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# Training and LOO predictive densities



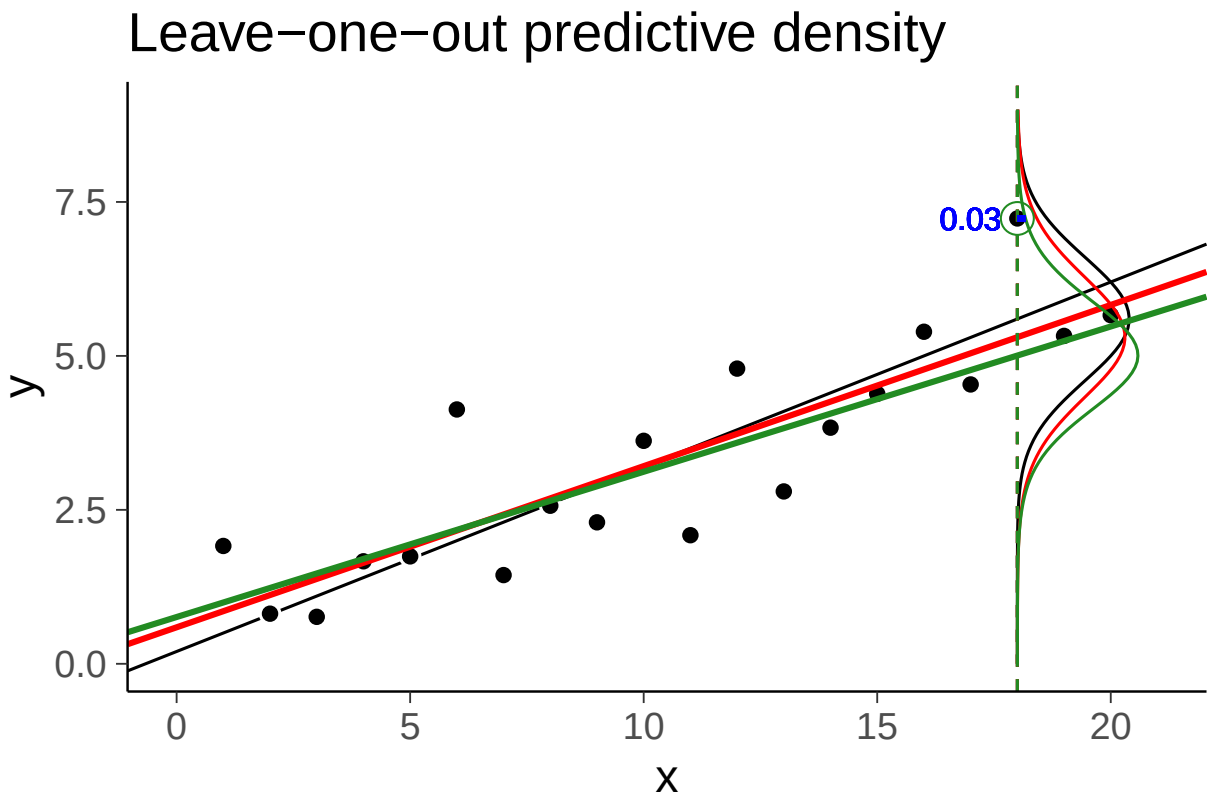

Training density: $p(y_{18} \mid \tilde{x} = 18, x, y) \approx 0.07$.



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# Training and LOO predictive densities




Training density: $p(y_{18} \mid \tilde{x} = 18, x, y) \approx 0.07$.

**LOO density:** $p(y_{18} \mid \tilde{x} = 18, x_{-18}, y_{-18}) \approx 0.03$.

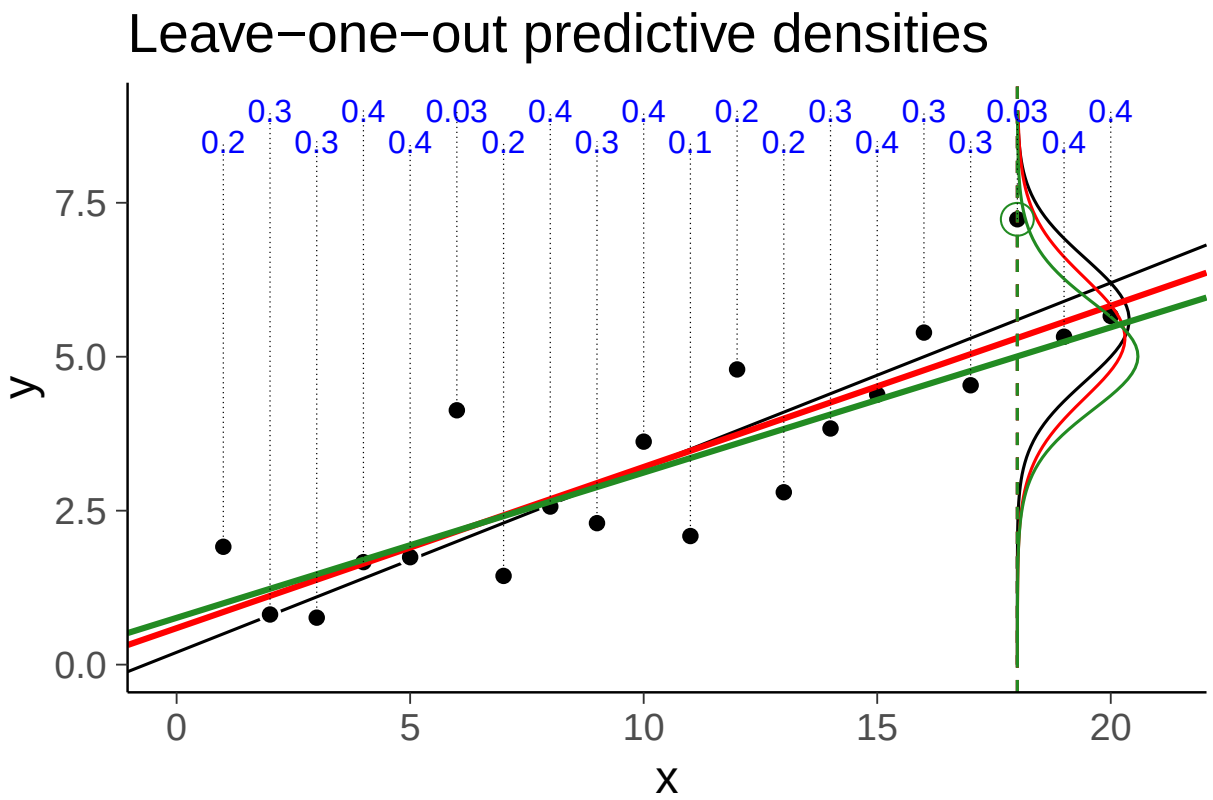
These are density heights, not probabilities of an exact outcome.



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# From pointwise densities to ELPD



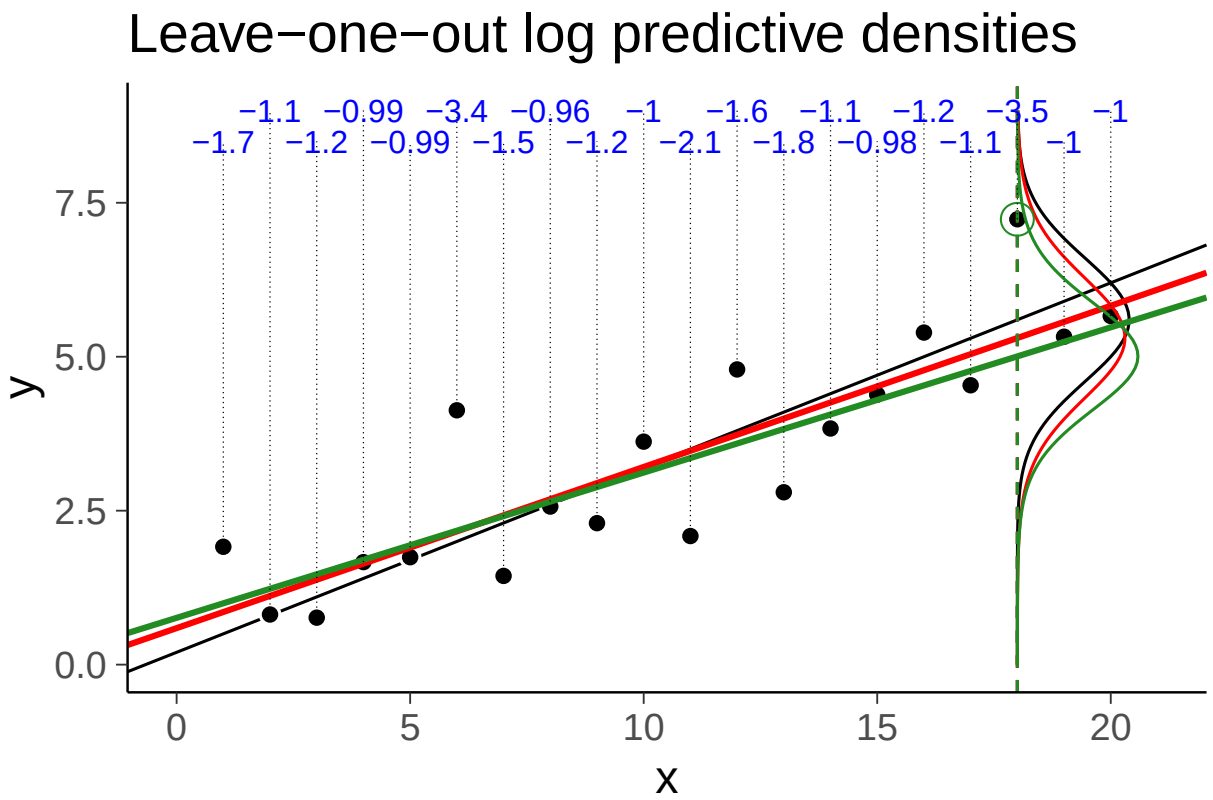

$p(y_i|x_i, x_{-i}, y_{-i}), \quad i = 1, \ldots, 20$



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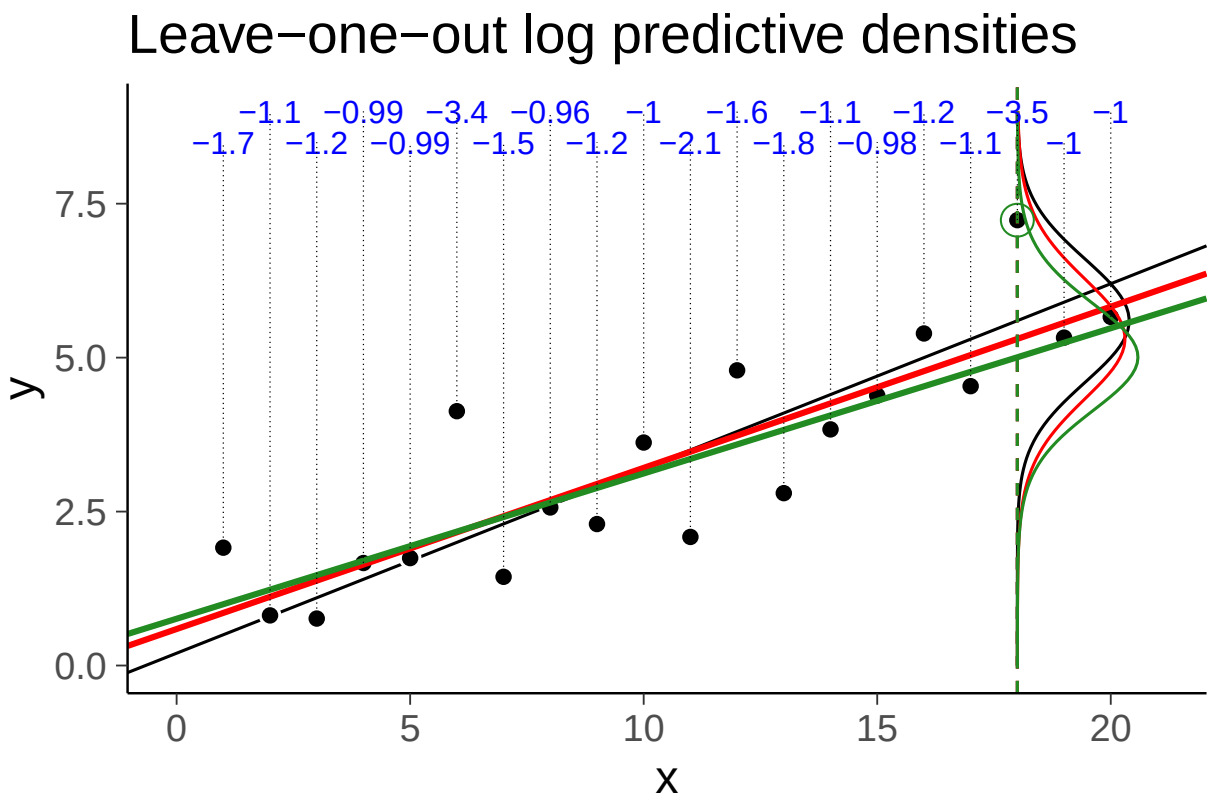

$\log p(y_i|x_i, x_{-i}, y_{-i}), \quad i = 1, \ldots, 20$



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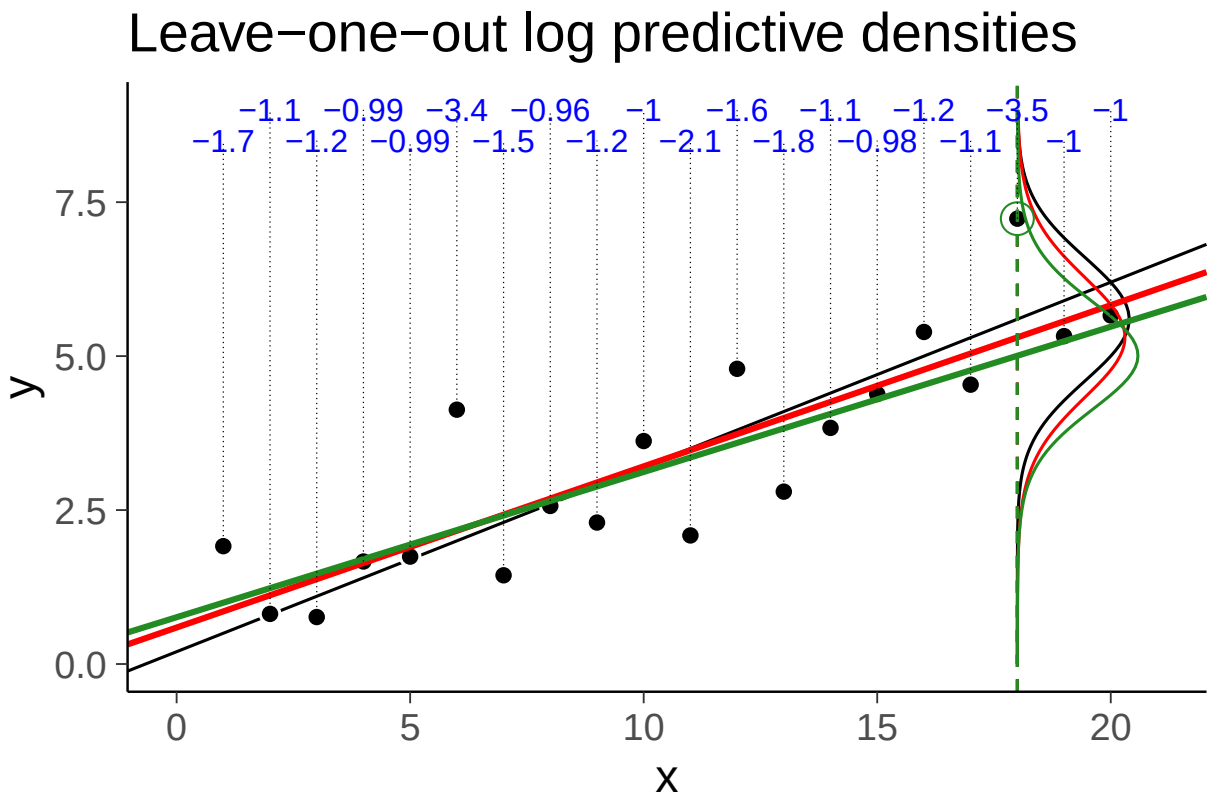

$\sum_{i=1}^{20} \log p(y_i|x_i, x_{-i}, y_{-i}) \approx -29.5$



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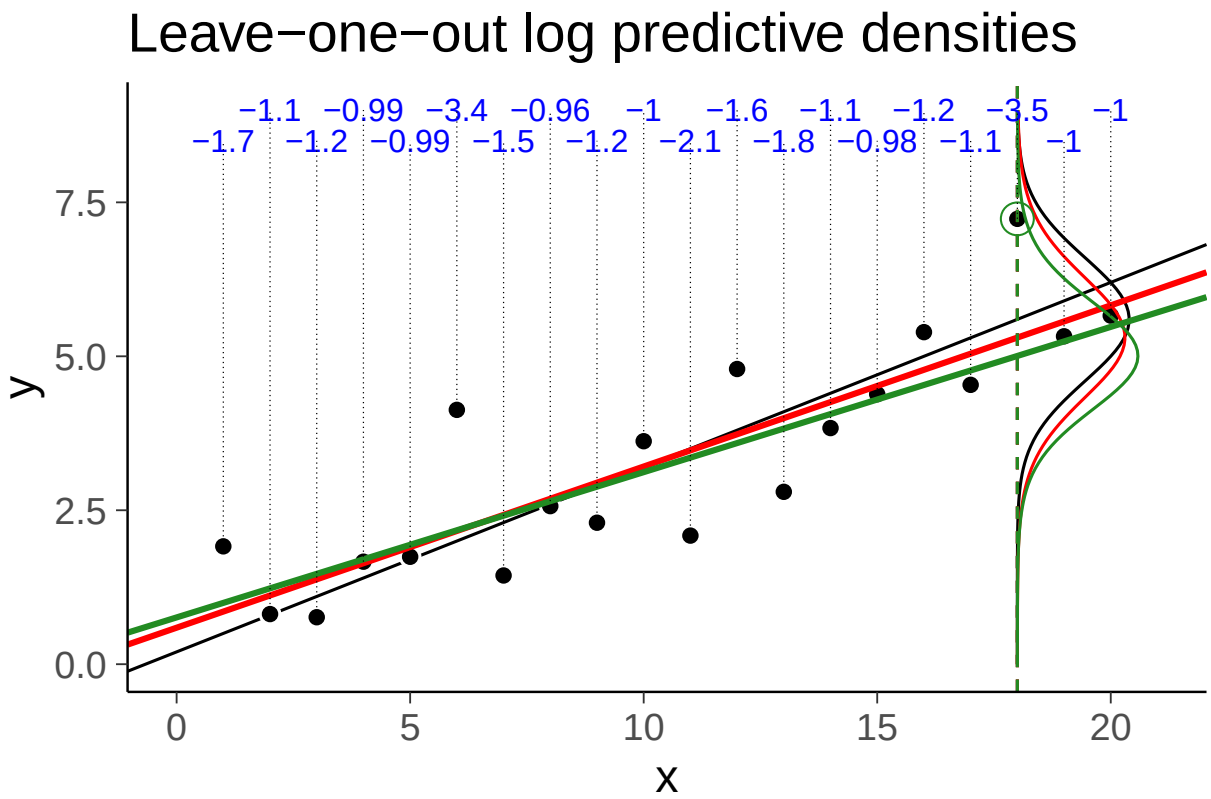

$$\text{elpd_loo} = \sum_{i=1}^{20} \log p(y_i|x_i, x_{-i}, y_{-i}) \approx -29.5$$



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$\text{elpd_loo} = \sum_{i=1}^{20} \log p(y_i|x_i, x_{-i}, y_{-i}) \approx -29.5$

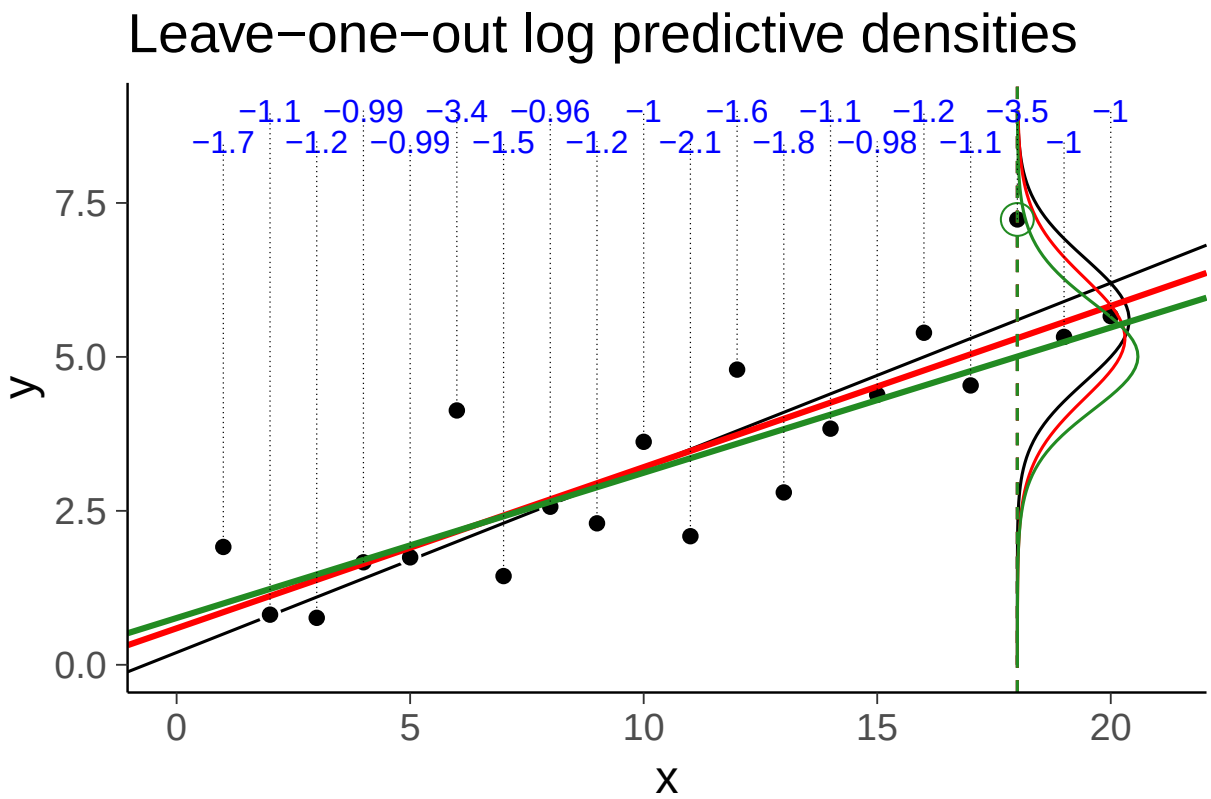
estimate of out-of-sample predictive performance



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$$\text{elpd_loo} = \sum_{i=1}^{20} \log p(y_i|x_i, x_{-i}, y_{-i}) \approx -29.5$$

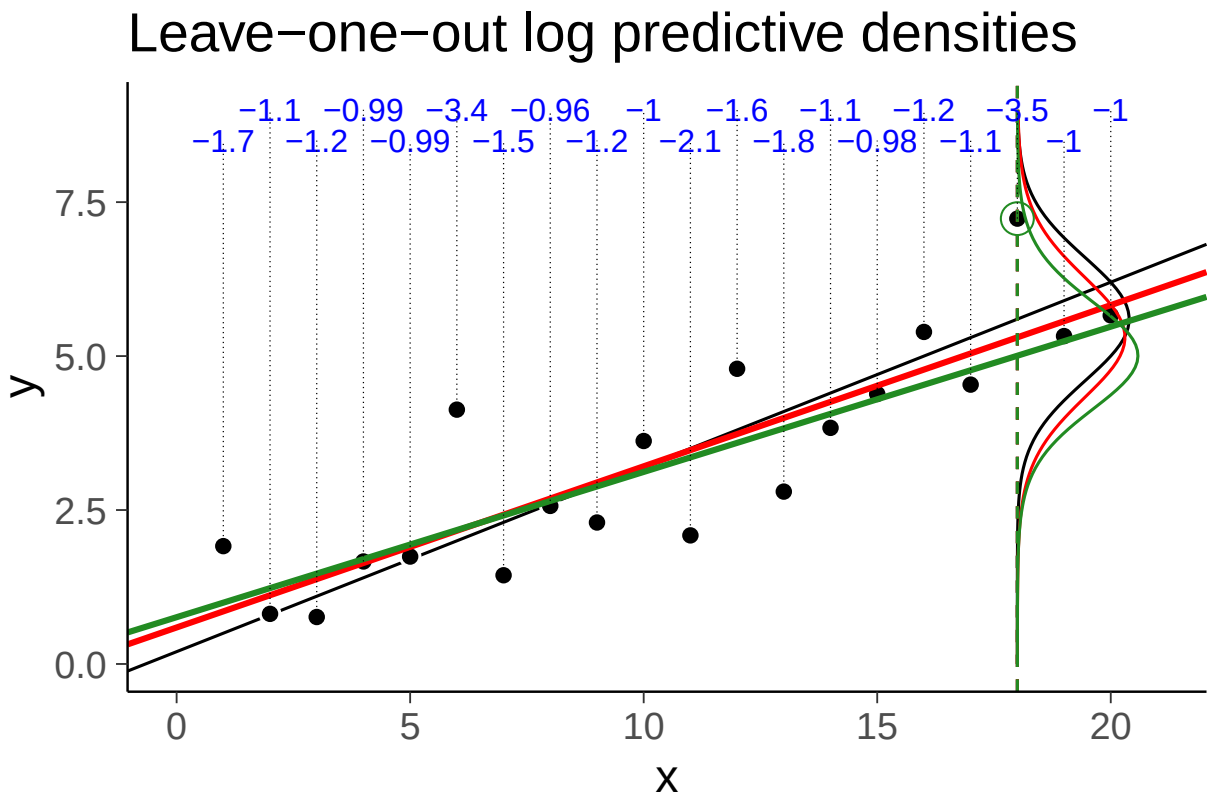
$$\text{lppd} = \sum_{i=1}^{20} \log p(y_i|x_i, x, y) \approx -26.8$$



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# From pointwise densities to ELPD




$$\text{elpd_loo} = \sum_{i=1}^{20} \log p(y_i|x_i, x_{-i}, y_{-i}) \approx -29.5$$

$$\text{lppd} = \sum_{i=1}^{20} \log p(y_i|x_i, x, y) \approx -26.8$$

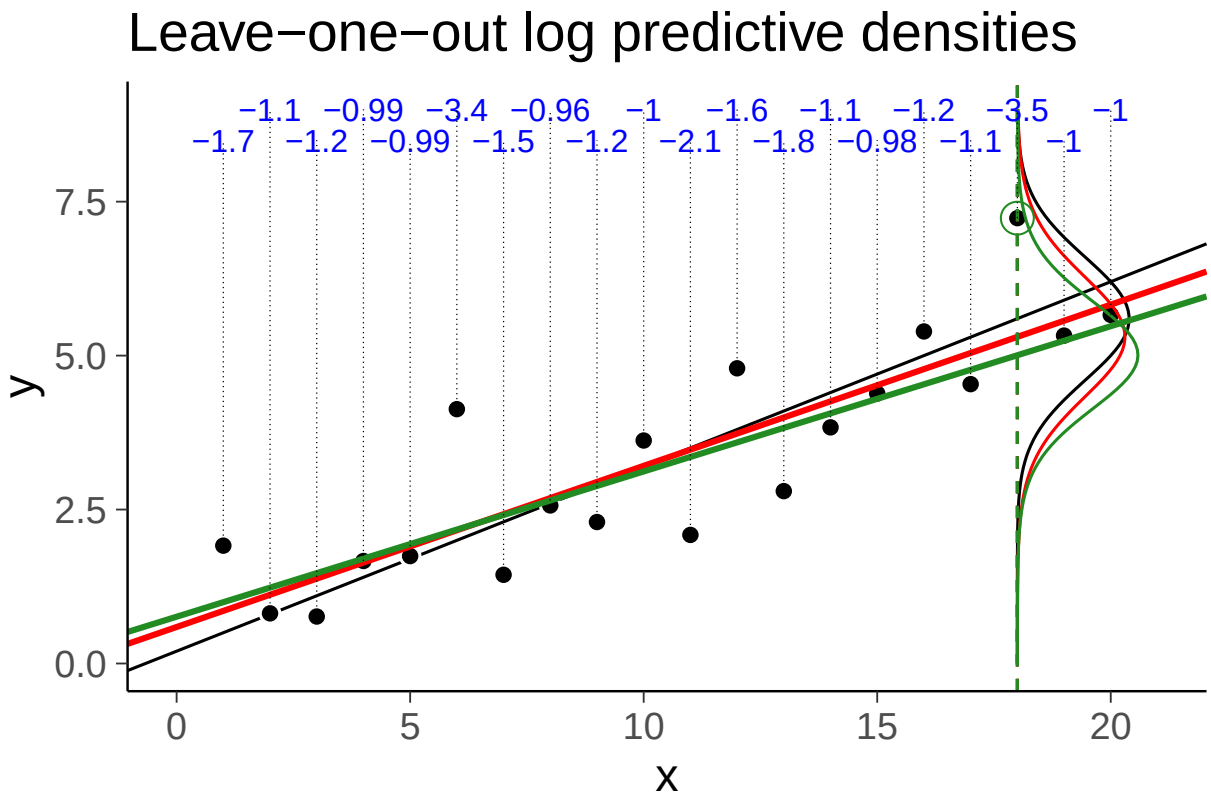
$$\text{p_loo} = \text{lppd} - \text{elpd_loo} \approx 2.7$$



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# From pointwise densities to ELPD




$$\text{elpd_loo} = \sum_{i=1}^{20} \log p(y_i|x_i, x_{-i}, y_{-i}) \approx -29.5$$

$$\text{SE} = \text{sd}(\log p(y_i|x_i, x_{-i}, y_{-i})) \cdot \sqrt{20} \approx 3.3$$



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# Stan and `loo` package

```
Computed from 4000 by 20 log-likelihood matrix

          Estimate  SE
elpd_loo    -29.5 3.3
p_loo         2.7 1.0
------

Monte Carlo SE of elpd_loo is 0.1.

Pareto k diagnostic values:
                         Count Pct.    Min. n_eff
(-Inf, 0.5]   (good)     18    90.0%   899
 (0.5, 0.7]   (ok)        2    10.0%   459
   (0.7, 1]   (bad)       0     0.0%   <NA>
   (1, Inf)   (very bad)  0     0.0%   <NA>

All Pareto k estimates are ok (k < 0.7).
See help('pareto-k-diagnostic') for details.
```



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Subsection 1

# When is LOO applicable



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# Pro and cons with LOO-CV

\+ Intuitive

\+ Robust

\+ Good theoretical properties



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# Pro and cons with LOO-CV

\+ Intuitive

\+ Robust

\+ Good theoretical properties

\- Can be costly (naive LOO-CV mean $n$ posterior computations)

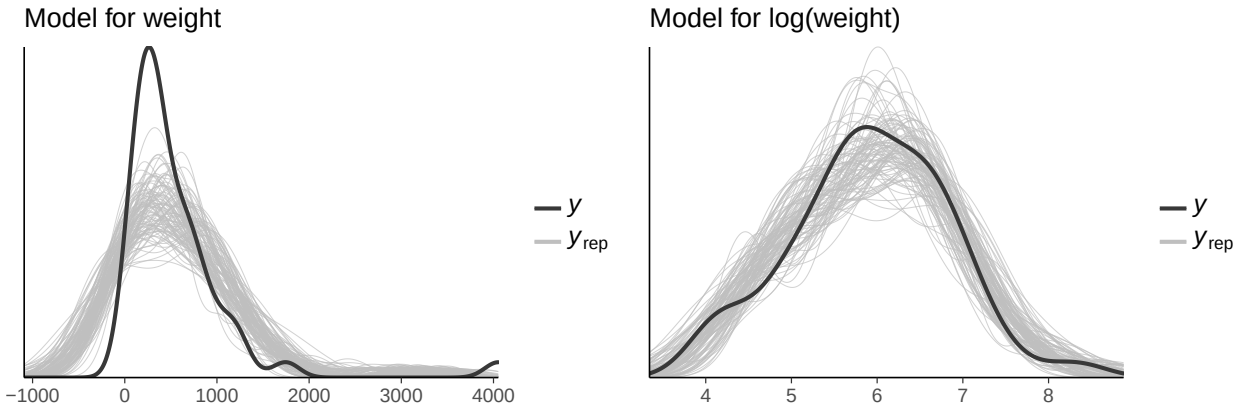


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# Sometimes cross-validation is not needed

- Posterior predictive checking can be sufficient




Predicting the yields of mesquite bushes.

Gelman, Hill & Vehtari (2020, Ch. 11)

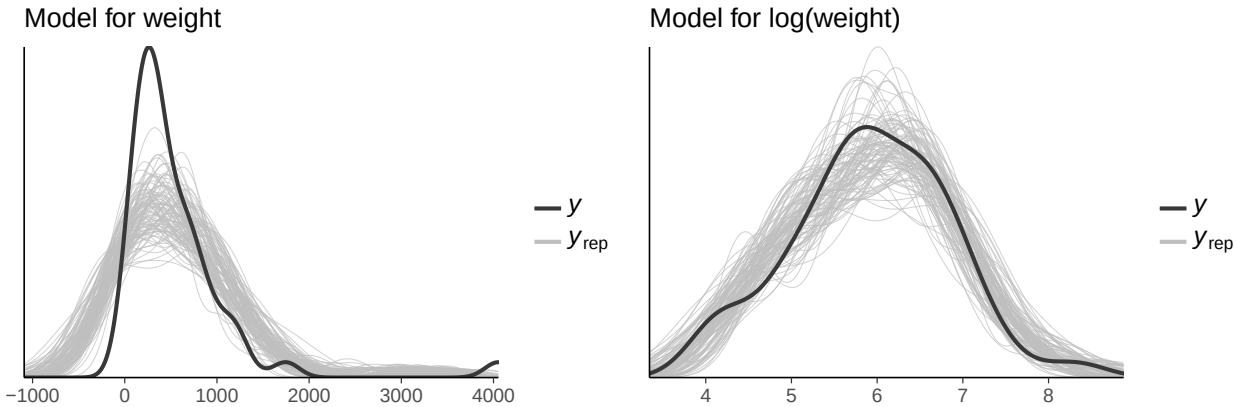


UPPSALA UNIVERSITET


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# Sometimes cross-validation is not needed

- Posterior predictive checking can be sufficient




Predicting the yields of mesquite bushes.

Gelman, Hill & Vehtari (2020, Ch. 11)

- In nested case, often easier and more accurate to analyse posterior distribution of more complex model directly



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# Data generating mechanisms and prediction tasks

1. You have to make some assumptions on data generating mechanism $p_{\text{true}}$

$$\text{elpd}_M = \sum_{i=1}^{n} \int \log p_M(\tilde{y}_i \mid y) p_{\text{true}}(\tilde{y}_i) d\tilde{y}_i \,,$$

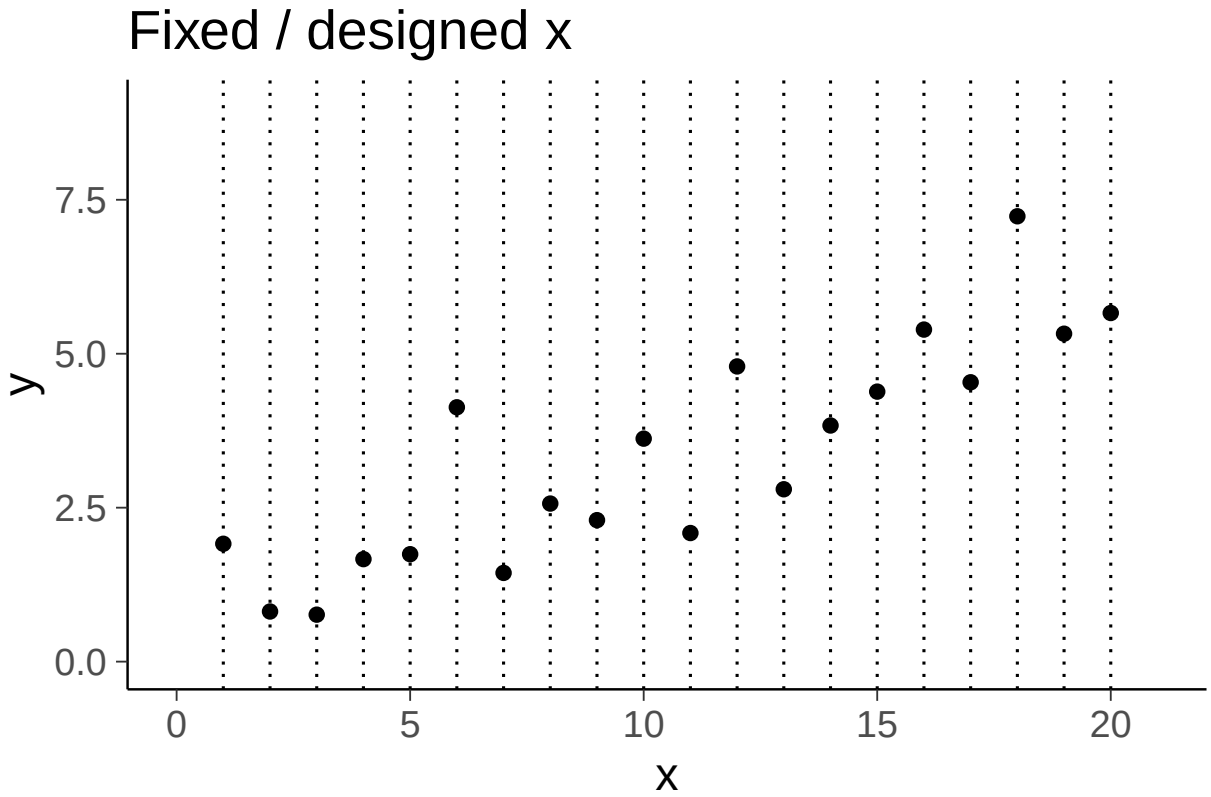
2. Use the knowledge of the prediction task if available
3. Cross-validation can be used to analyse different parts, even if there is no clear prediction task



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# $x$ in $p_{\text{true}}$



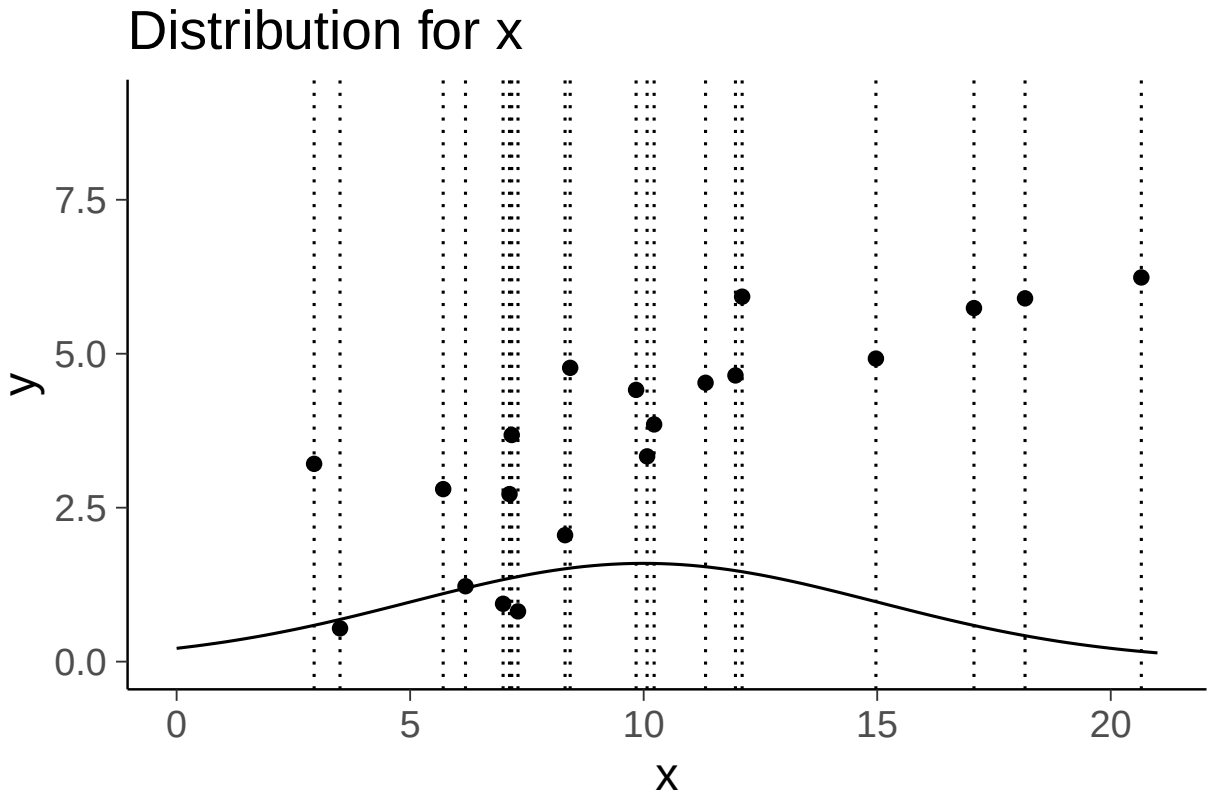

LOO is ok for fixed / designed $x$: $p_{\text{true}}(y|x)$



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# $x$ in $p_{\text{true}}$



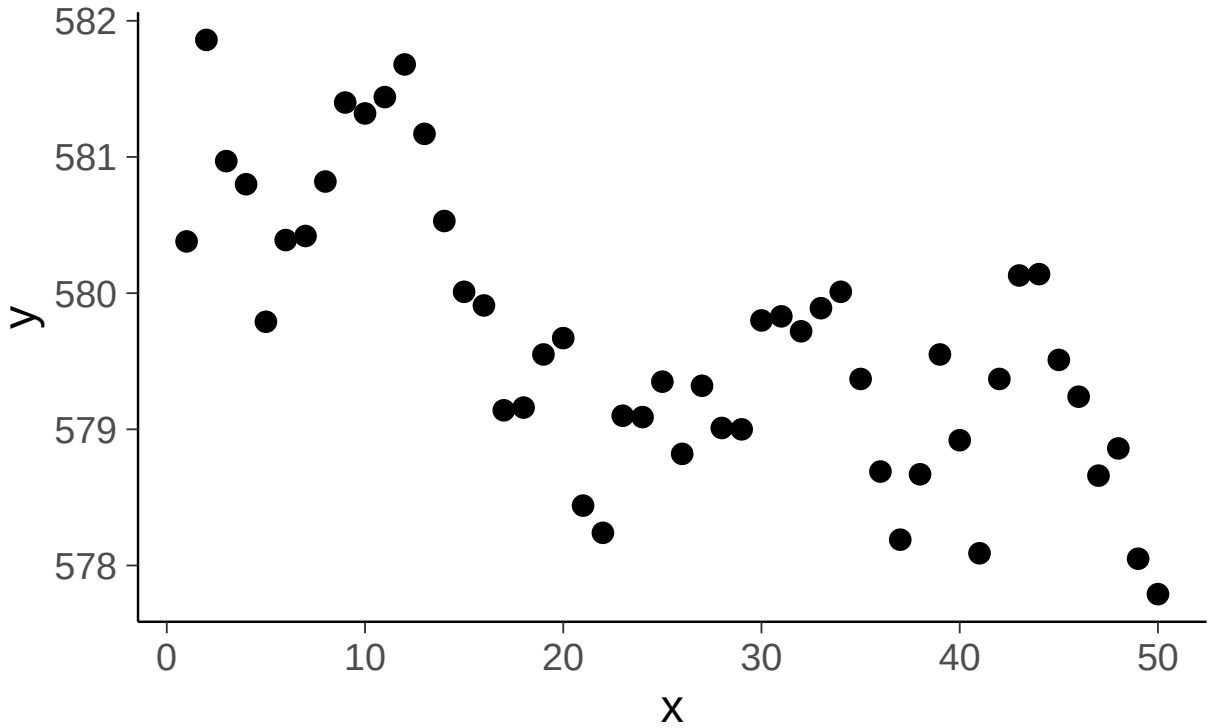

LOO is ok for random $x$. $p_{\text{true}}(y, x)$



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# $p_{\text{true}}$ extrapolation

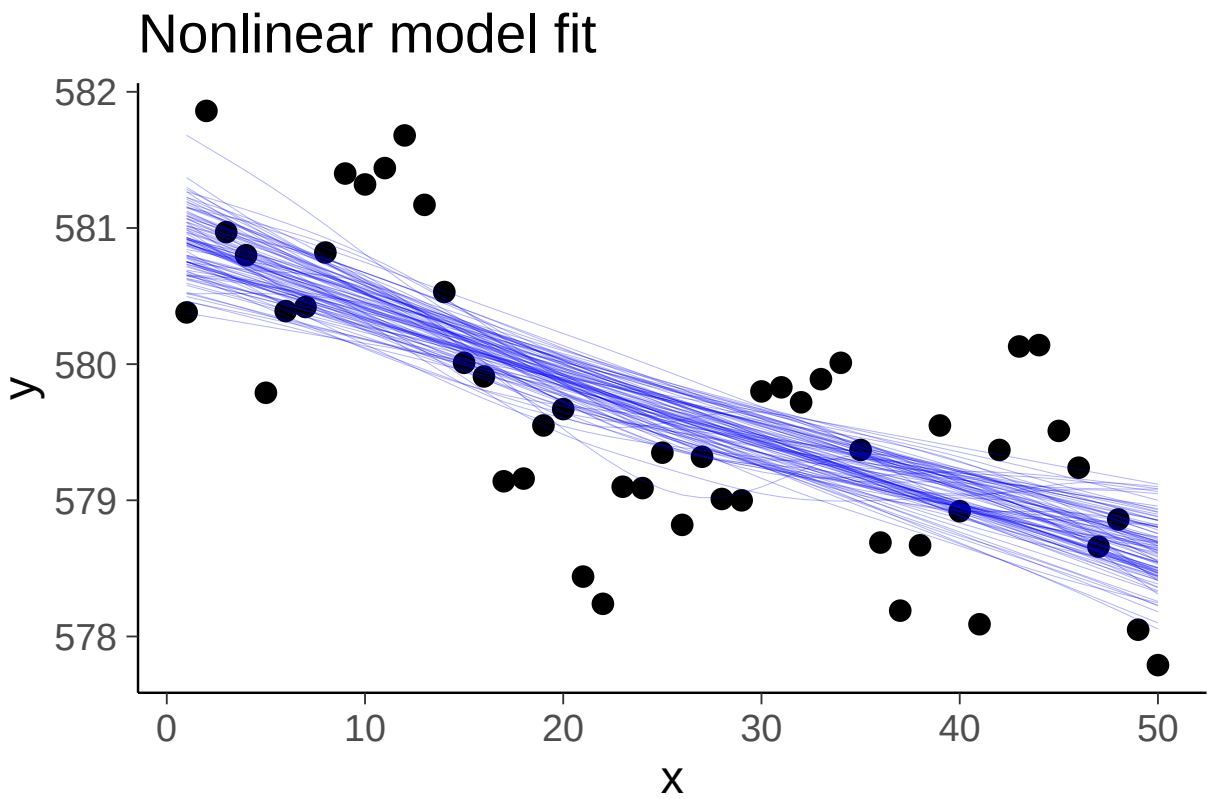




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# $p_{\text{true}}$ extrapolation

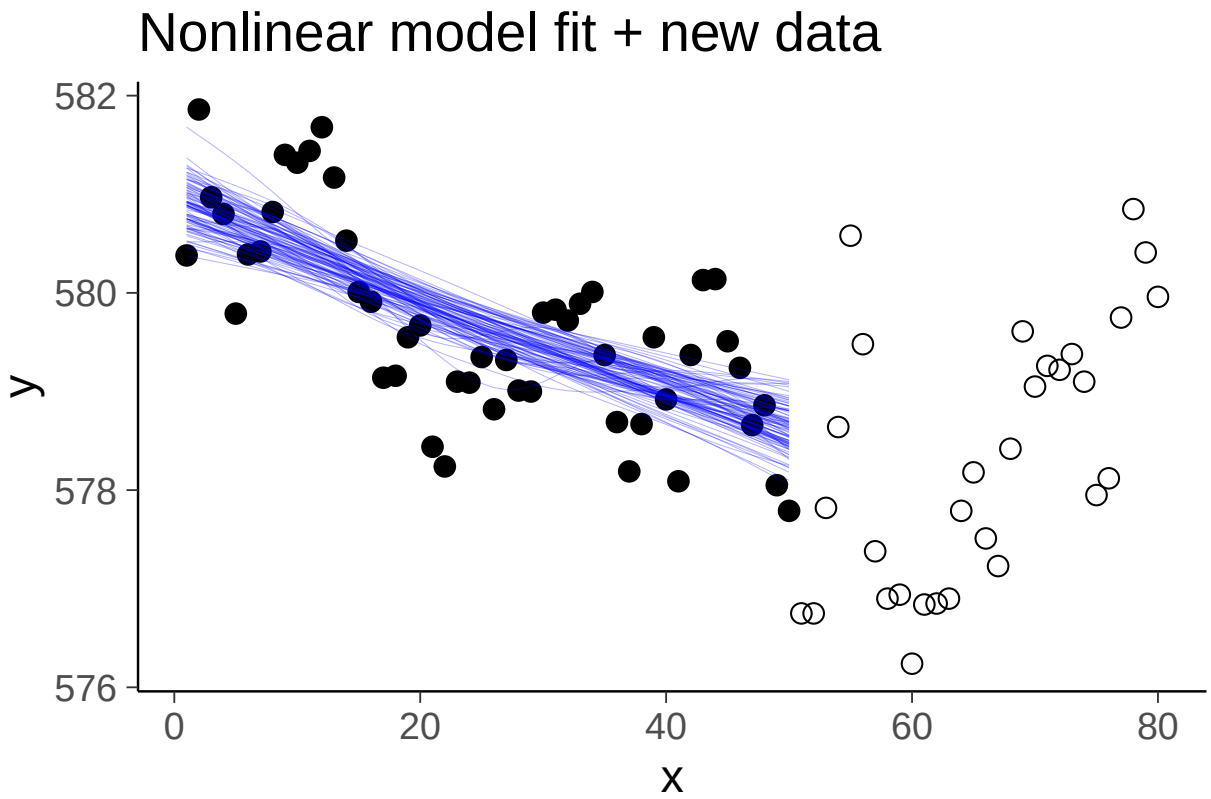




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# $p_{\text{true}}$ extrapolation



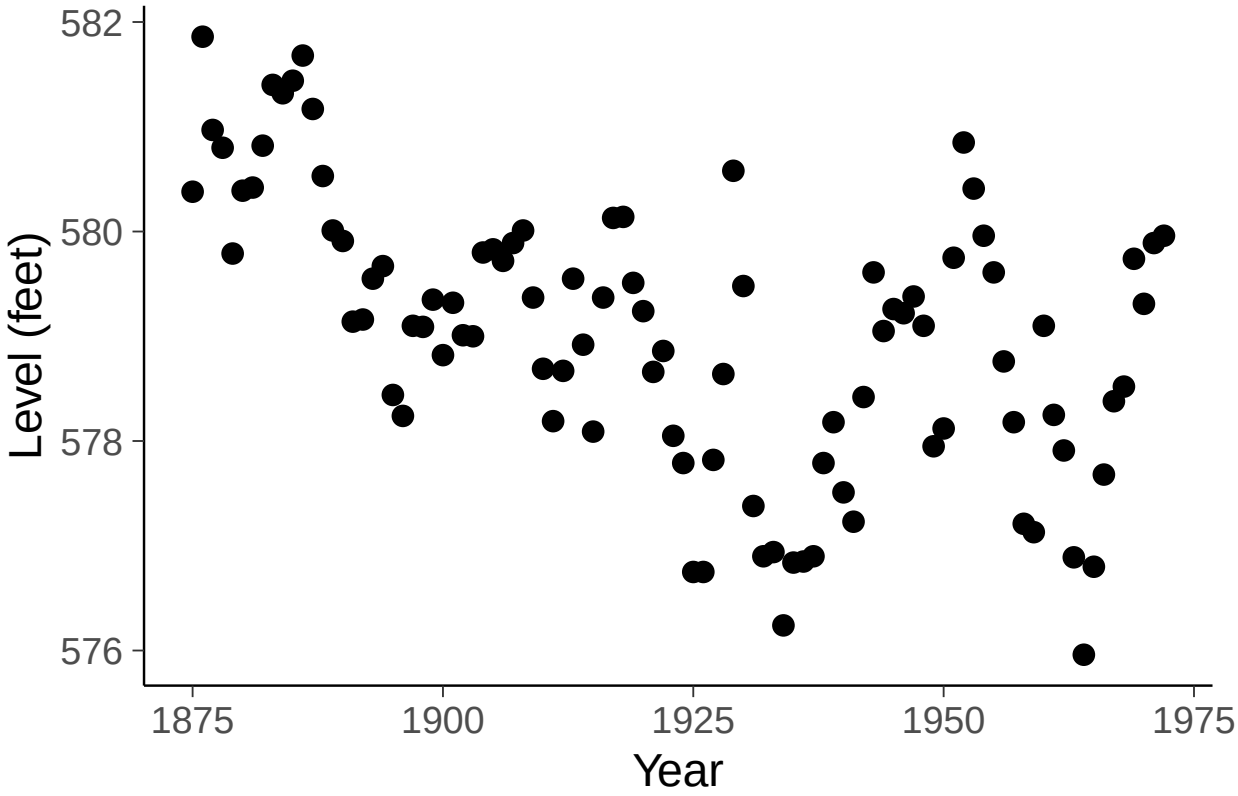

Extrapolation is more difficult



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# LOO for time series data



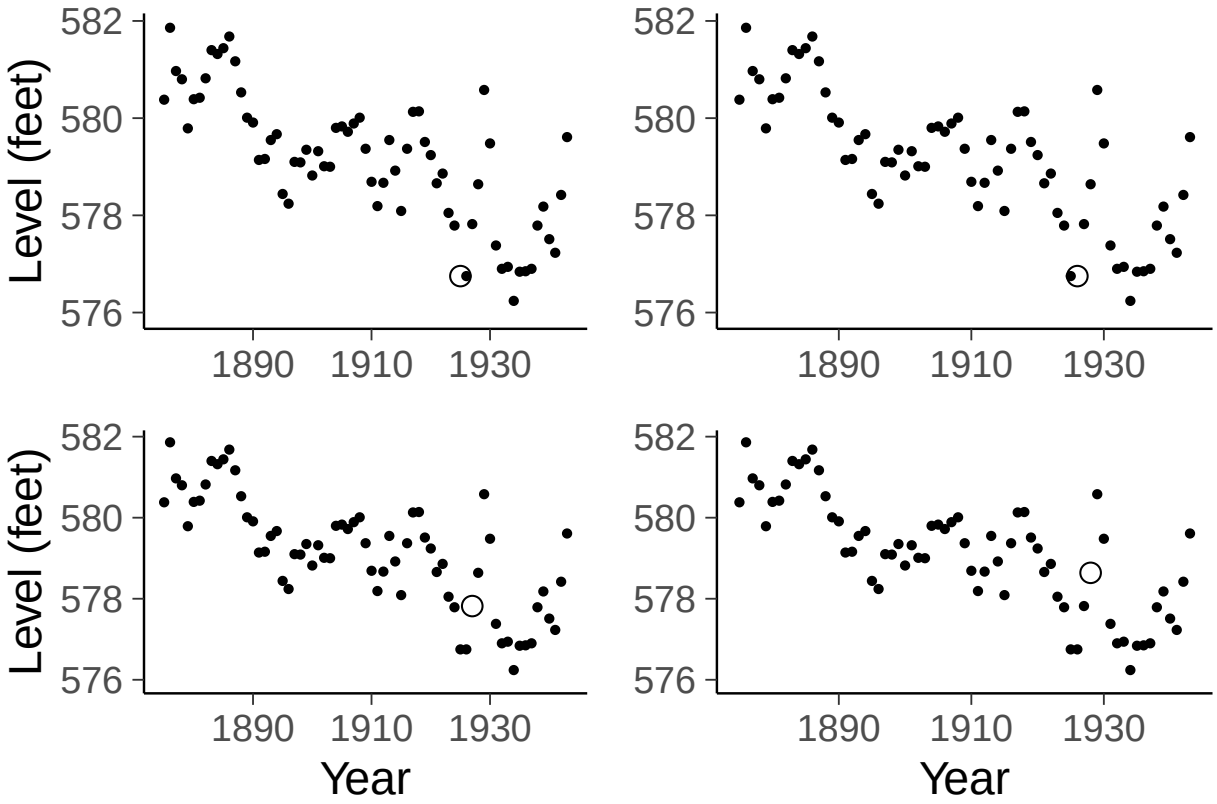

$p_{\text{true}}$ for time series?



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# LOO for time series



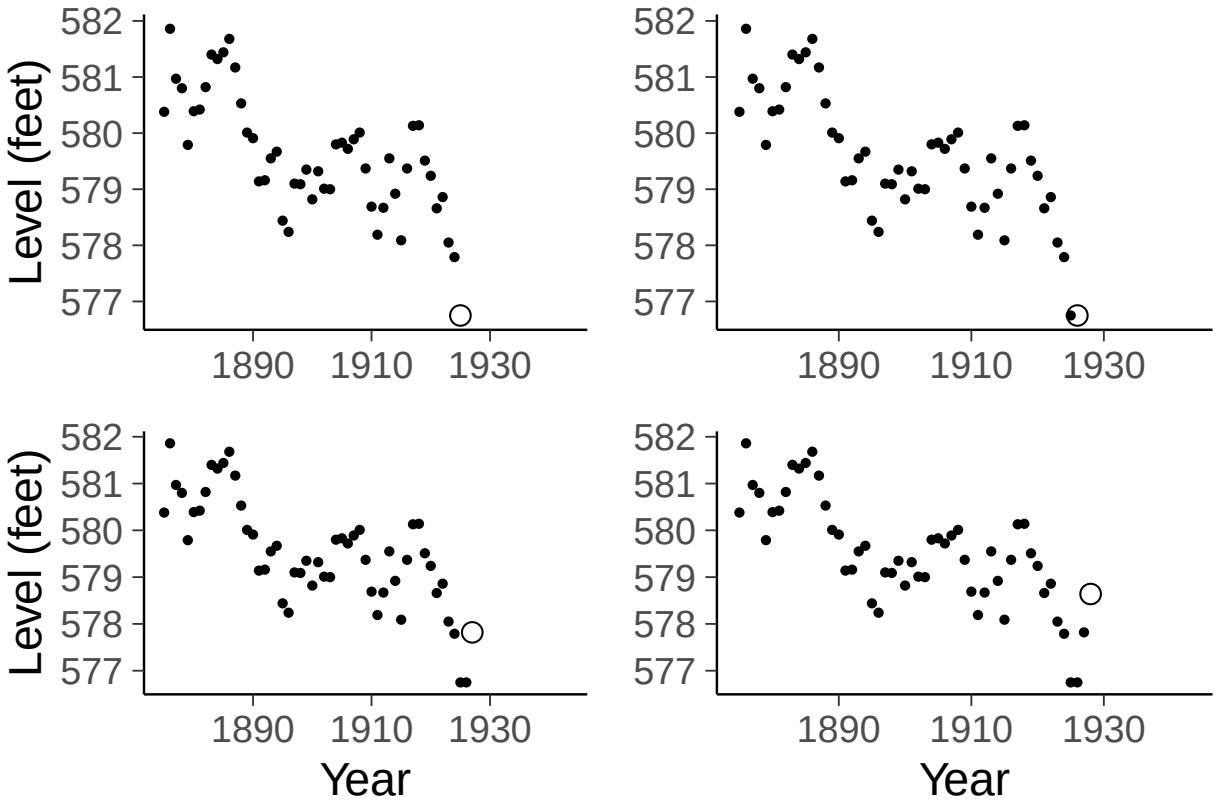

Leave-one-out cross-validation is ok for assessing conditional model



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# LOO for time series



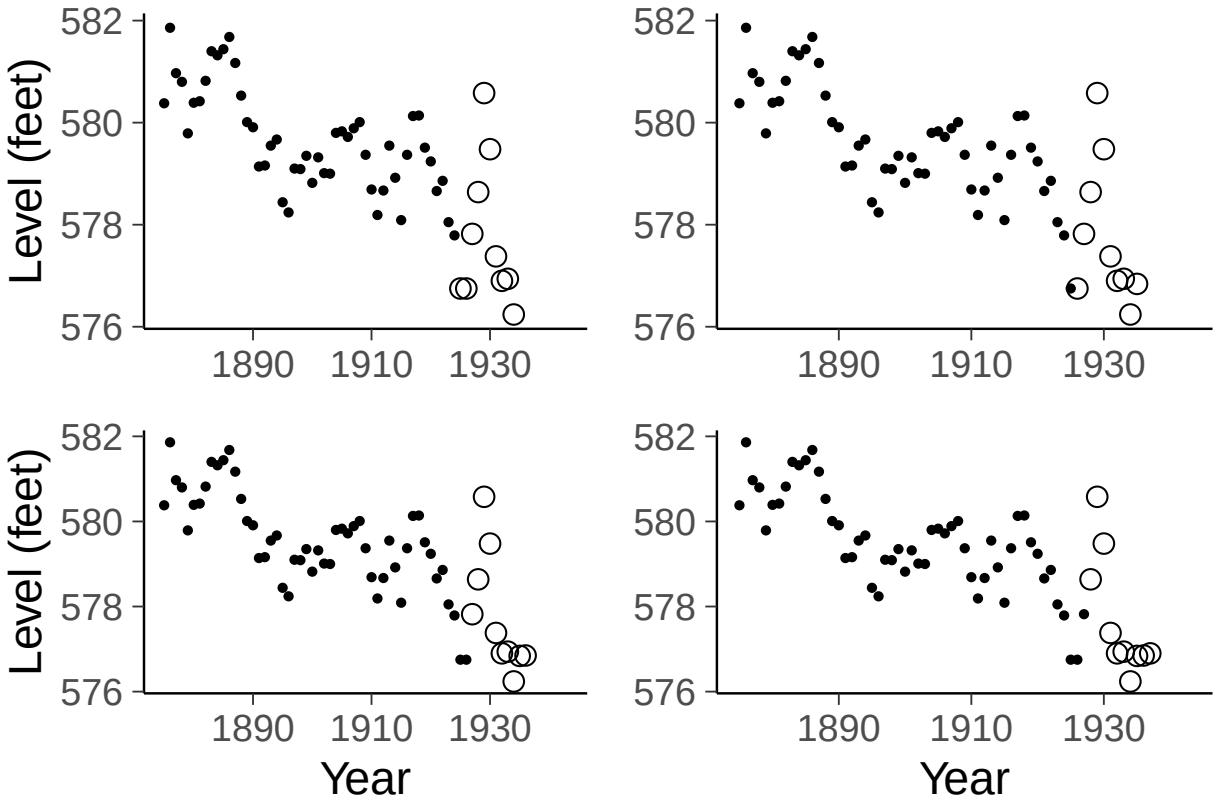

Leave-future-out cross-validation is better for predicting future



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# LOO for time series



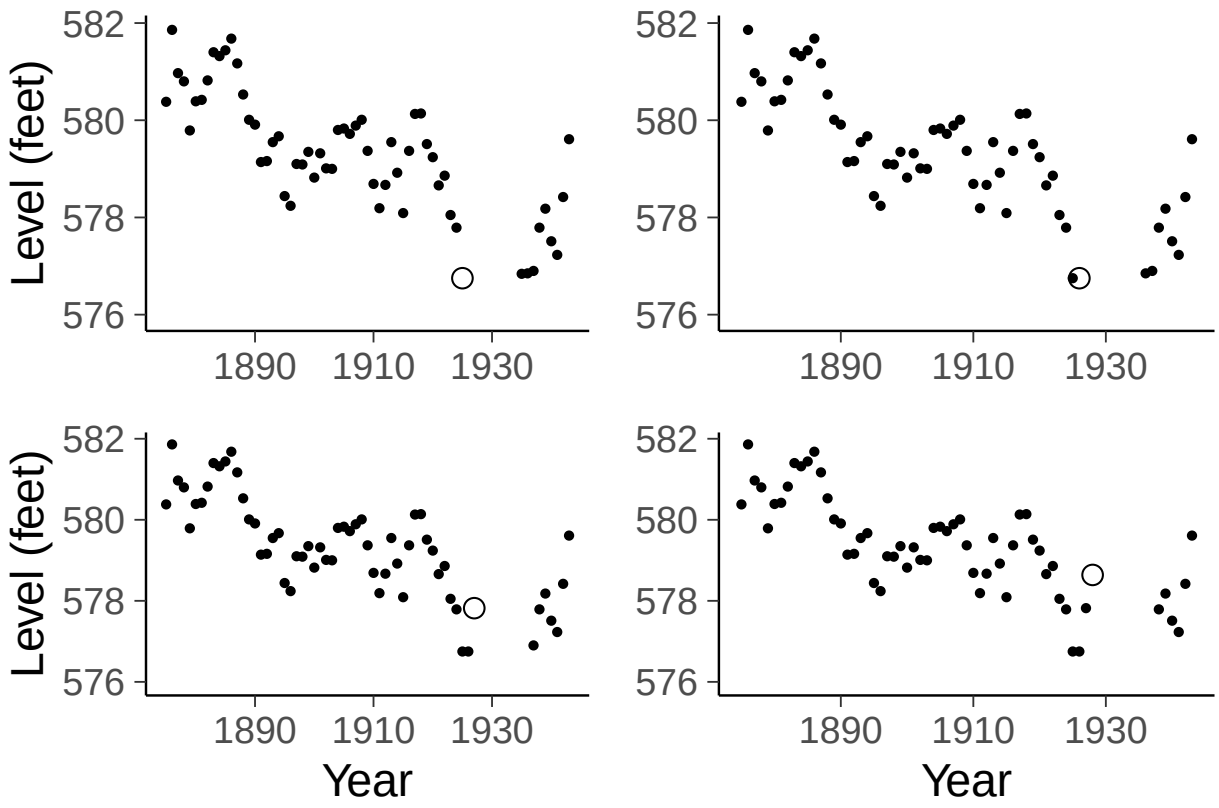

*m*-step-ahead cross-validation is better for predicting further future



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# LOO for time series



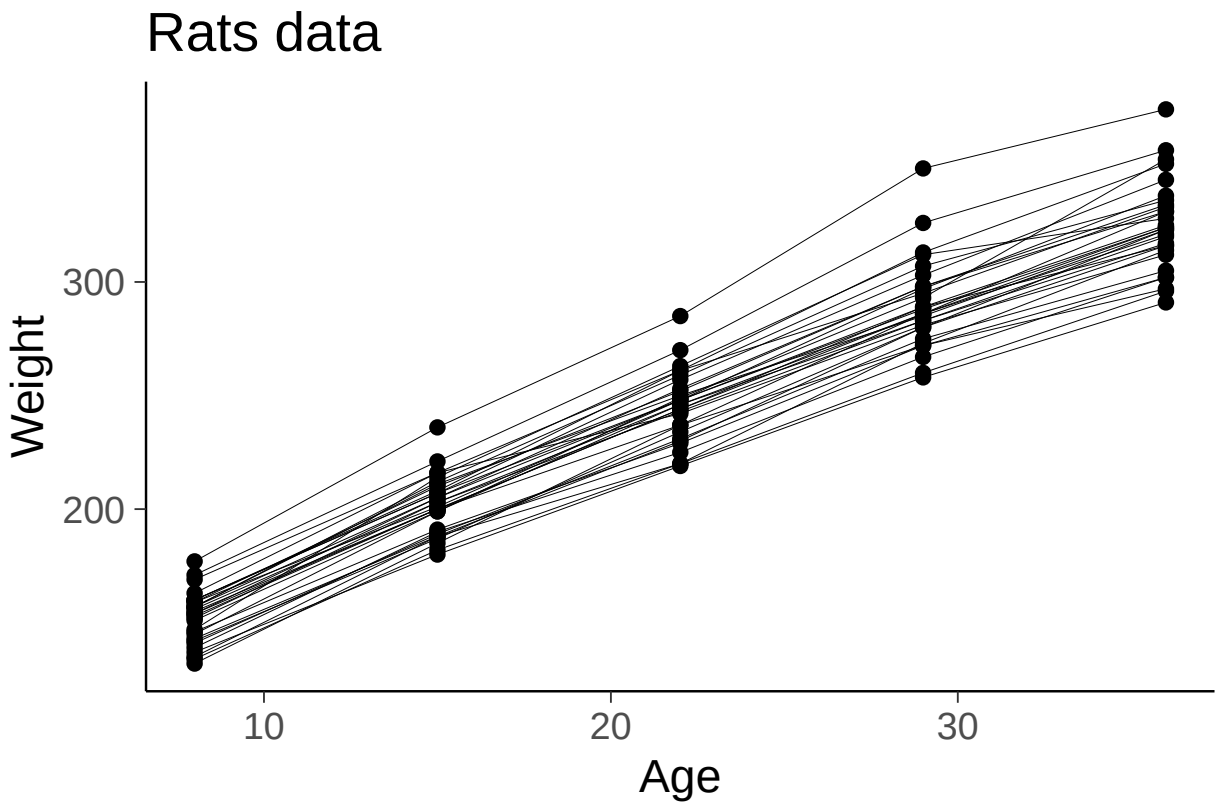

*m*-step-ahead leave-a-block-out cross-validation



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# LOO for hierarchical data



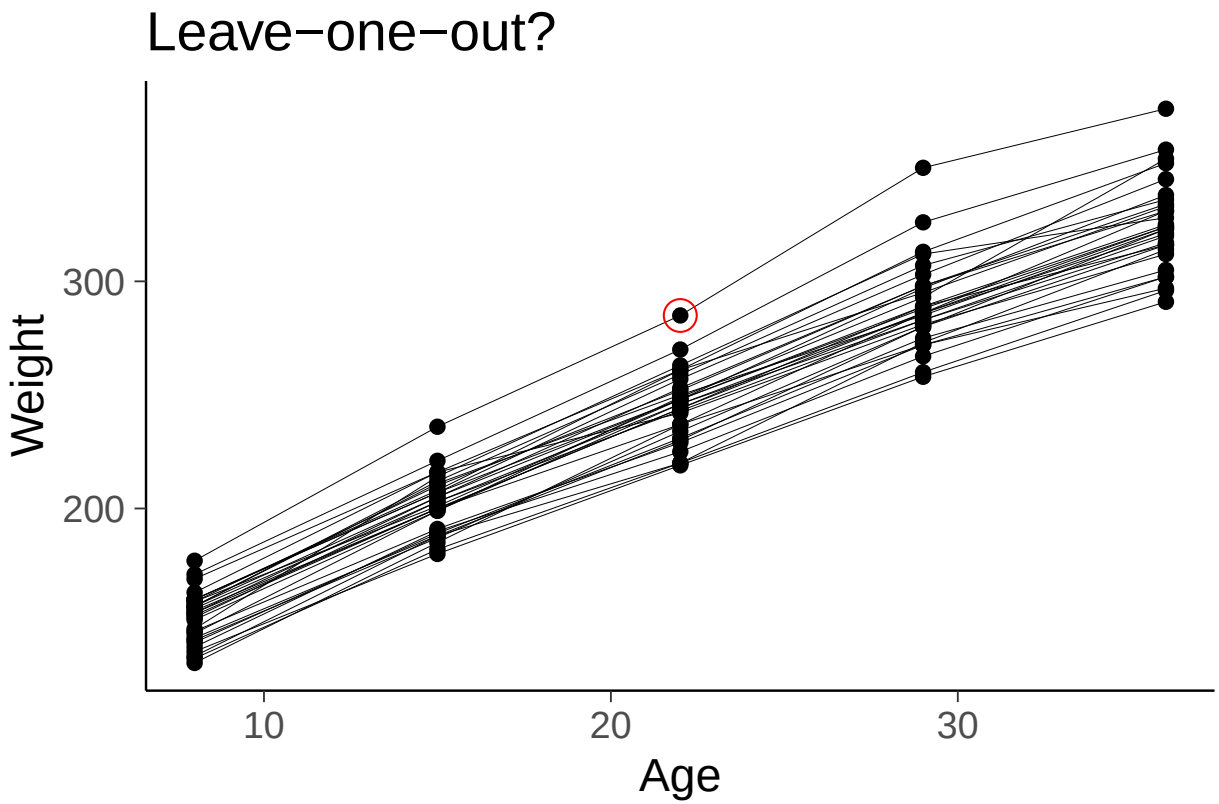

Can LOO or other cross-validation be used with hierarchical data?



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# LOO for hierarchical data



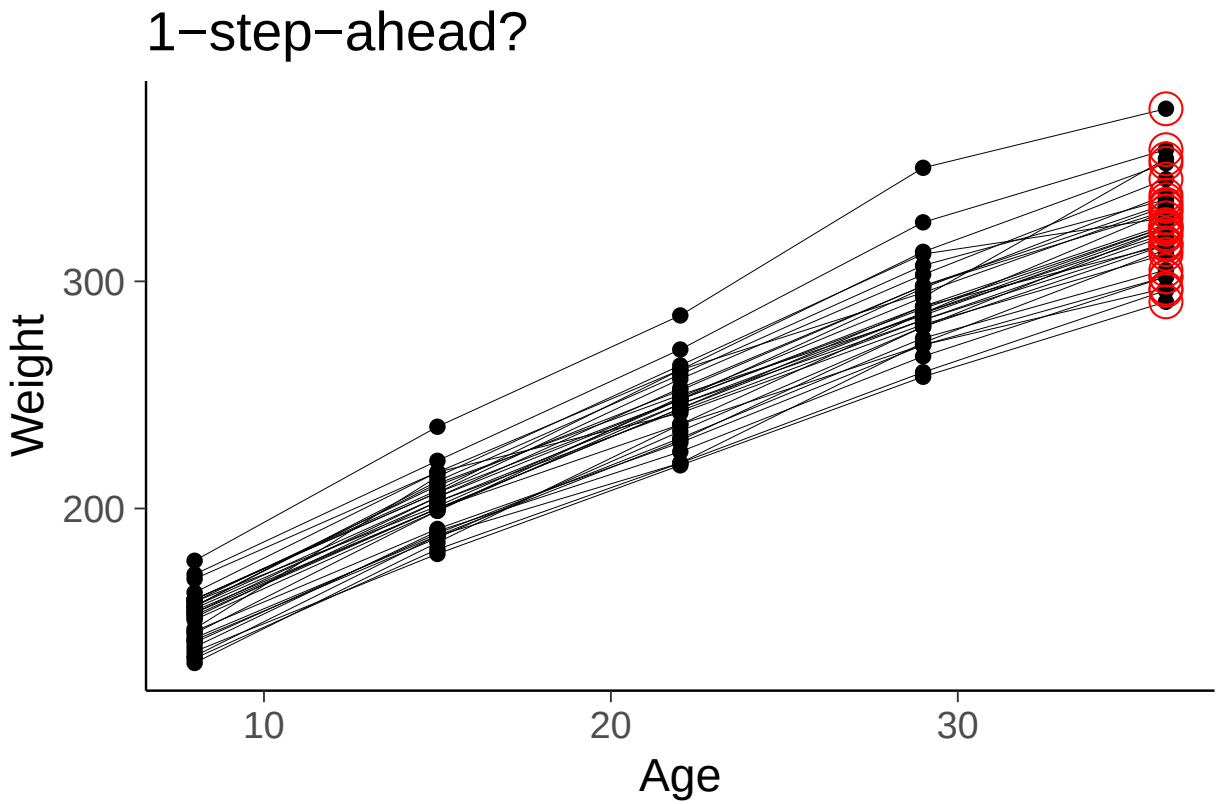

Yes!



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# LOO for hierarchical data



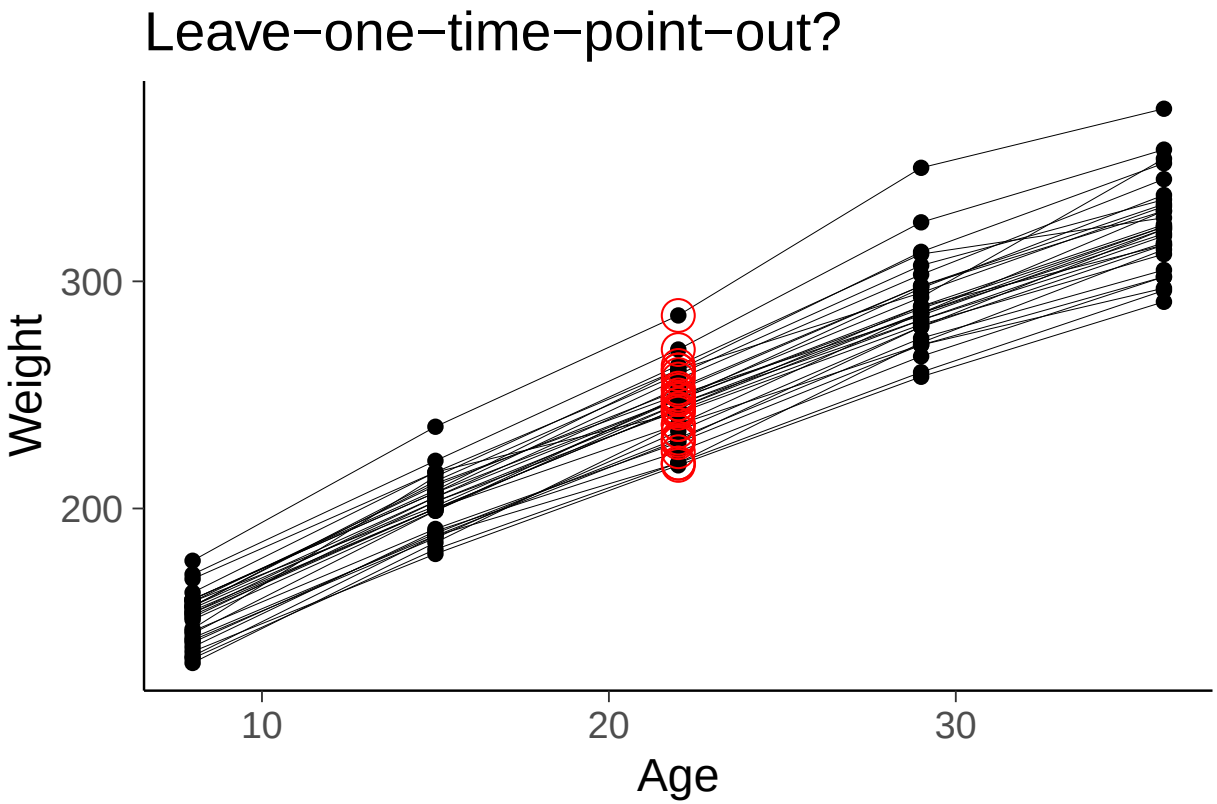

Yes!



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# LOO for hierarchical data



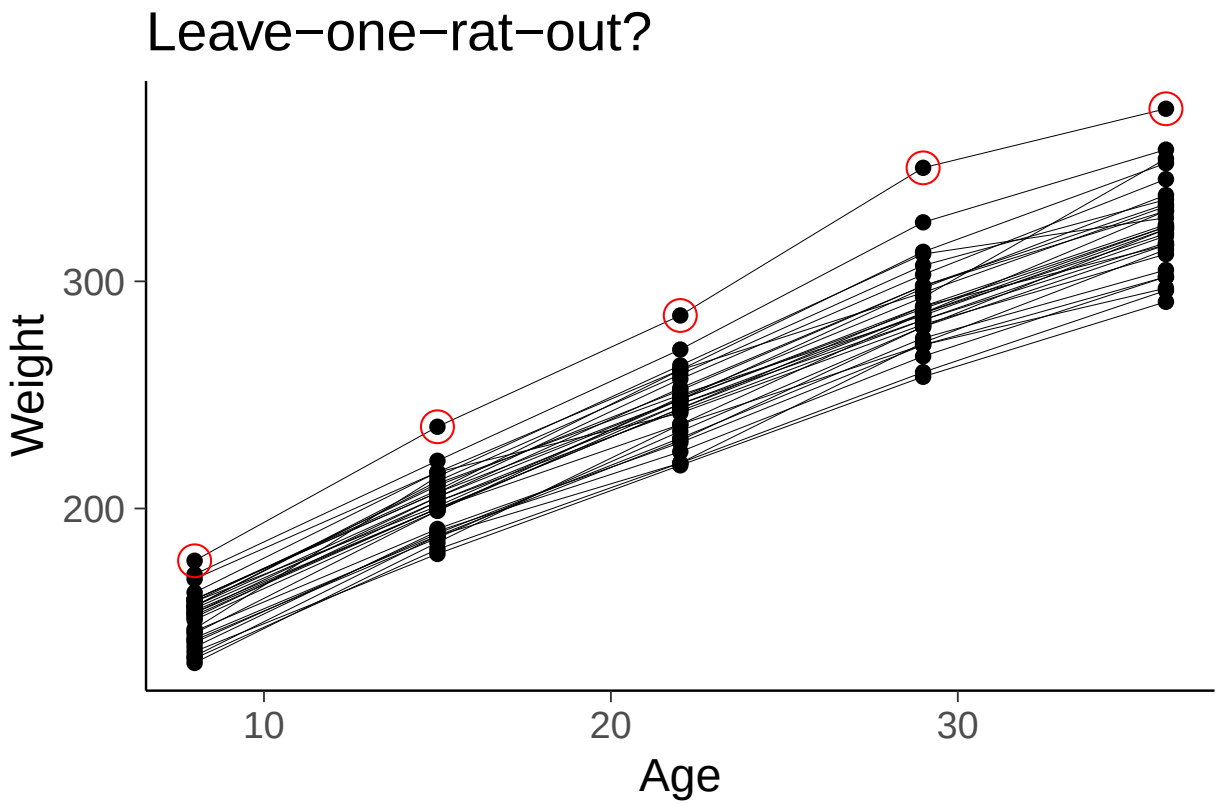

Yes!



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# LOO for hierarchical data




Yes!

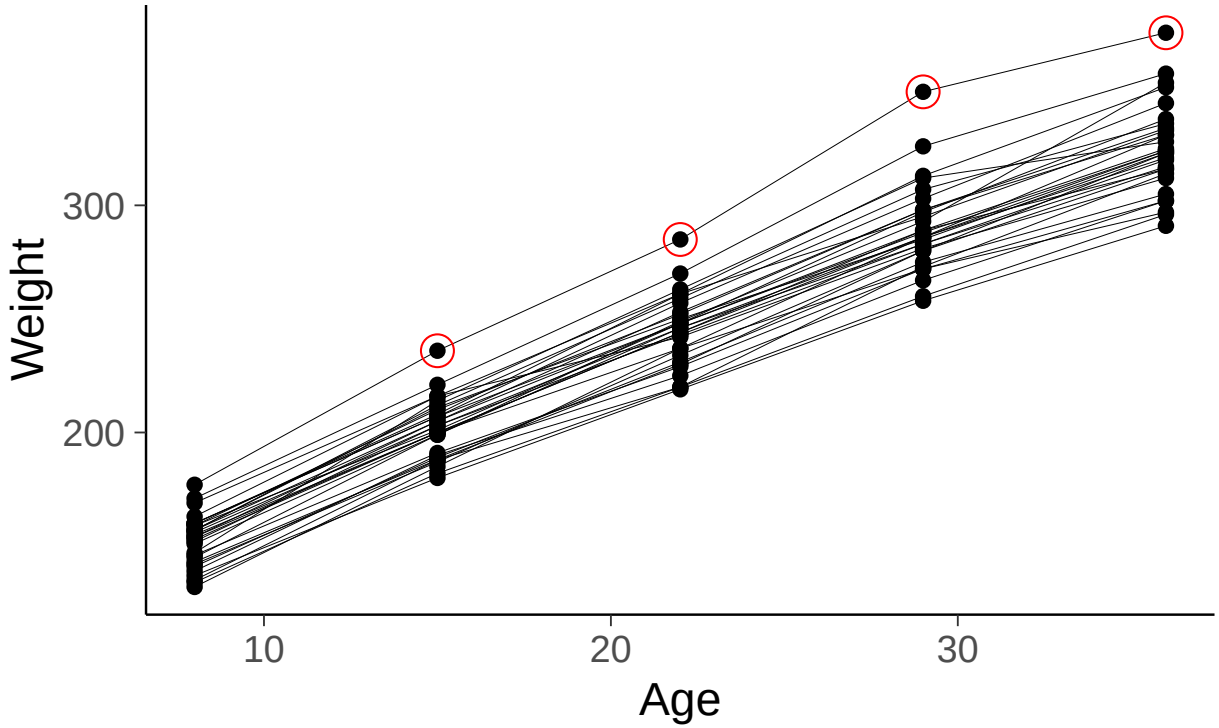


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# LOO for hierarchical data

Predict given initial weight?




Yes!



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# Pair discussion: what should we leave out?

**Discuss in pairs (1–2 minutes).**

Recall the factory example, with several measurements from each machine. We want to predict:

1. a new measurement from an existing machine;
2. a measurement from a new, similar machine, with no data yet.

**What should we leave out in each case?**



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# Pair discussion: what should we leave out?

**Discuss in pairs (1–2 minutes).**

Recall the factory example, with several measurements from each machine. We want to predict:

1. a new measurement from an existing machine;
2. a measurement from a new, similar machine, with no data yet.

**What should we leave out in each case?**

**Answer:** For an existing machine, leave out one measurement and retain the others from that machine.

For a new machine, leave out *all* measurements from one machine. The split should match the information available at prediction time.



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# Fast cross-validation

1. Pareto smoothed importance sampling LOO (PSIS-LOO)



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# Fast cross-validation

1. Pareto smoothed importance sampling LOO (PSIS-LOO)
2. K-fold cross-validation



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Subsection 2

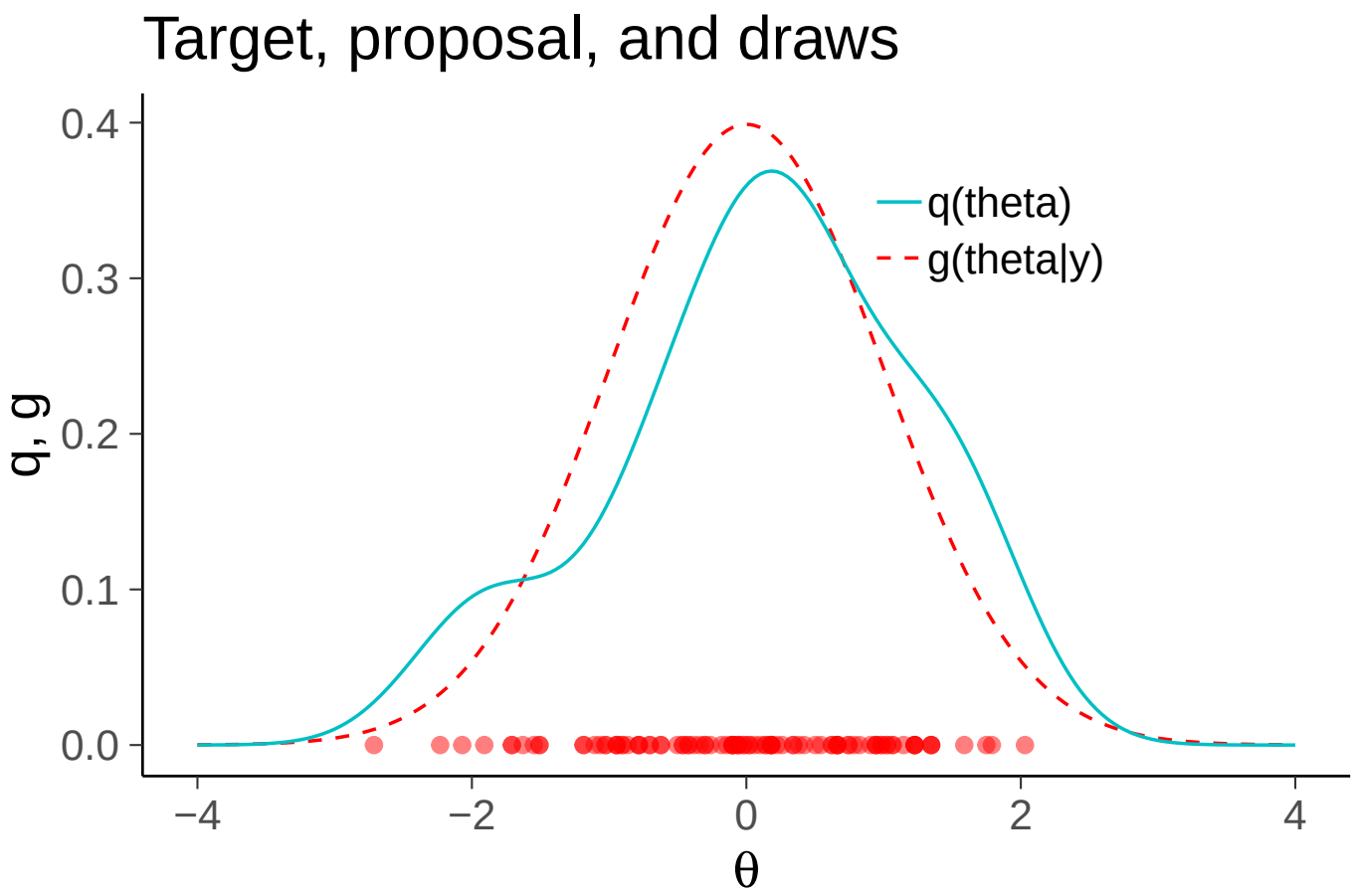
# PSIS LOO-CV



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# Recap: Importance sampling

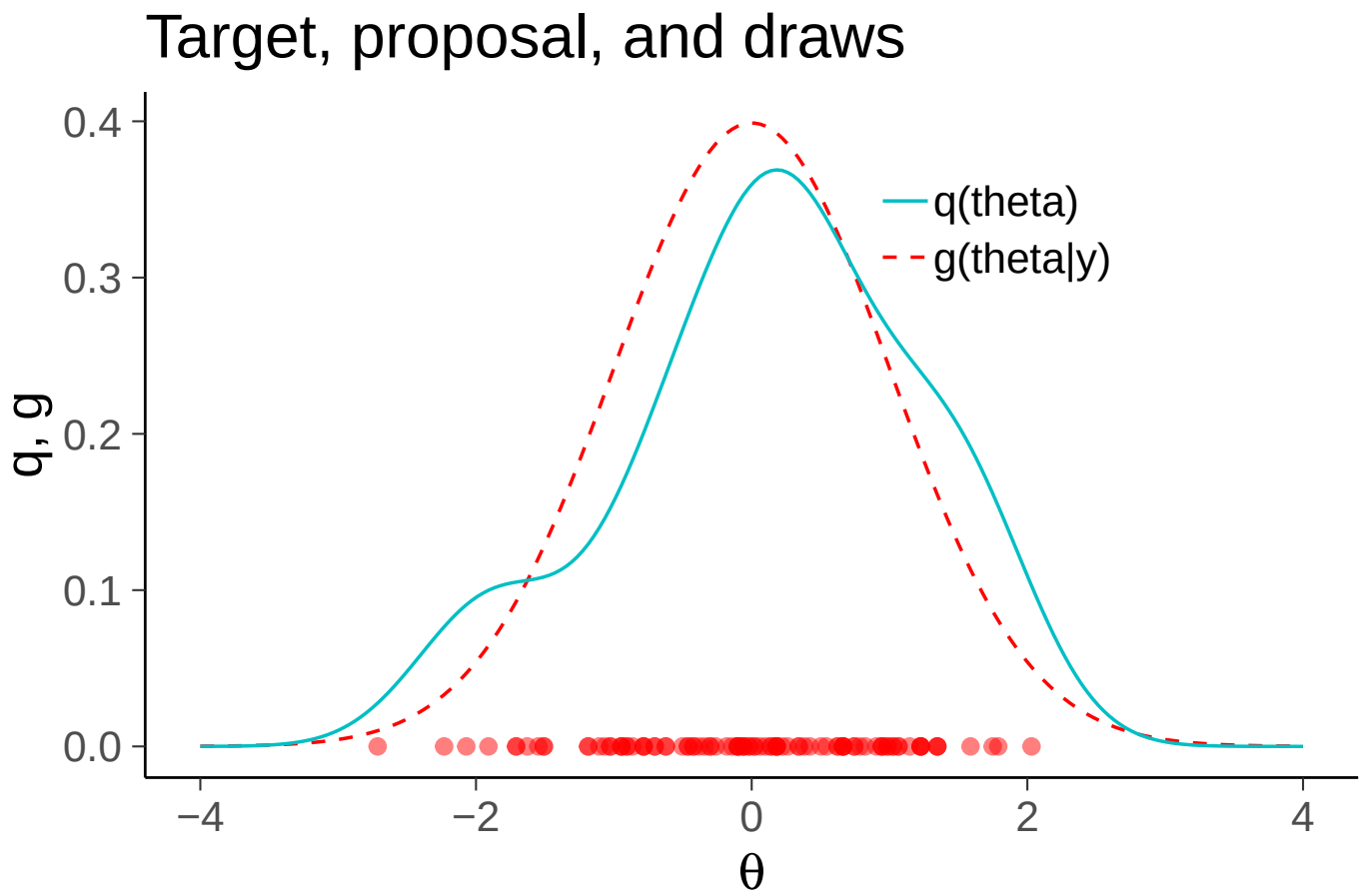




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# Recap: Importance sampling



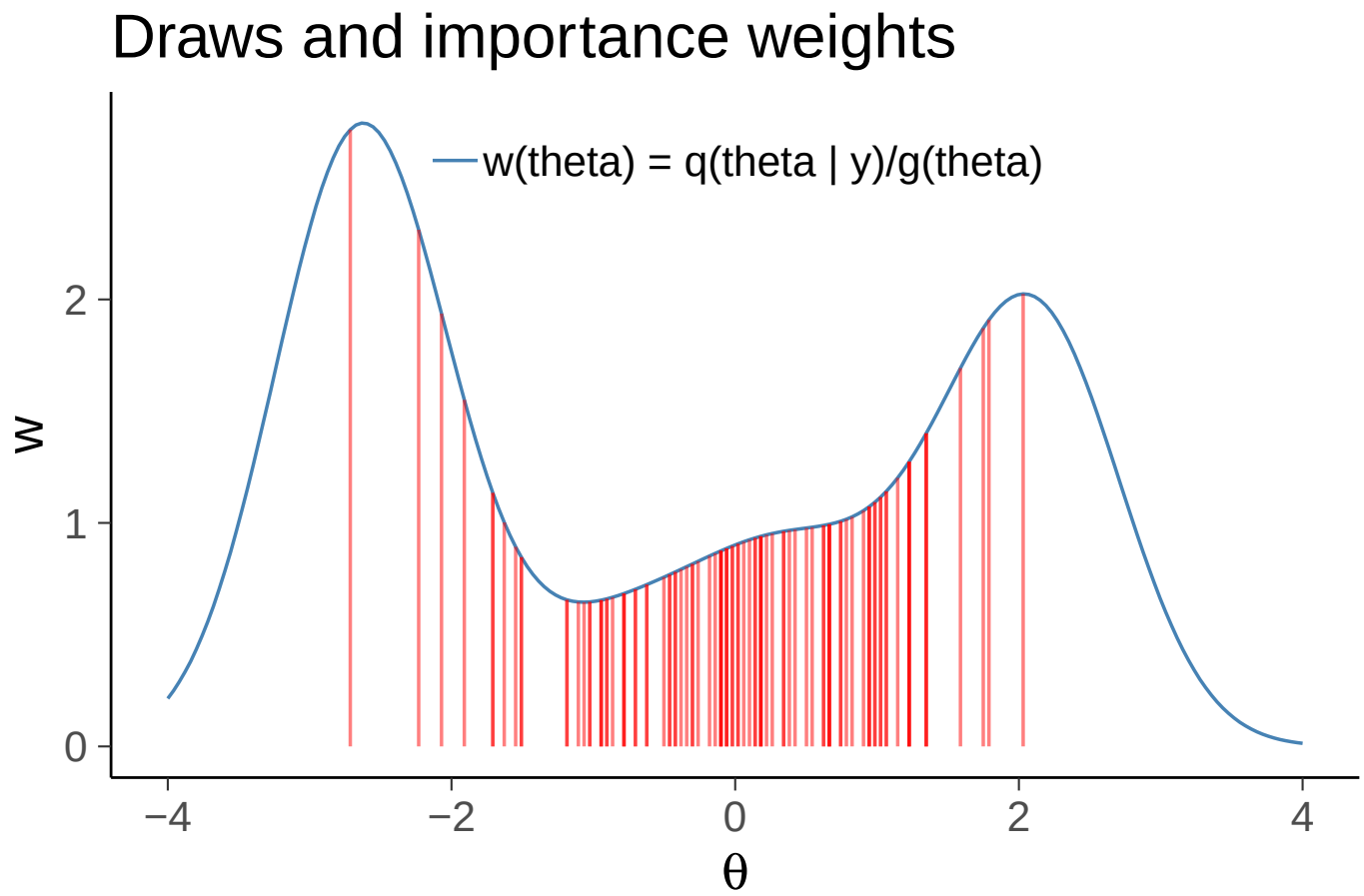

$$\mathrm{E}[f(\theta)] \approx \frac{\sum_s w_s f(\theta^{(s)})}{\sum_s w_s}, \qquad \text{where} \quad w_s = \frac{q(\theta^{(s)})}{g(\theta^{(s)})}$$



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# Recap: Importance sampling




$$\mathrm{E}[f(\theta)] \approx \frac{\sum_s w_s f(\theta^{(s)})}{\sum_s w_s}, \qquad \text{where} \quad w_s = \frac{q(\theta^{(s)})}{g(\theta^{(s)})}$$



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# Importance sampling for LOO-CV

- Let $f(\theta) = p(\tilde{y}_i \mid \theta)$, with proposal $g(\theta) = p(\theta \mid y)$ and target $q(\theta) = p(\theta \mid y_{-i})$.



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# Importance sampling for LOO-CV

- Let $f(\theta) = p(\tilde{y}_i \mid \theta)$, with proposal $g(\theta) = p(\theta \mid y)$ and target $q(\theta) = p(\theta \mid y_{-i})$.
- Using draws $\theta^{(s)} \sim p(\theta \mid y)$:

$$p(\tilde{y}_i \mid y_{-i}) \approx \frac{\sum_{s=1}^{S} p(\tilde{y}_i \mid \theta^{(s)}) r_i^{(s)}}{\sum_{s=1}^{S} r_i^{(s)}},$$

where the raw importance weights need only be proportional to

$$r_i^{(s)} \propto \frac{p(\theta^{(s)} \mid y_{-i})}{p(\theta^{(s)} \mid y)}.$$



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# Importance sampling for LOO-CV

- For conditionally independent observations, $p(y \mid \theta) = \prod_j p(y_j \mid \theta)$:

$$\frac{p(\theta^{(s)} \mid y_{-i})}{p(\theta^{(s)} \mid y)} = \frac{p(y)}{p(y_{-i})} \frac{1}{p(y_i \mid \theta^{(s)})} \propto \frac{1}{p(y_i \mid \theta^{(s)})}.$$

- Define $r_i^{(s)} = 1/p(y_i \mid \theta^{(s)})$, so $\log r_i^{(s)} = -\log p(y_i \mid \theta^{(s)})$.



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# Importance sampling for LOO-CV

- For conditionally independent observations, $p(y \mid \theta) = \prod_j p(y_j \mid \theta)$:

$$\frac{p(\theta^{(s)} \mid y_{-i})}{p(\theta^{(s)} \mid y)} = \frac{p(y)}{p(y_{-i})} \frac{1}{p(y_i \mid \theta^{(s)})} \propto \frac{1}{p(y_i \mid \theta^{(s)})}.$$

- Define $r_i^{(s)} = 1/p(y_i \mid \theta^{(s)})$, so $\log r_i^{(s)} = -\log p(y_i \mid \theta^{(s)})$.
- Evaluating at $\tilde{y}_i = y_i$ gives the raw IS estimate:

$$p(y_i \mid y_{-i}) \approx \frac{\sum_{s=1}^{S} p(y_i \mid \theta^{(s)}) r_i^{(s)}}{\sum_{s=1}^{S} r_i^{(s)}} = \frac{S}{\sum_{s=1}^{S} \frac{1}{p(y_i \mid \theta^{(s)})}}.$$

Each term in the numerator is 1, so the numerator is $S$.



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# Pair discussion: LOO importance weights

**Discuss in pairs (1–2 minutes).**

Three draws from the full posterior give likelihood values for $y_i$:

$$p(y_i \mid \theta^{(1)}) = 0.2, \quad p(y_i \mid \theta^{(2)}) = 0.1, \quad p(y_i \mid \theta^{(3)}) = 0.05.$$

**Which draw receives the greatest importance weight when we remove $y_i$? Why?**



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**Which draw receives the greatest importance weight when we remove $y_i$? Why?**

**Answer:** The third draw. The raw weights are

$$r_i^{(s)} = \frac{1}{p(y_i \mid \theta^{(s)})} : \quad 5,\ 10,\ 20, \qquad \text{normalised: } \frac{1}{7},\ \frac{2}{7},\ \frac{4}{7}.$$

Observation $y_i$ penalised the third draw most strongly in the full posterior. Removing $y_i$ reverses that likelihood contribution.



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# LOO importance sampling: worked example

The three likelihood values are 0.2, 0.1, 0.05. Their raw weights are 5, 10, 20, with sum 35.

$$\overline{r}_i^{(1)} = \frac{1}{7}, \qquad \overline{r}_i^{(2)} = \frac{2}{7}, \qquad \overline{r}_i^{(3)} = \frac{4}{7}.$$

Three draws illustrate the calculation; they are not enough for reliable estimation.



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The weighted predictive density is

$$\begin{aligned}\widehat{p}(y_i \mid y_{-i}) &= \frac{1}{7}(0.2) + \frac{2}{7}(0.1) + \frac{4}{7}(0.05) \\ &= \frac{3}{35} \approx 0.0857.\end{aligned}$$

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The unweighted training density is $(0.2 + 0.1 + 0.05)/3 \approx 0.1167$. Removing $y_i$ increases the relative weight of draws that predicted it poorly.

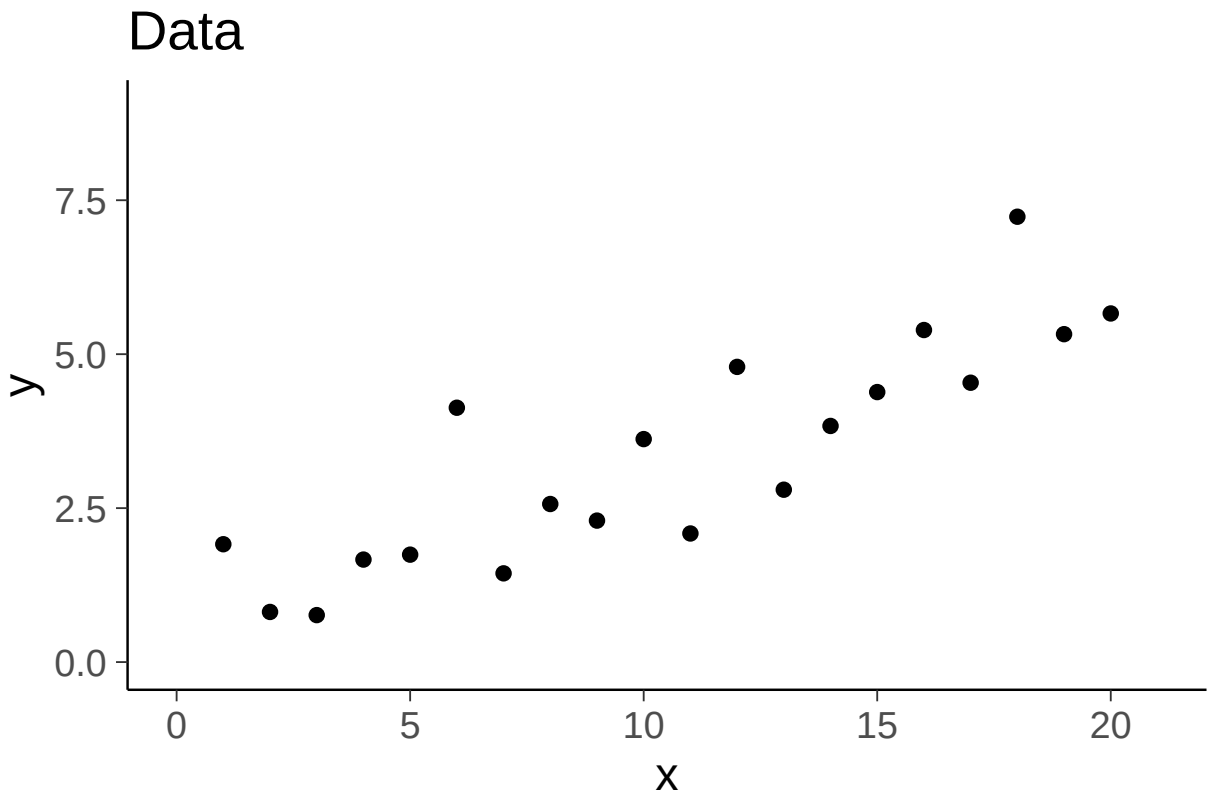
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# PSIS-LOO: reweighting posterior draws

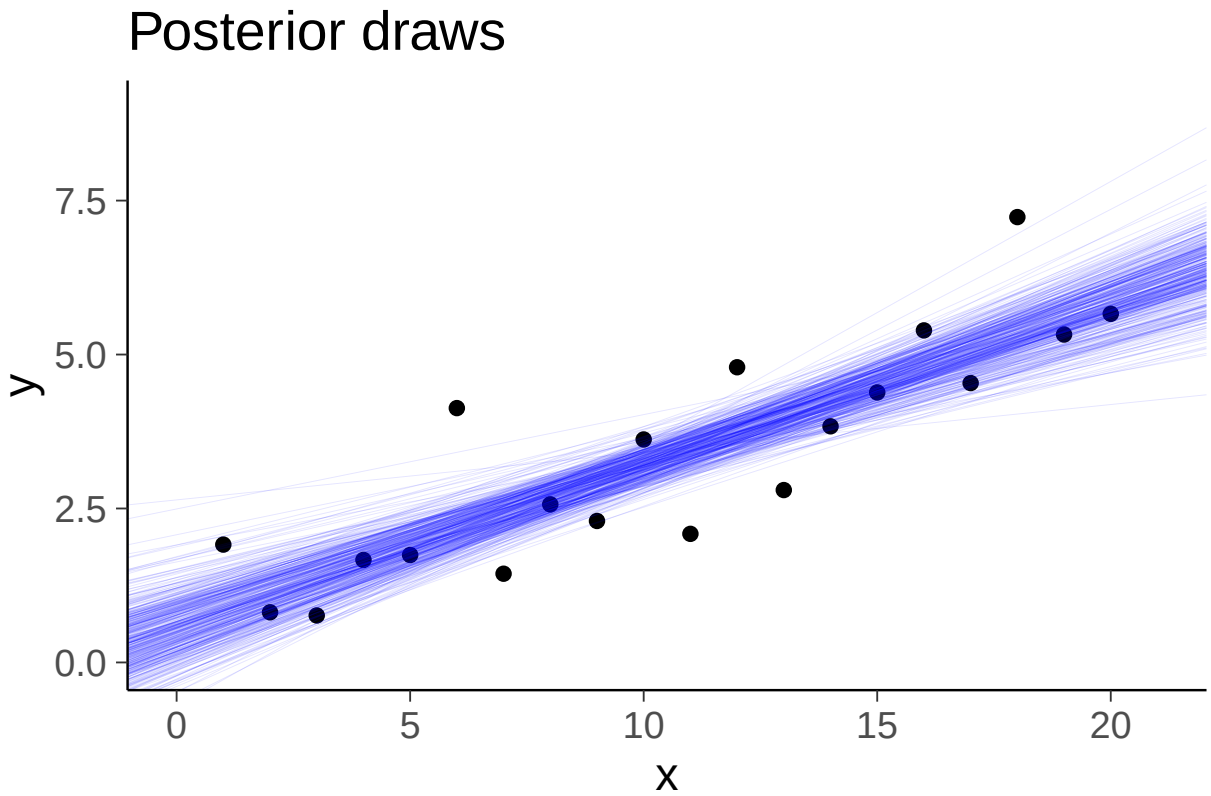




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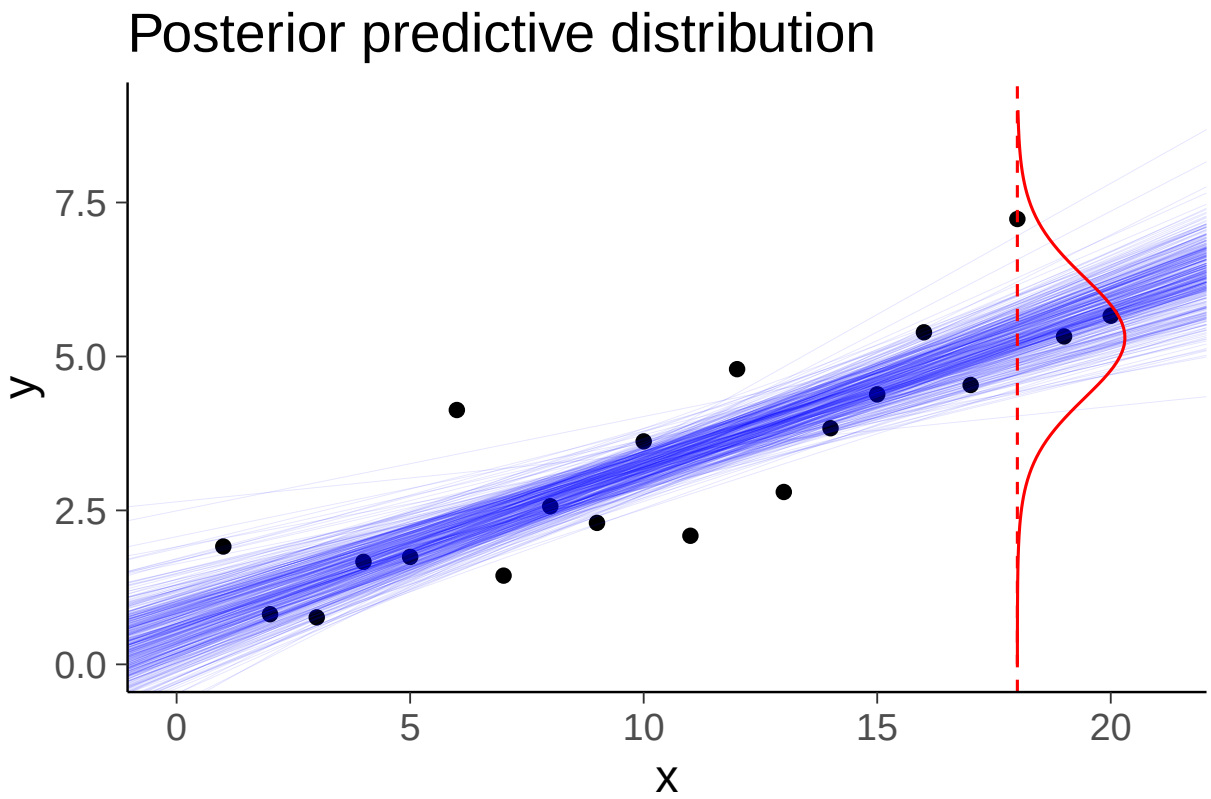

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# PSIS-LOO: reweighting posterior draws




$$\theta^{(s)} \sim p(\theta|x,y)$$

# PSIS-LOO: reweighting posterior draws



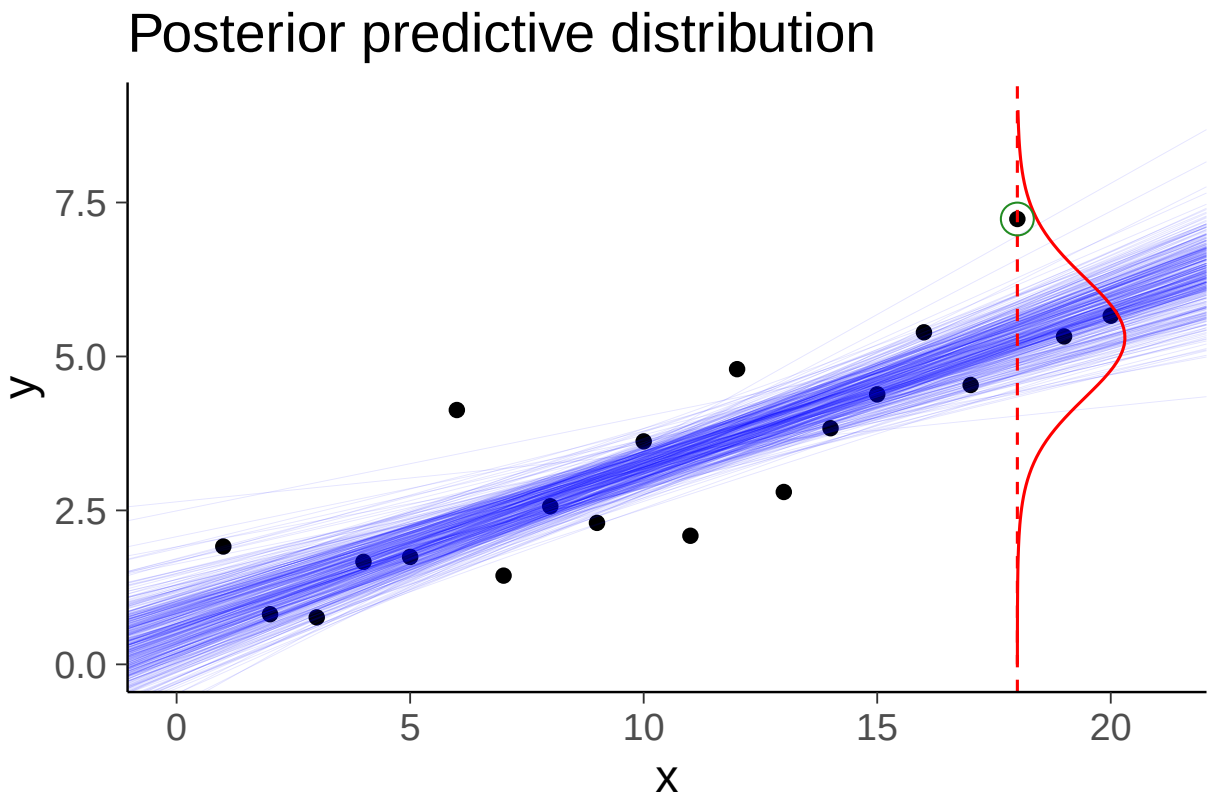

$$\theta^{(s)} \sim p(\theta|x,y), \quad p(\tilde{y}|\tilde{x},x,y) \approx \frac{1}{S}\sum_{s=1}^{S} p(\tilde{y}|\tilde{x},\theta^{(s)})$$



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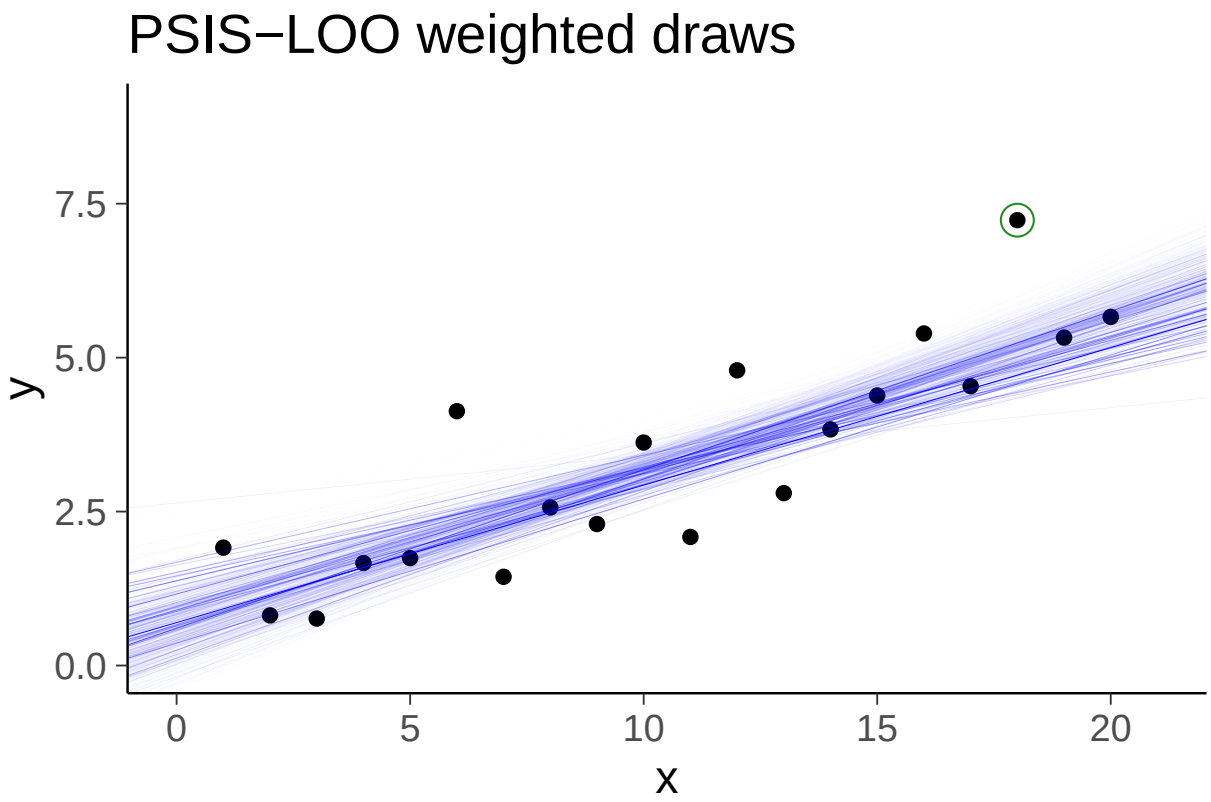

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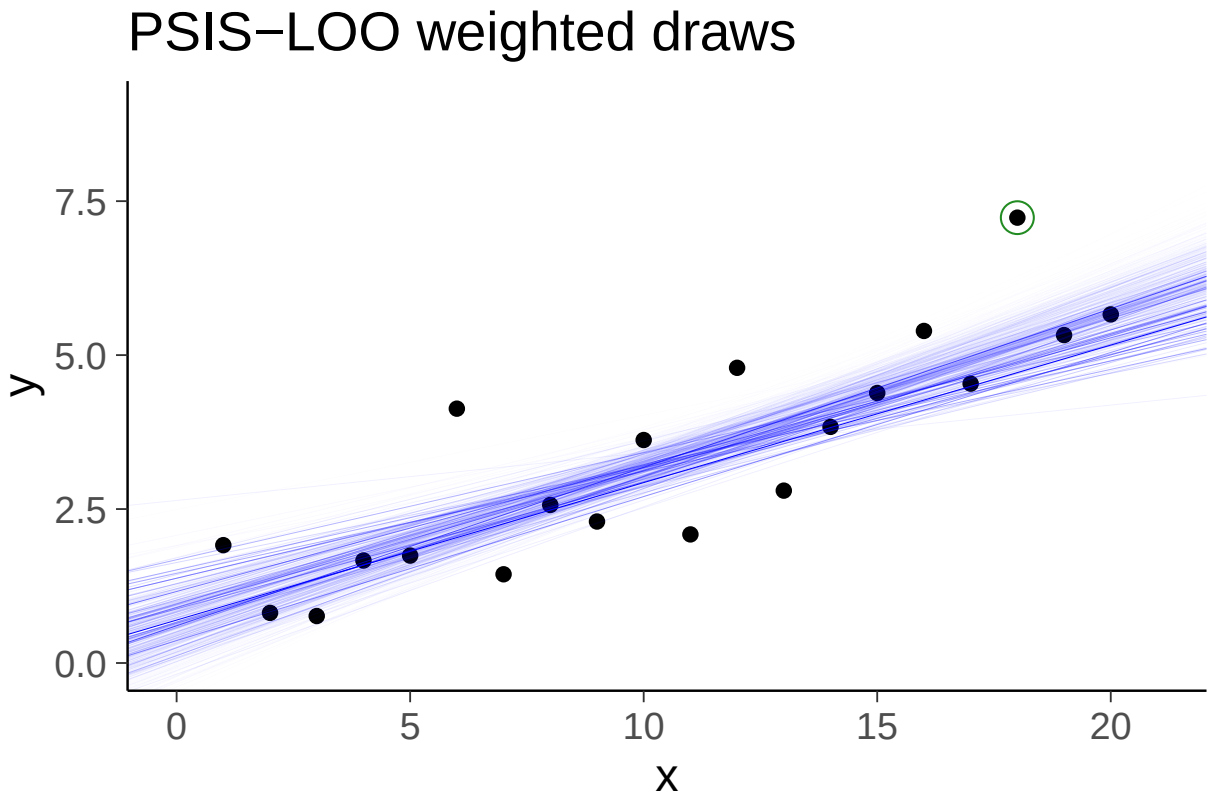
$$r_i^{(s)} \propto p(\theta^{(s)}|x_{-i},y_{-i})/p(\theta^{(s)}|x,y)$$



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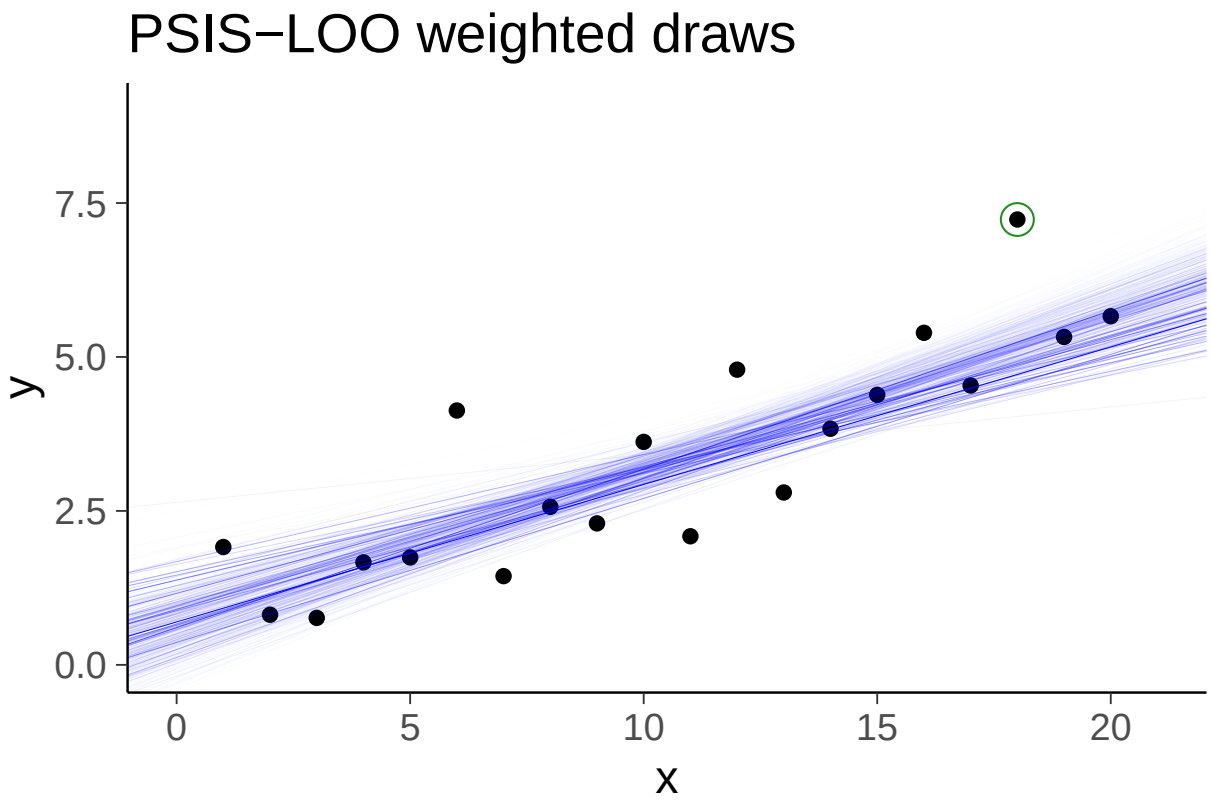
$$r_i^{(s)} = 1/p(y_i|x_i,\theta^{(s)})$$



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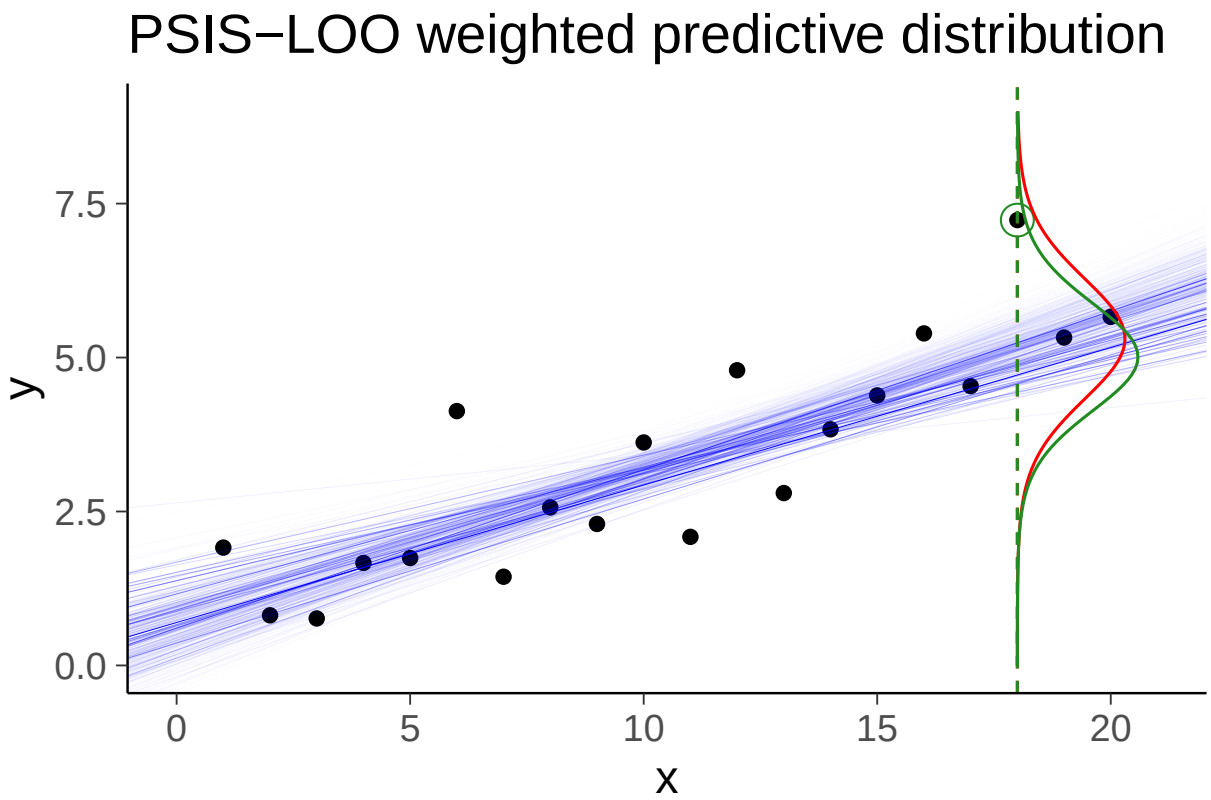
$$\log(1/p(y_i|x_i,\theta^{(s)})) = -\text{log_lik}[i]$$



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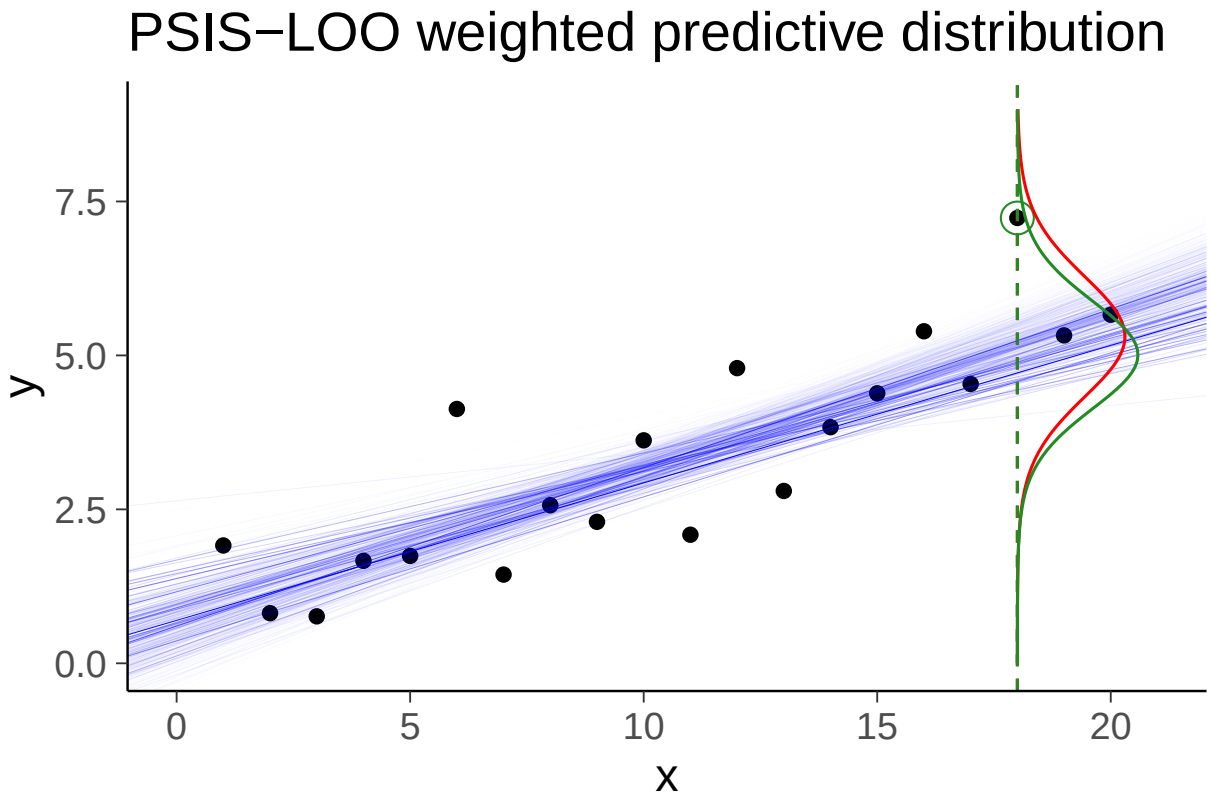

PSIS smooths the largest raw weights $r_i^{(s)}$ to give $w_i^{(s)}$.



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# PSIS-LOO: reweighting posterior draws




Normalise the smoothed weights: $\bar{w}_i^{(s)} = w_i^{(s)} / \sum_{t=1}^{S} w_i^{(t)}$.

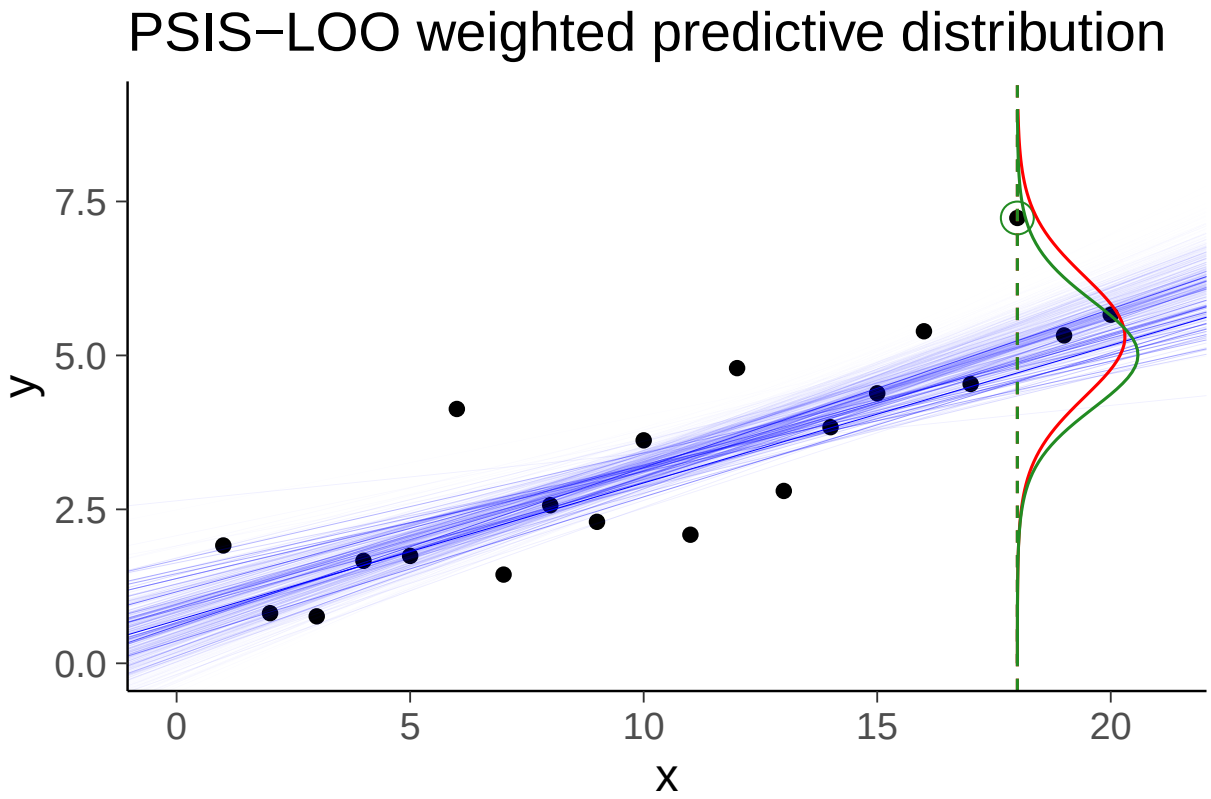
$p(y_i|x_i, x_{-i}, y_{-i}) \approx \sum_{s=1}^{S} \bar{w}_i^{(s)} p(y_i|x_i, \theta^{(s)})$.



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# Recap: Pareto smoothed importance sampling

- PSIS fits a generalized Pareto distribution to the largest raw weights and smooths these extreme weights.
- The estimated tail shape $\widehat{k}$ measures how heavy that tail is: larger values mean a few draws can dominate the estimate.



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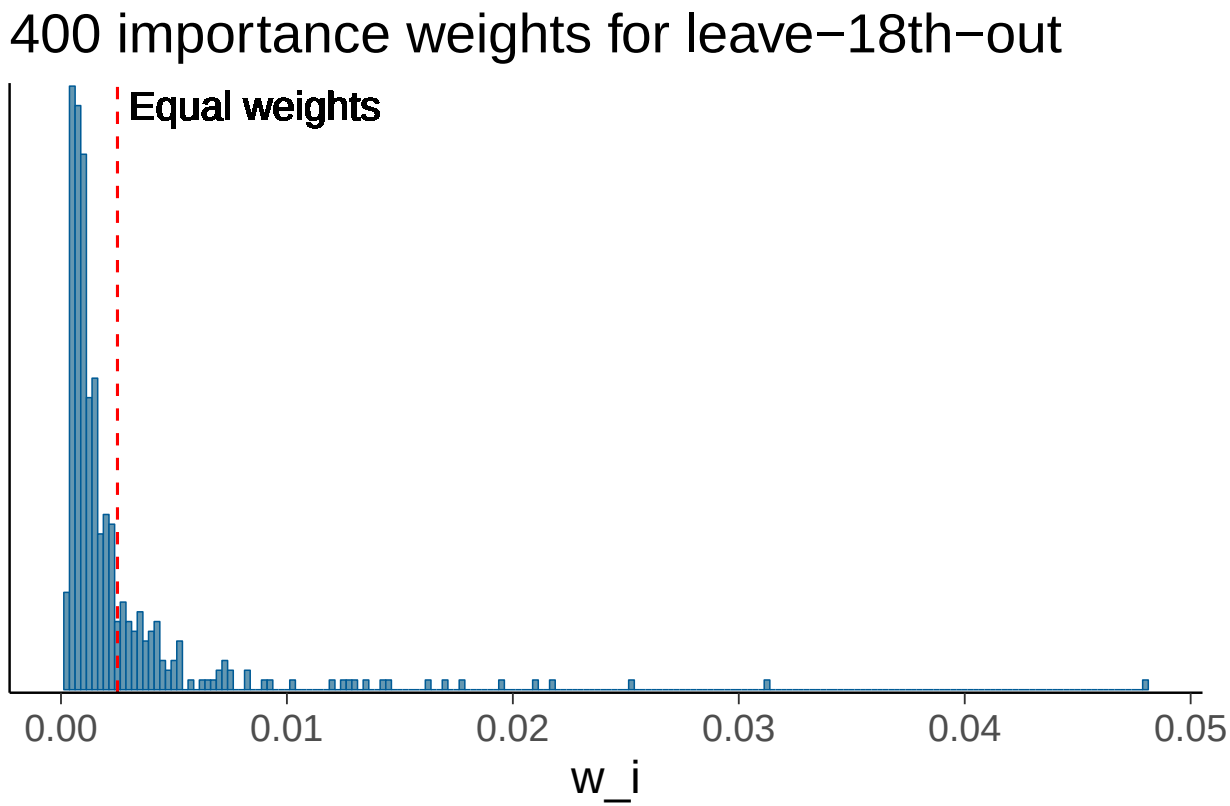
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- Software may use a stricter threshold when fewer posterior draws are available. Check its warnings.
- $\widehat{k} > 0.7$ warns that the PSIS approximation is unreliable. It does not mean that exact leave-one-out prediction is impossible.



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# PSIS weights and tail behaviour

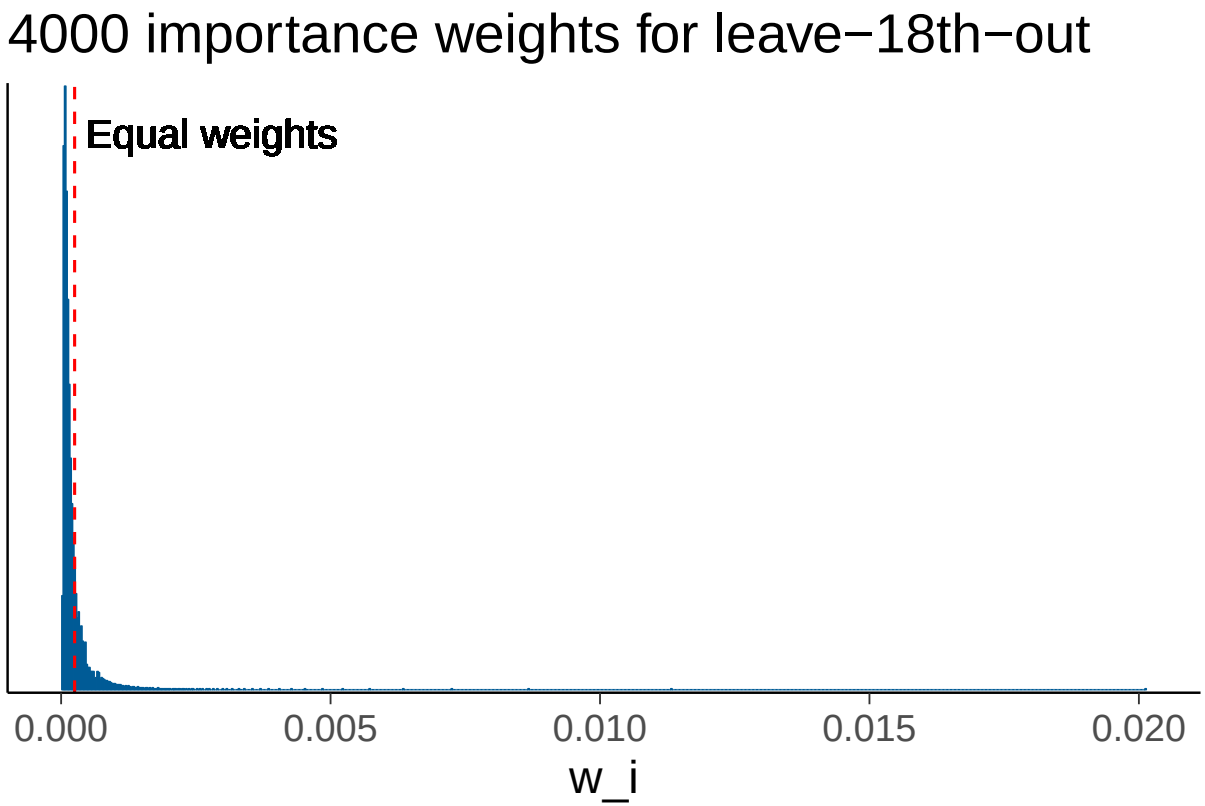




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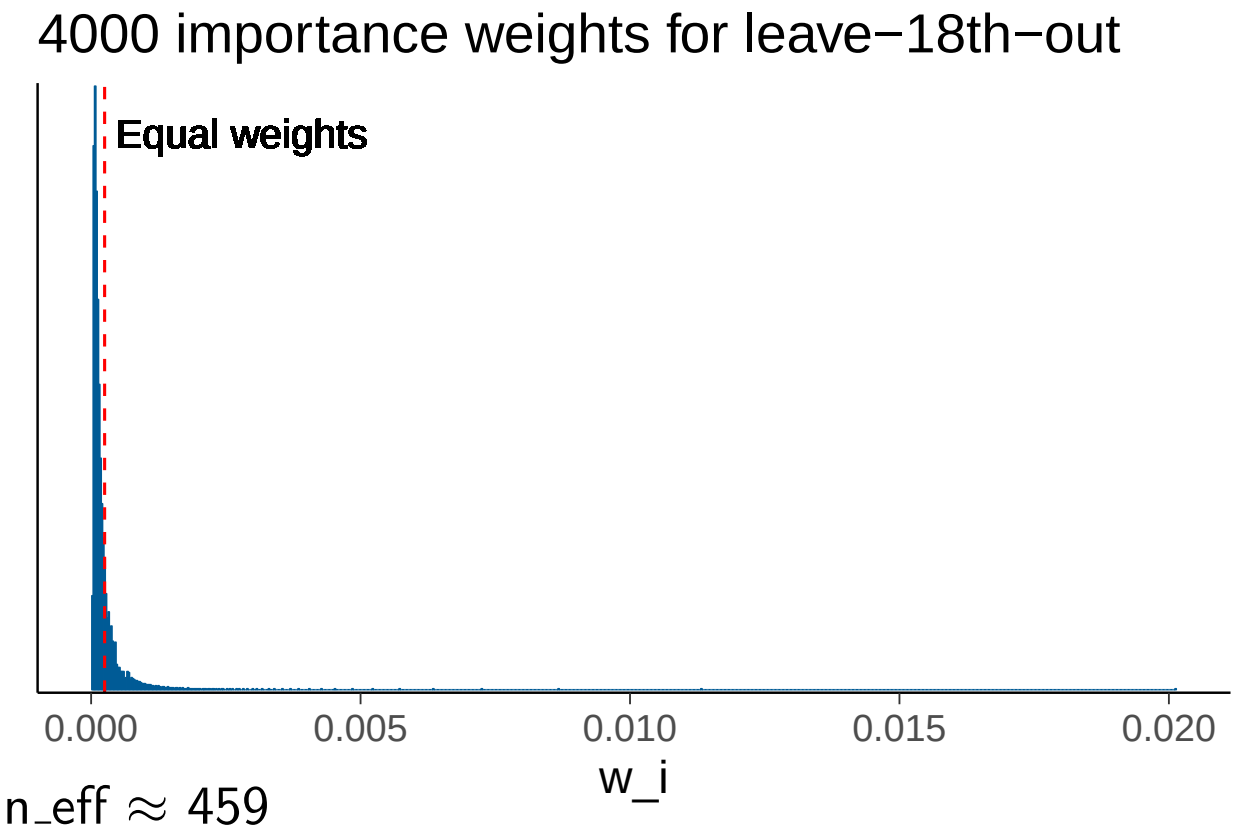




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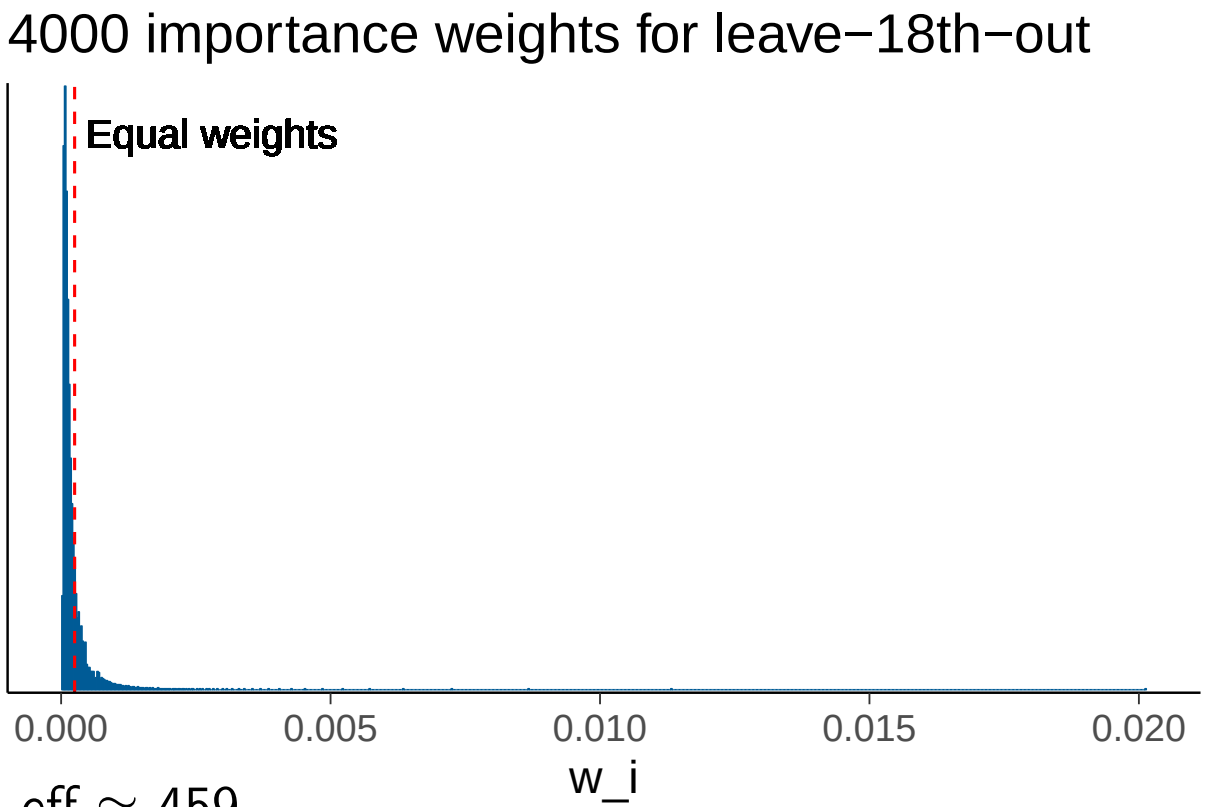

n_eff $\approx$ 459



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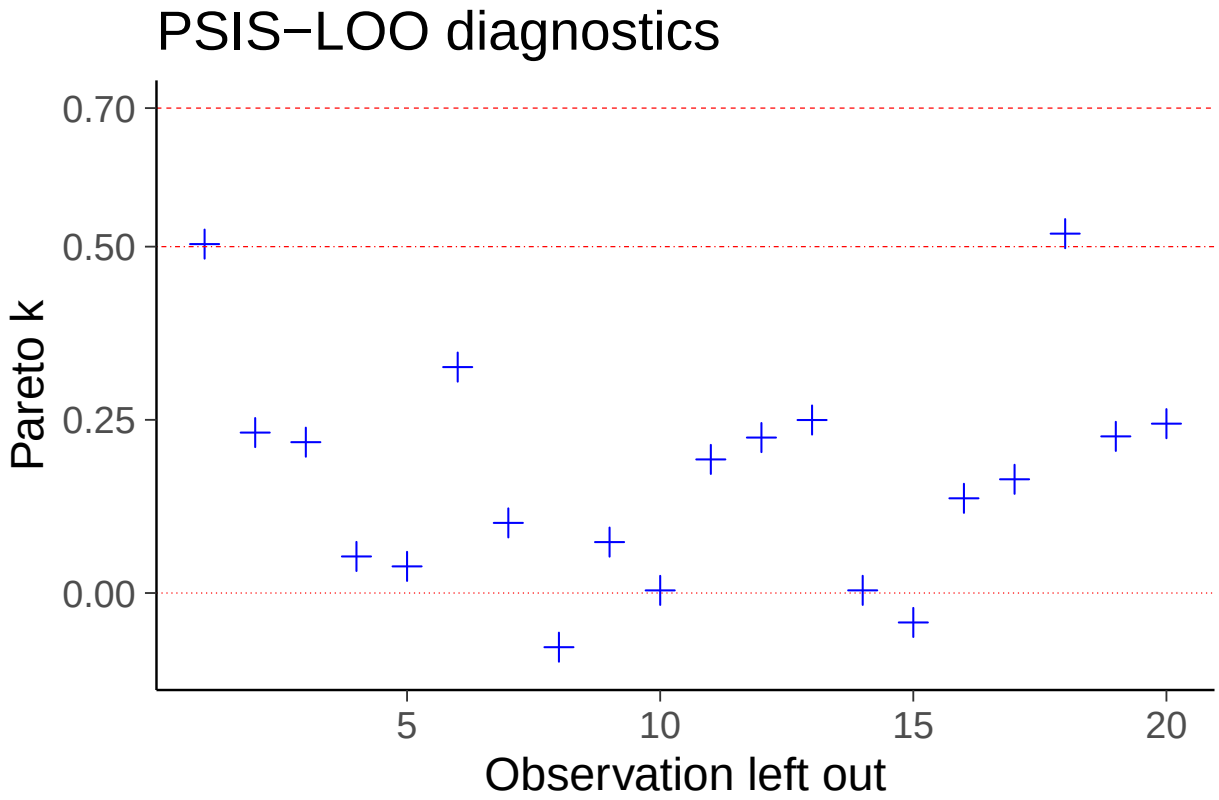
Pareto $\hat{k} \approx 0.52$ Pareto $\hat{k}$ estimates the tail shape which determines the convergence rate of PSIS. Less than 0.7 is ok.



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# Pareto diagnostics for each observation

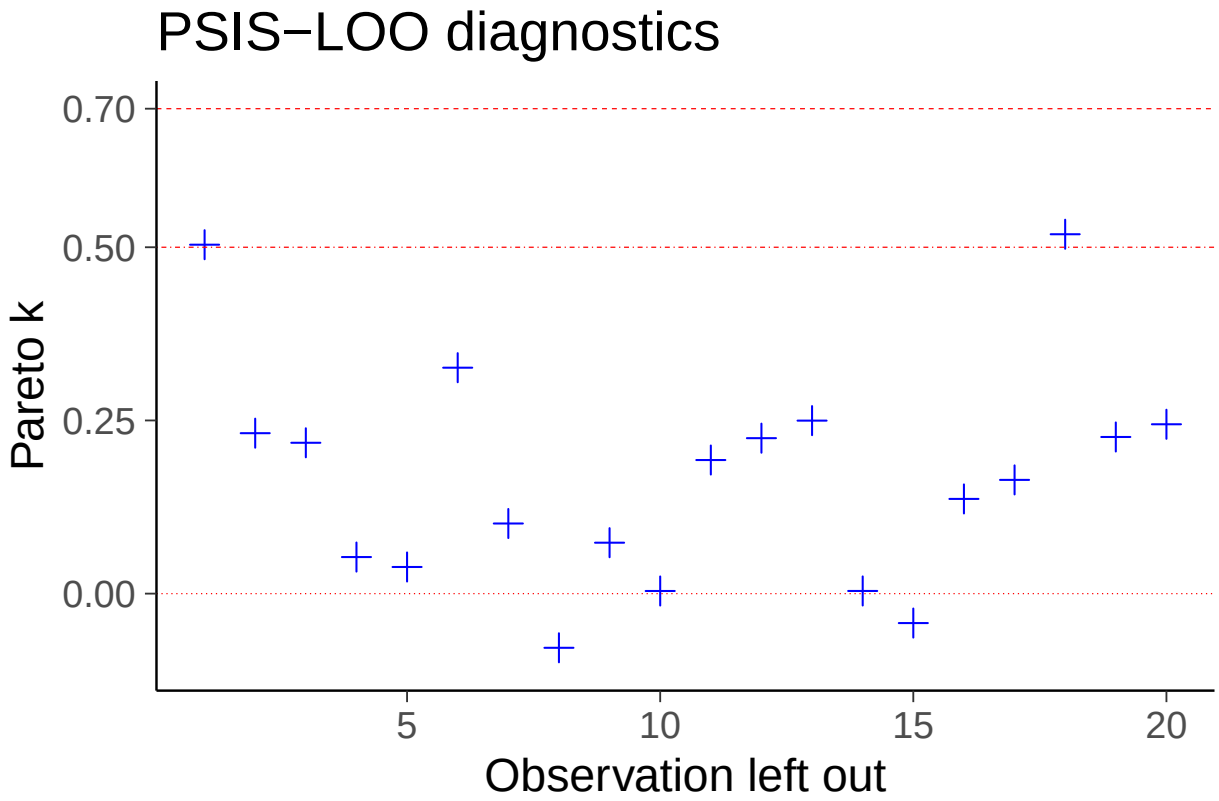




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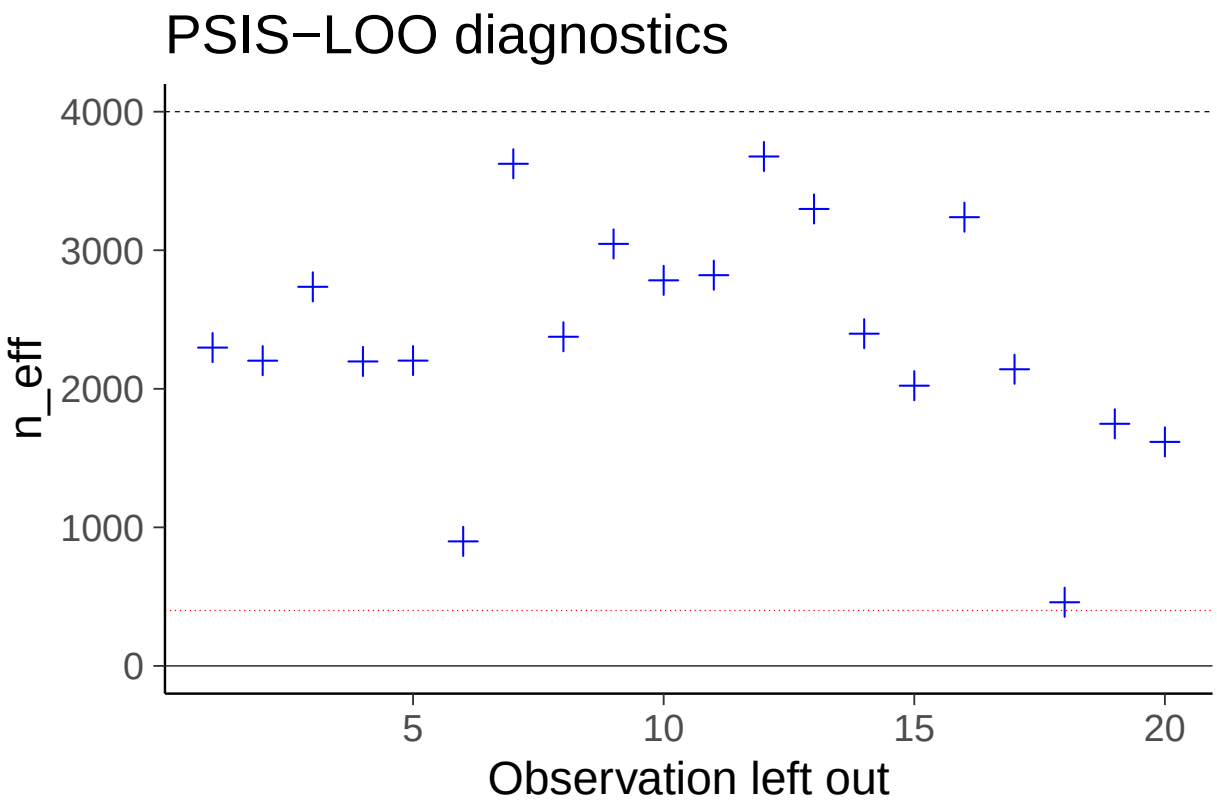

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# Pareto diagnostics and PSIS ESS




```
Pareto k diagnostic values:
                         Count Pct.    Min. n_eff
(-Inf, 0.5]   (good)     18    90.0\%   899
 (0.5, 0.7]   (ok)        2    10.0\%   459
   (0.7, 1]   (bad)       0     0.0\%   <NA>
   (1, Inf)   (very bad)  0     0.0\%   <NA>
```

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# Pair discussion: Pareto diagnostics

**Discuss in pairs (1–2 minutes).**

- Model A has a higher estimated ELPD, but several Pareto diagnostics have $\hat{k} > 1$.
- Model B has reliable PSIS diagnostics.

**Can we conclude that A predicts better? What should we do next?**



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- Model B has reliable PSIS diagnostics.

**Can we conclude that A predicts better? What should we do next?**

**Answer:** Not yet. A's PSIS estimate is unreliable. Reliable diagnostics for B do not establish that B predicts better.

Refit A leaving out each problematic observation in turn to compute those LOO predictions directly, or evaluate both models with suitable K-fold CV. Then compare the estimates and their uncertainty.



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# What if $\widehat{k}$ is large?

1. Inspect the influential observations and model assumptions. Check for data errors and whether the model represents plausible variation.

loo: Pareto diagnostic guidance.



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# What if $\widehat{k}$ is large?

1. Inspect the influential observations and model assumptions. Check for data errors and whether the model represents plausible variation.
2. For a few problematic observations, refit the model without each one and compute its LOO predictive density directly.

loo: Pareto diagnostic guidance.



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# What if $\widehat{k}$ is large?

1. Inspect the influential observations and model assumptions. Check for data errors and whether the model represents plausible variation.
2. For a few problematic observations, refit the model without each one and compute its LOO predictive density directly.
3. For many problematic observations, use suitable K-fold validation or revise the model and evaluate it again.

loo: Pareto diagnostic guidance.



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# What if $\widehat{k}$ is large?

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2. For a few problematic observations, refit the model without each one and compute its LOO predictive density directly.
3. For many problematic observations, use suitable K-fold validation or revise the model and evaluate it again.

Large $\widehat{k}$ is a warning about the approximation, not an instruction to delete an observation. Match the validation split to the prediction task.

loo: Pareto diagnostic guidance.



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# Stan code

$$\log(r_i^{(s)}) = \log(1/p(y_i|x_i, \theta^{(s)})) = -\text{log_lik}[i]$$



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# Stan code

$$\log(r_i^{(s)}) = \log(1/p(y_i|x_i, \theta^{(s)})) = -\text{log_lik}[i]$$

```
...
model {
  alpha ~ normal(pmualpha, psalpha);
  beta ~ normal(pmubeta, psbeta);
  y ~ normal(mu, sigma);
}
generated quantities {
  vector[N] log_lik;
  for (i in 1:N)
    log_lik[i] = normal_lpdf(y[i] | mu[i], sigma);
}
```



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# Stan and `loo` package

```
Computed from 4000 by 20 log-likelihood matrix

          Estimate  SE
elpd_loo    -29.5 3.3
p_loo         2.7 1.0
------

Monte Carlo SE of elpd_loo is 0.1.

Pareto k diagnostic values:
                         Count Pct.    Min. n_eff
(-Inf, 0.5]   (good)     18    90.0%   899
 (0.5, 0.7]   (ok)        2    10.0%   459
   (0.7, 1]   (bad)       0     0.0%   <NA>
   (1, Inf)   (very bad)  0     0.0%   <NA>

All Pareto k estimates are ok (k < 0.7).
See help('pareto-k-diagnostic') for details.
```



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Subsection 3

## K-fold CV



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# K-fold cross-validation

- Split observations into $K$ folds $I_1, \ldots, I_K$, each containing approximately $1/K$ of the observations.



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# K-fold cross-validation

- Split observations into $K$ folds $I_1, \ldots, I_K$, each containing approximately $1/K$ of the observations.
  1. For $k = 1, \ldots, K$:
     - 1.1 hold out $y_{I_k}$; use $y_{-I_k}$ as the training data
     - 1.2 fit $p(\theta \mid y_{-I_k})$
     - 1.3 score only the held-out observations:

$$L_k = \sum_{i \in I_k} \log \int p(y_i \mid \theta) p(\theta \mid y_{-I_k}) d\theta$$

  2. sum the scores: $\text{elpd}_{K\text{-fold}} = \sum_{k=1}^{K} L_k$



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$$L_k = \sum_{i \in I_k} \log \int p(y_i \mid \theta) p(\theta \mid y_{-I_k}) d\theta$$

  2. sum the scores: $\text{elpd}_{K\text{-fold}} = \sum_{k=1}^{K} L_k$
- Fit $K$ models, each with fewer training observations than in LOO when $K < n$.



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# K-fold cross-validation

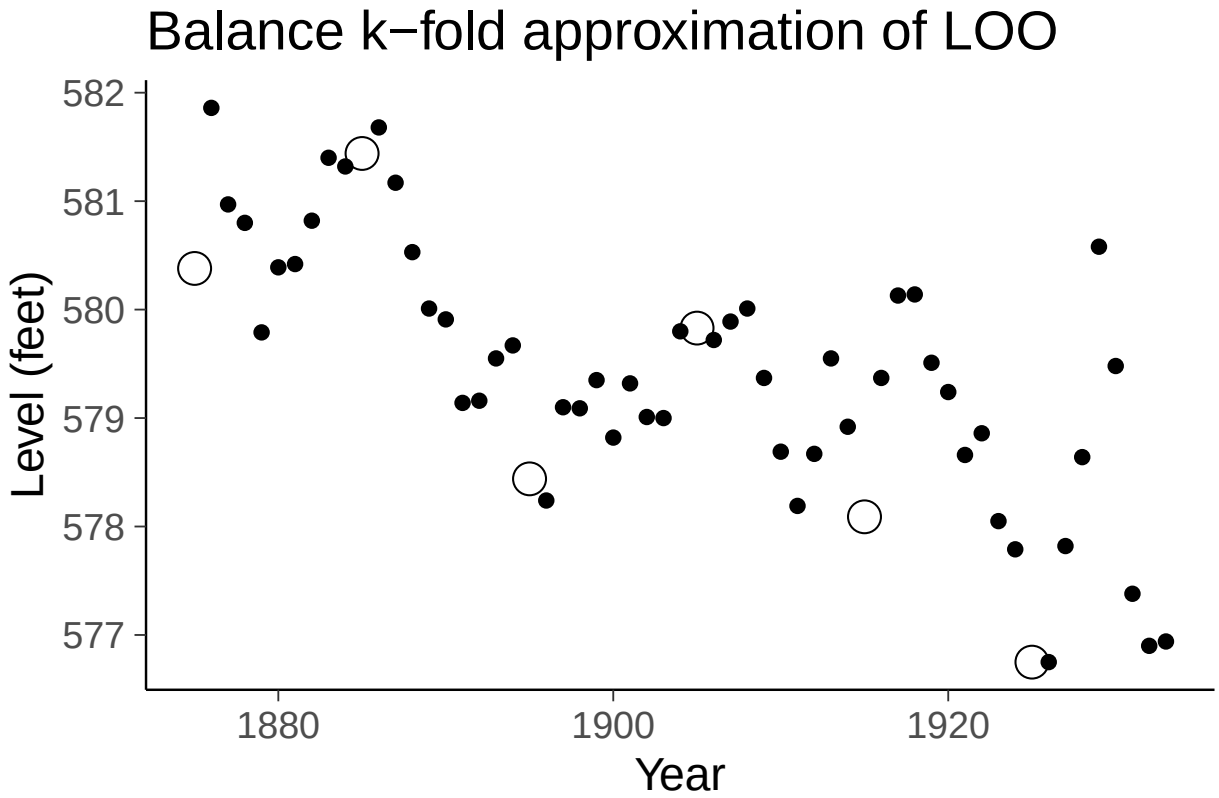
1. K-fold cross-validation can approximate LOO
   1.1 can be used all uses for LOO
2. K-fold cross-validation can be used for hierarchical models
   2.1 good for leave-one-group-out
3. K-fold cross-validation can be used for time series
   3.1 with leave-block-out



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# K-fold splits for different prediction tasks

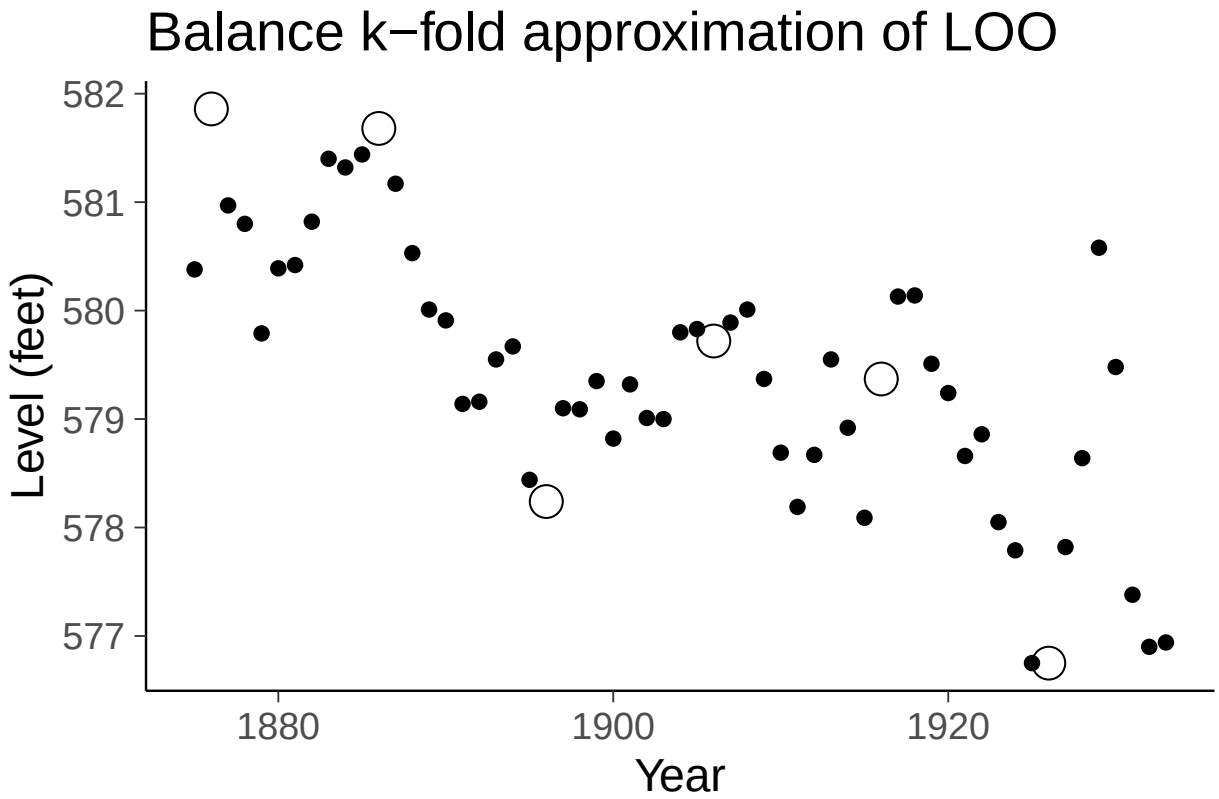




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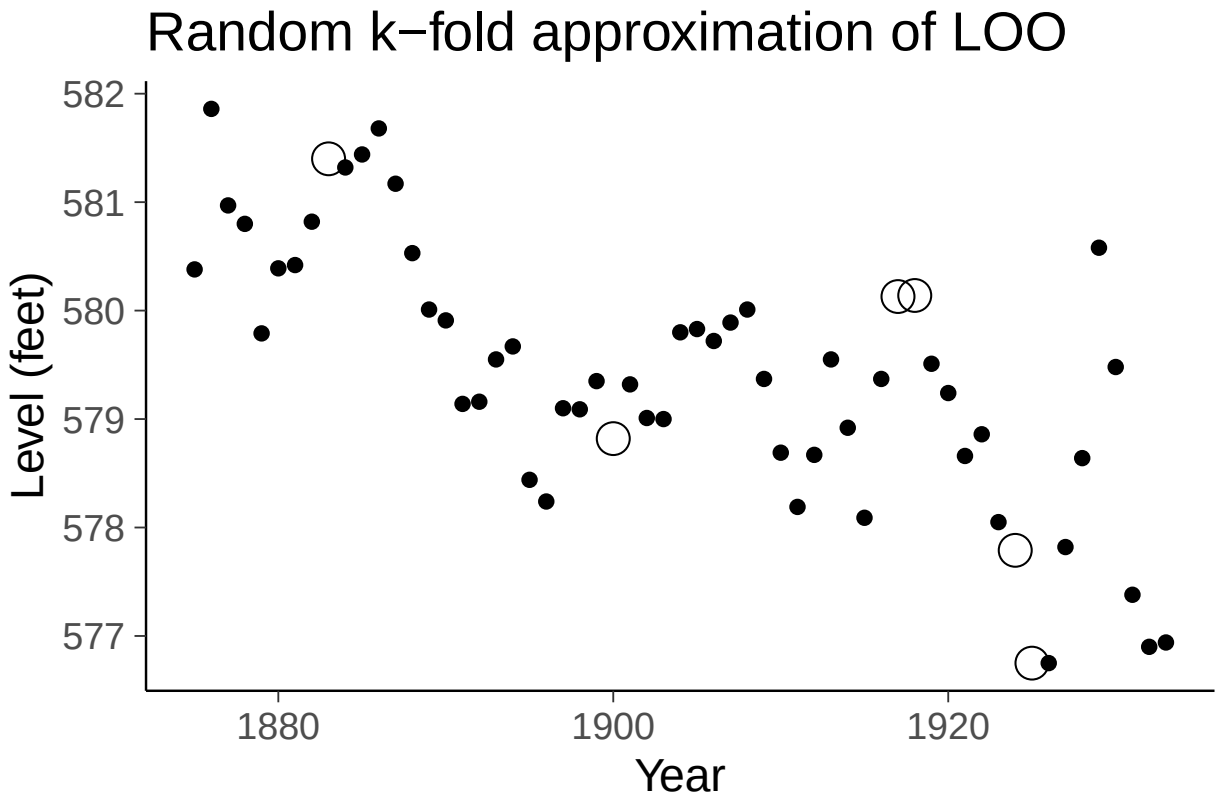




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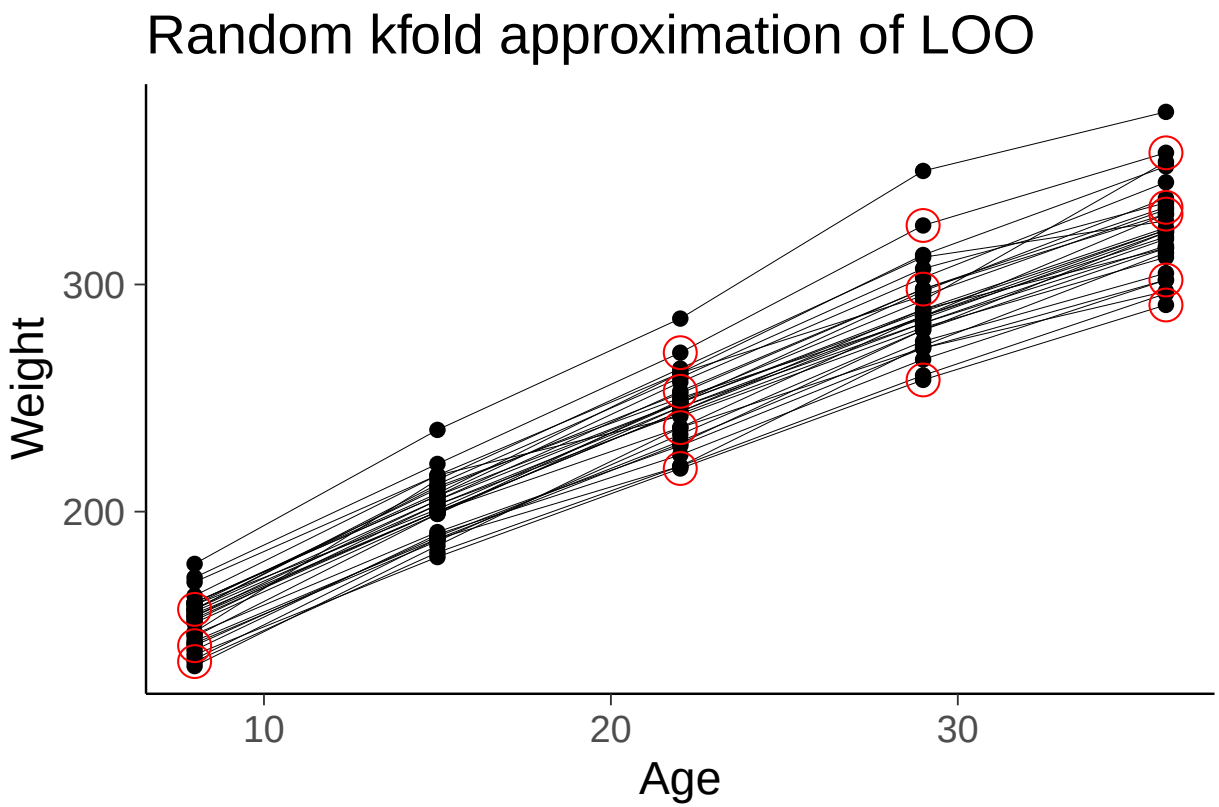




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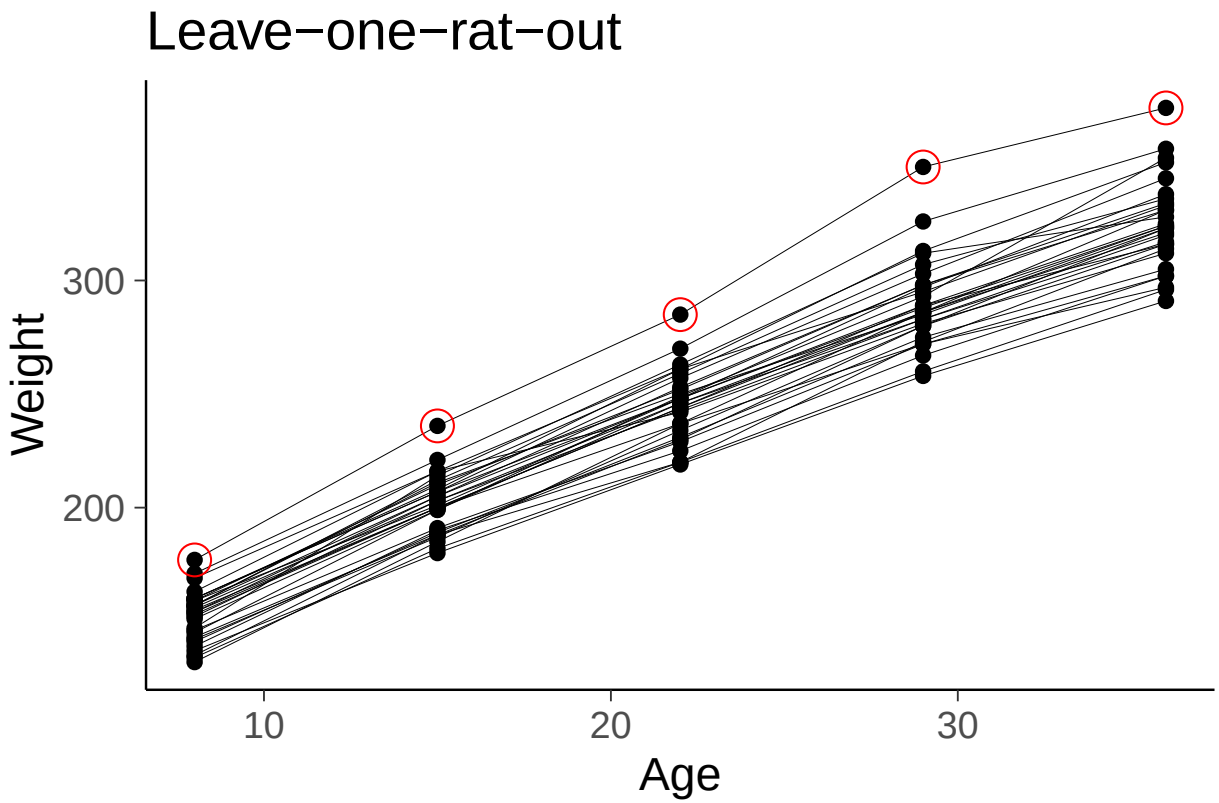




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# K-fold splits for different prediction tasks





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# Cross-validation for model assessment

1. CV is good for model assessment when application specific utility/cost functions are used



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# Cross-validation for model assessment

1. CV is good for model assessment when application specific utility/cost functions are used
2. Also useful in model checking in similar way as posterior predictive checking (PPC)
   - 2.1 model misspecification diagnostics (e.g. Pareto-$k$ and p_loo)
   - 2.2 checking calibration of leave-one-out predictive posteriors (ppc_loo_pit in bayesplot)



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Subsection 4

# Model Comparison



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# Paired model comparison

Score both models on the same held-out observations:

$$\ell_{mi} = \log p_m(y_i \mid y_{-i}), \qquad d_i = \ell_{1i} - \ell_{2i}.$$

$$\widehat{\Delta} = \sum_{i=1}^{n} d_i = \mathrm{elpd}_{\mathrm{loo},1} - \mathrm{elpd}_{\mathrm{loo},2}.$$



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- Positive differences favour model 1; negative differences favour model 2.



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- Positive differences favour model 1; negative differences favour model 2.
- Use the same prediction task, outcome units and validation split. Pairing accounts for observations that are easy or hard for both models.



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# Uncertainty in a model comparison

For approximately independent held-out units,

$$\bar{d} = \frac{1}{n}\sum_i d_i, \quad s_d^2 = \frac{1}{n-1}\sum_i (d_i - \bar{d})^2, \quad \mathrm{SE}(\widehat{\Delta}) \approx \sqrt{n\, s_d^2}.$$

loo: model comparison.



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- Use the variability of the paired differences. Do not subtract the two models' standard errors.

loo: model comparison.



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- Use the variability of the paired differences. Do not subtract the two models' standard errors.
- This is an approximation. Small samples, influential observations or dependent validation units require care.

loo: model comparison.



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- Use the variability of the paired differences. Do not subtract the two models' standard errors.
- This is an approximation. Small samples, influential observations or dependent validation units require care.
- `loo_compare()` reports each model minus the model with the largest ELPD. The best row is 0; other rows are negative. `se_diff` is the paired standard error.

loo: model comparison.



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# Model comparison

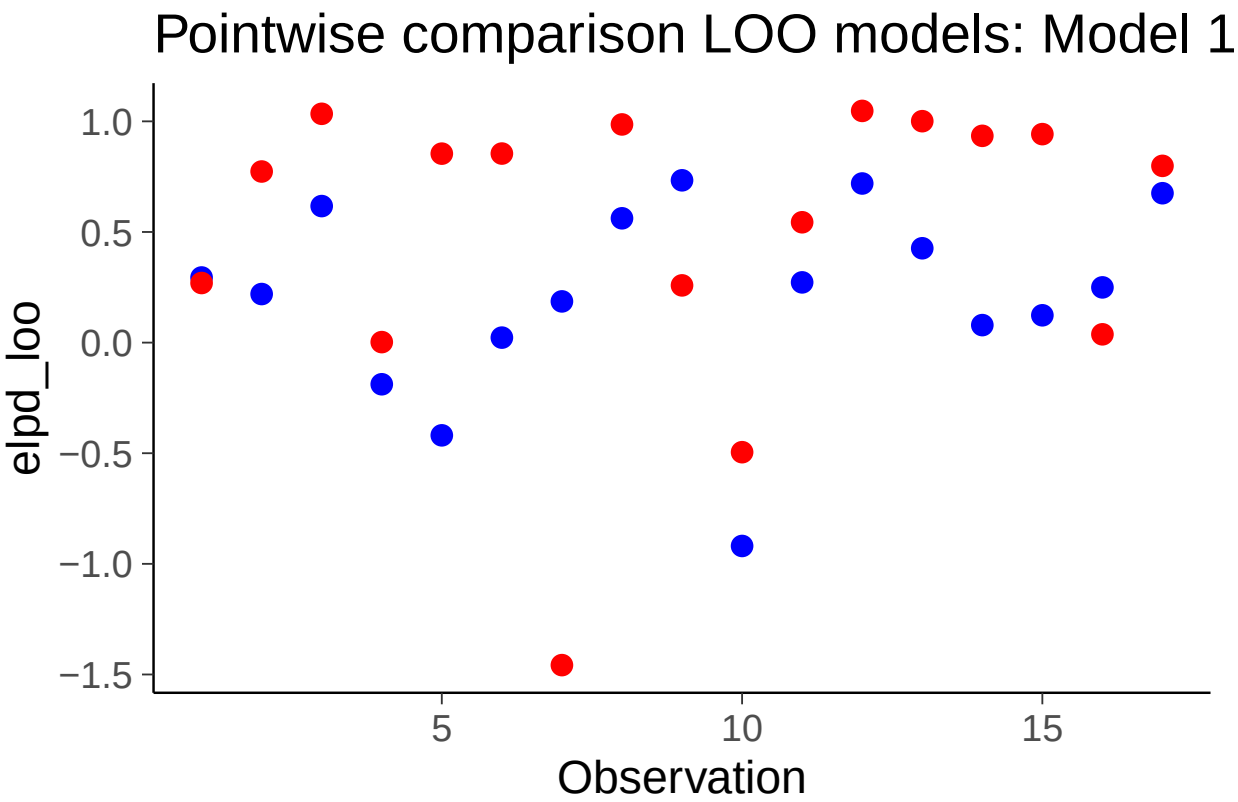
1. "A popular hypothesis has it that primates with larger brains produce more energetic milk, so that brains can grow quickly" (from Statistical Rethinking)
   - 1.1 Model 1 (blue): formula = kcal.per.g $\sim$ neocortex
   - 1.2 Model 2 (red): formula = kcal.per.g $\sim$ neocortex + log(mass)



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# Milk example: pointwise log scores

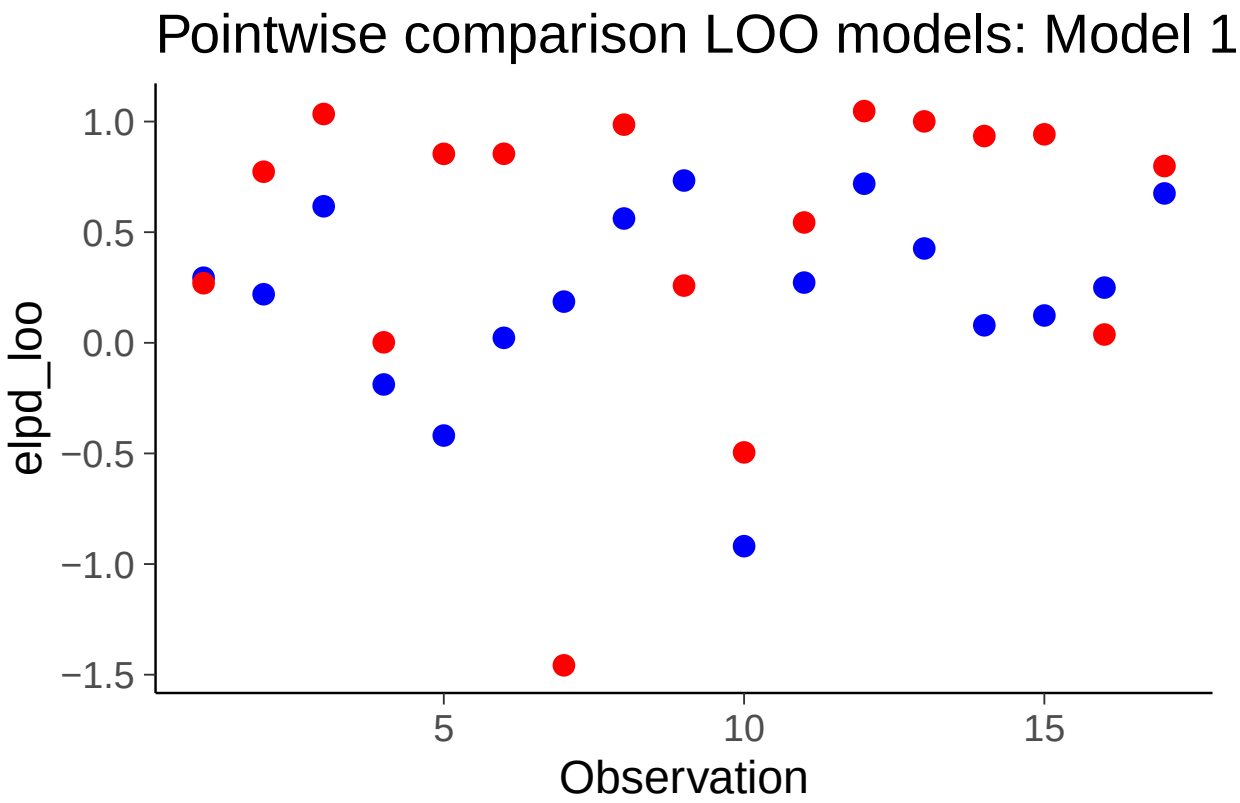




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# Milk example: pointwise log scores



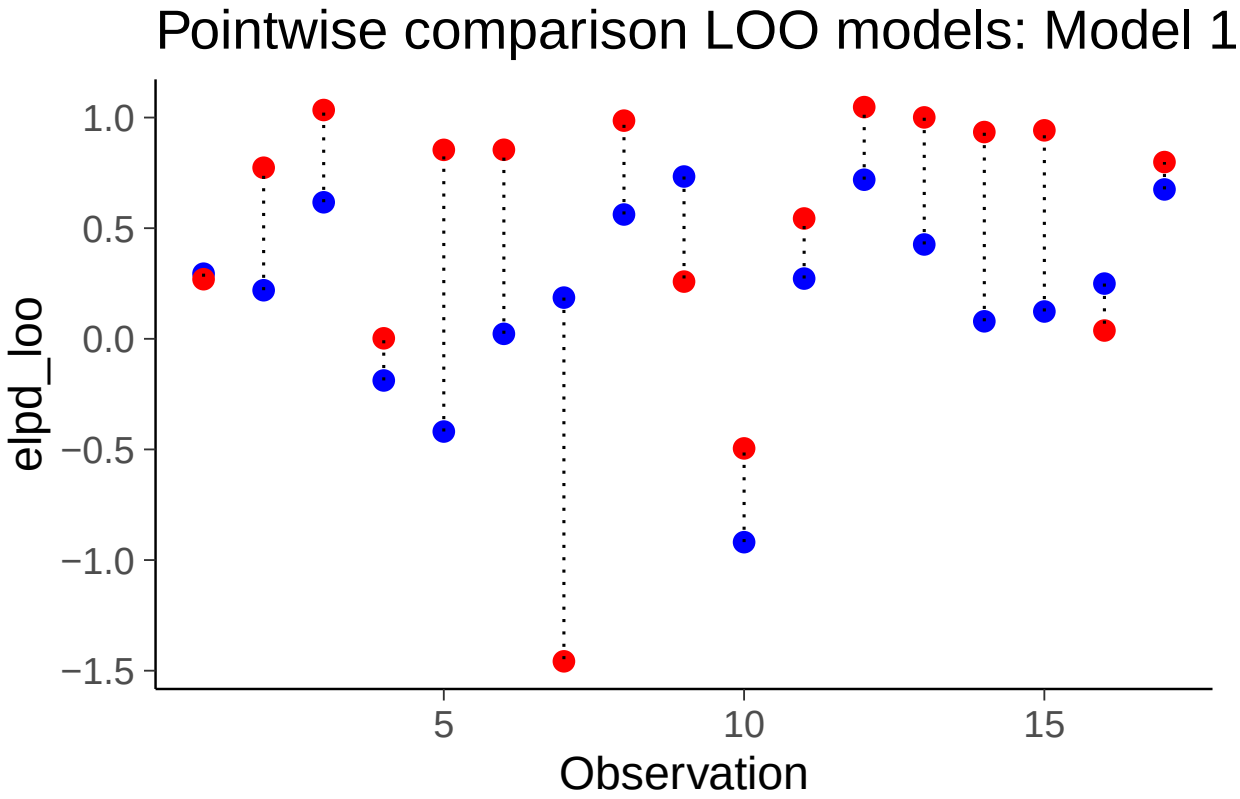

Model 1 elpd_loo $\approx$ 3.7, SE=1.8
Model 2 elpd_loo $\approx$ 8.4, SE=2.8



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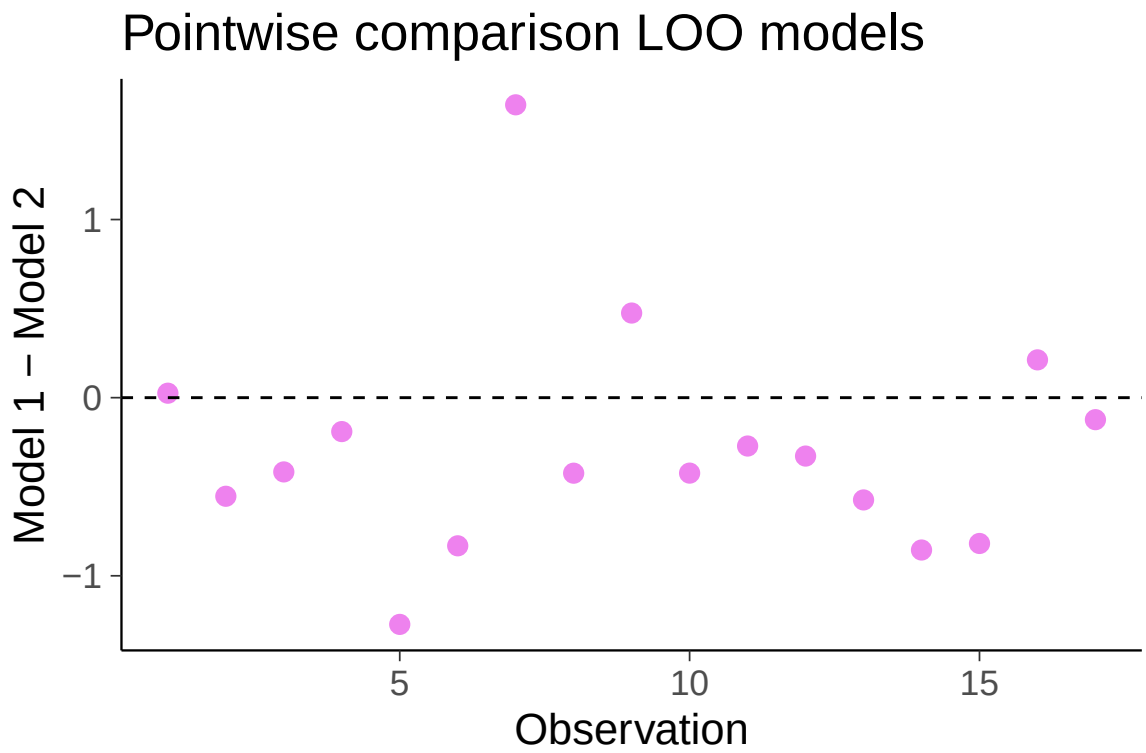

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# Milk example: paired differences




```
> loo_compare(model1, model2)
       elpd_diff se_diff
model2    0.0       0.0
model1   −4.7       2.7
```

Model 1 minus model 2: $\widehat{\Delta} = -4.7$, SE $= 2.7$. The estimate favours model 2, with substantial uncertainty.



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# Arsenic well example – Model comparison

1. Probability of switching well with high arsenic level in rural Bangladesh
   - 1.1 Model 1 covariates: log(arsenic) and distance
   - 1.2 Model 2 covariates: log(arsenic), distance and education level

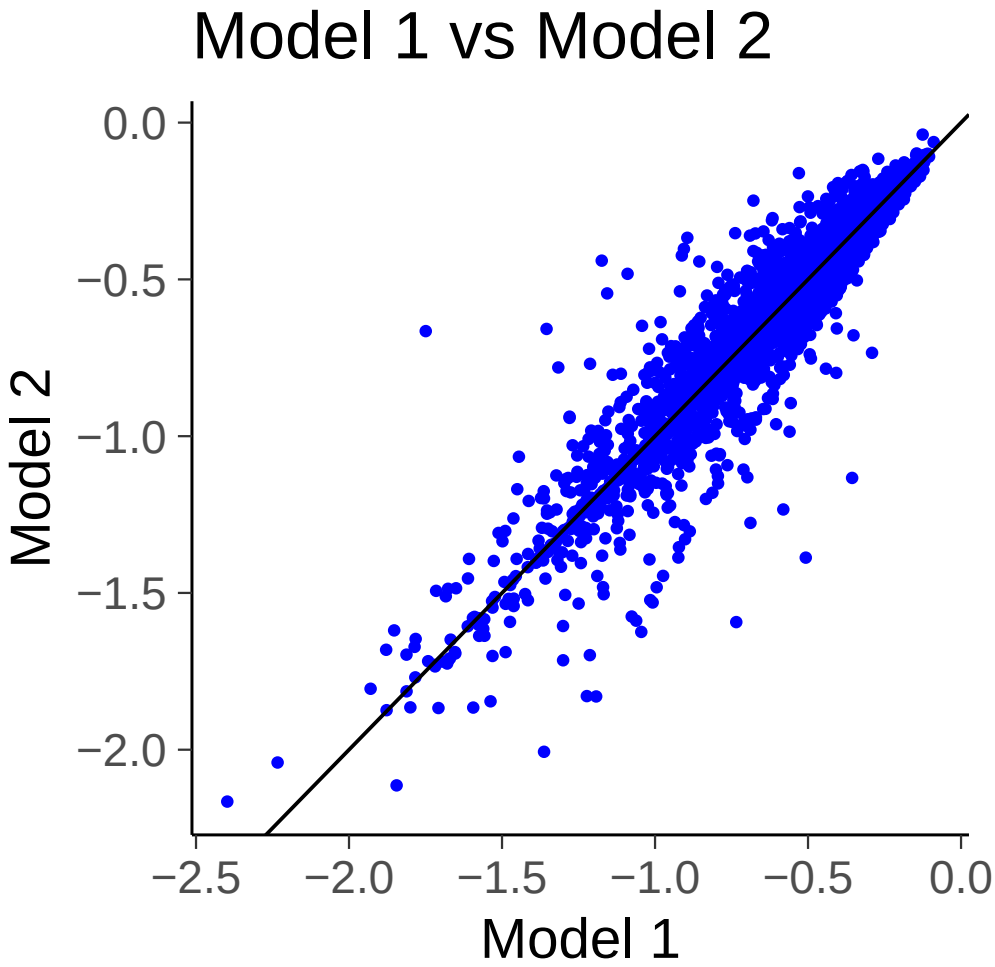
Gelman, Hill & Vehtari (2020): Regression and Other Stories, Chapter 13.



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# Arsenic well example – Model comparison




Model 1 elpd_loo ≈ -1952, SE=16

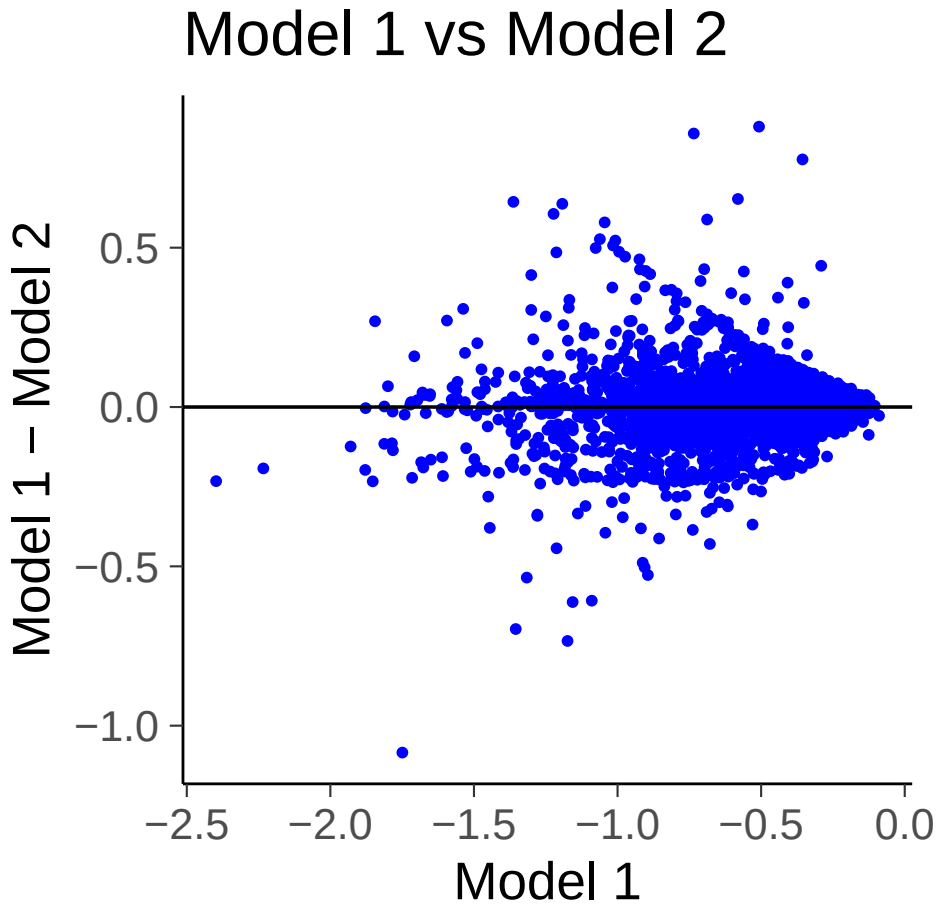
Model 2 elpd_loo ≈ -1938, SE=17



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# Arsenic well example – Model comparison




```
> loo_compare(model1, model2)
       elpd_diff se_diff
model2    0.0       0.0
model1  −14.4       6.1
```

Model 1 minus model 2: $\widehat{\Delta} = -14.4$, SE $= 6.1$.



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# Arsenic well example – Model comparison

```
> loo_compare(model1, model2)
       elpd_diff se_diff
model2    0.0       0.0
model1  -14.4       6.1
```

The estimate favours model 2 by 14.4 ELPD units. The paired SE is 6.1, not the difference between 17 and 16.

A normal approximation can be useful with enough observations and no severe misspecification. Small samples or influential observations can make it misleading.



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# What if one is not clearly better than others?



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# What if one is not clearly better than others?

1. Continuous expansion including all models?
   - 1.1 and then analyse the posterior distribution directly
   - 1.2 sparse priors like regularized horseshoe prior instead of variable selection



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# What if one is not clearly better than others?

1. Continuous expansion including all models?
   - 1.1 and then analyse the posterior distribution directly
   - 1.2 sparse priors like regularized horseshoe prior instead of variable selection
2. Model averaging with BMA or Bayesian stacking?



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2. Model averaging with BMA or Bayesian stacking?
3. In a nested case choose simpler if assuming some cost for extra parts?
4. A larger nested model can retain uncertainty about additional effects, especially with regularising priors. It does not account for all model uncertainty.



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# Cross-validation and model selection

1. Cross-validation can be used for model selection if
   1.1 small number of models
   1.2 the difference between models is clear



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# Cross-validation and model selection

1. Cross-validation can be used for model selection if
   - 1.1 small number of models
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2. Do not use cross-validation to choose from a large set of models
   - 2.1 selection process leads to overfitting



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3. Overfitting in selection process is not unique for cross-validation



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# Selection induced bias and overfitting

- Selection induced bias in cross-validation
  - same data is used to assess the performance and make the selection
  - the selected model fits more to the data
  - the CV estimate for the selected model is biased



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- Performance of the selection process itself can be assessed using two level cross-validation, but it does not help choosing better models



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  - same data is used to assess the performance and make the selection
  - the selected model fits more to the data
  - the CV estimate for the selected model is biased
- Performance of the selection process itself can be assessed using two level cross-validation, but it does not help choosing better models
- Bigger problem if there is a large number of models as in covariate selection



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# Looking ahead: model averaging

Model comparison need not end with selecting one model.

Lecture 9b introduces model averaging and stacking: combining predictive distributions from several candidate models.

Today's foundation is to define the prediction task, check the computations, and compare held-out performance with its uncertainty.



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# Section 3

# Summary



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# Take-home messages

1. It's good to think predictions of observables, because observables are the only ones we can observe
2. Cross-validation can simulate predicting and observing new data
3. Cross-validation is good if you don't trust your model
4. Different variants of cross-validation are useful in different scenarios



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# Section 4

# Additional material



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# PSIS for groups, dependence and time series

1. PSIS-LOO for hierarchical models
   1.1 leave-one-group out is challenging for PSIS-LOO



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# PSIS for groups, dependence and time series

1. PSIS-LOO for hierarchical models
   1.1 leave-one-group out is challenging for PSIS-LOO
2. PSIS-LOO for non-factorizable models



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# PSIS for groups, dependence and time series

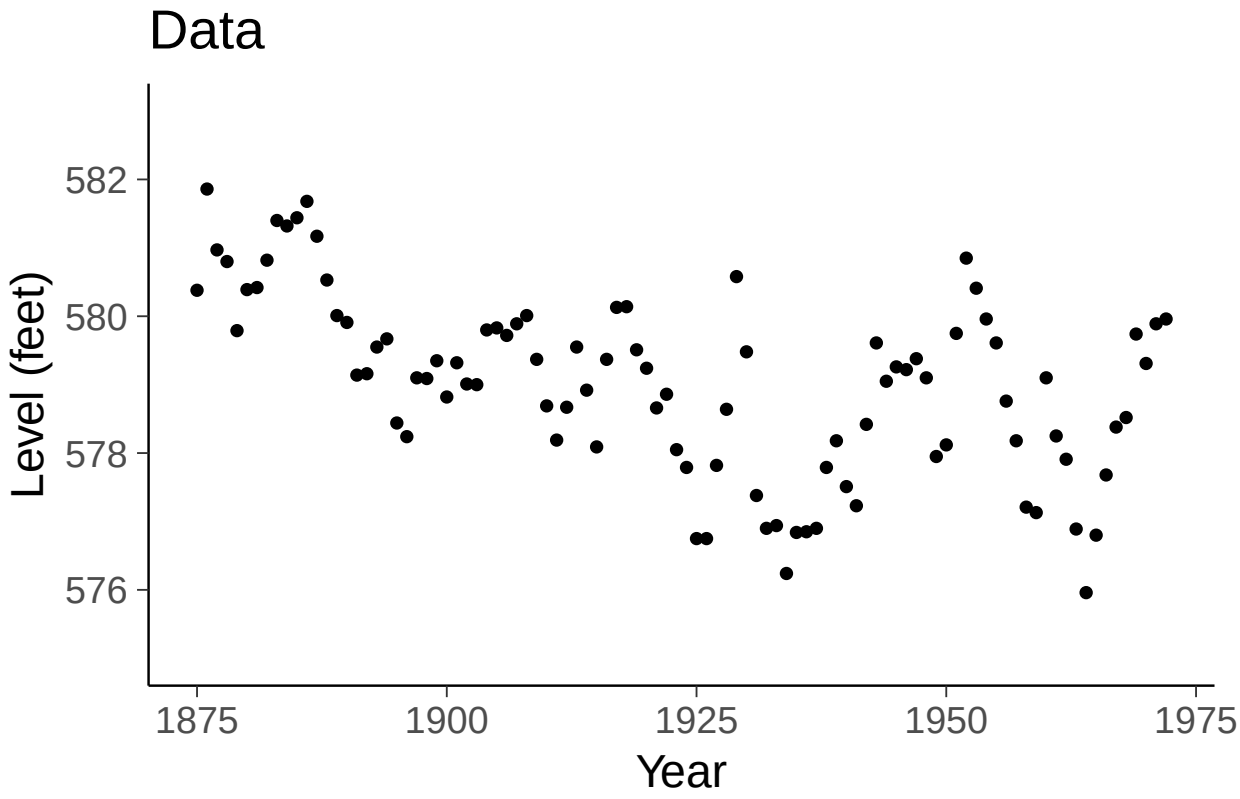
1. PSIS-LOO for hierarchical models
   1.1 leave-one-group out is challenging for PSIS-LOO
2. PSIS-LOO for non-factorizable models
3. PSIS-LOO for time series
   3.1 Approximate leave-future-out cross-validation



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# PSIS-LOO for time series

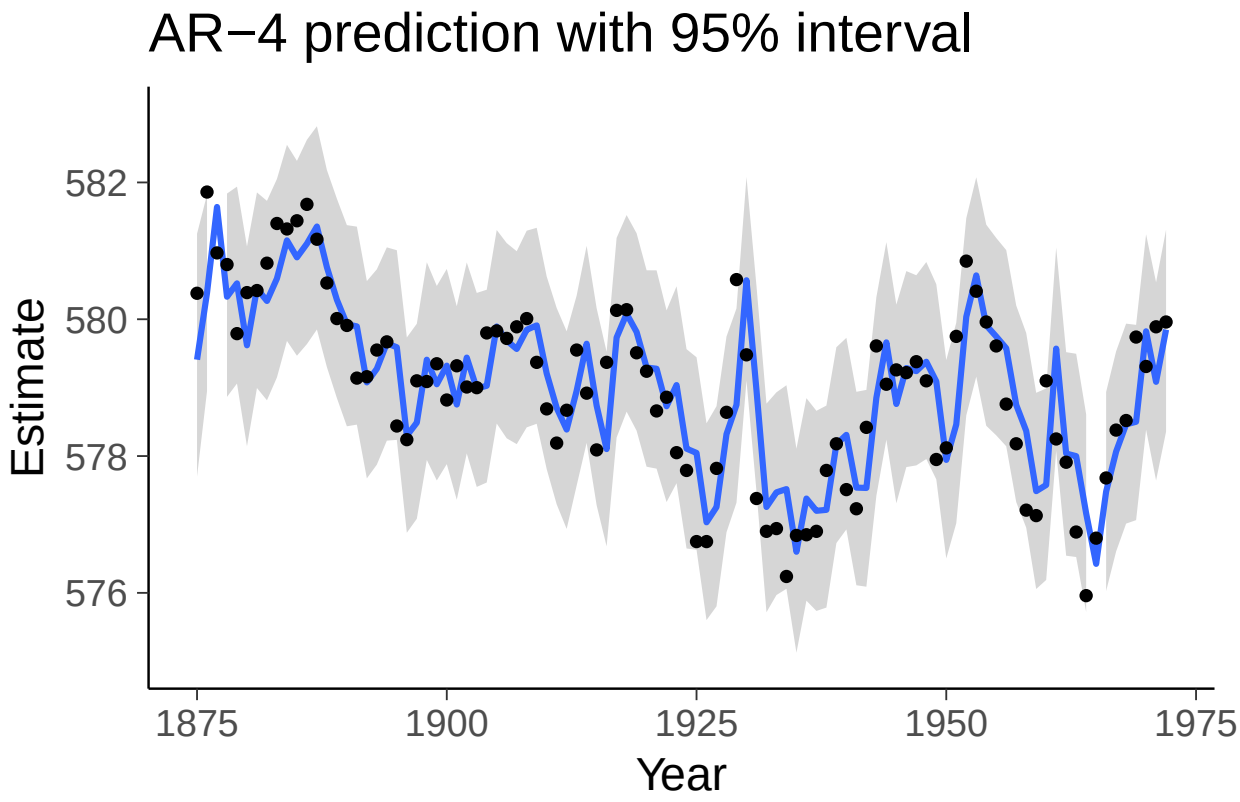




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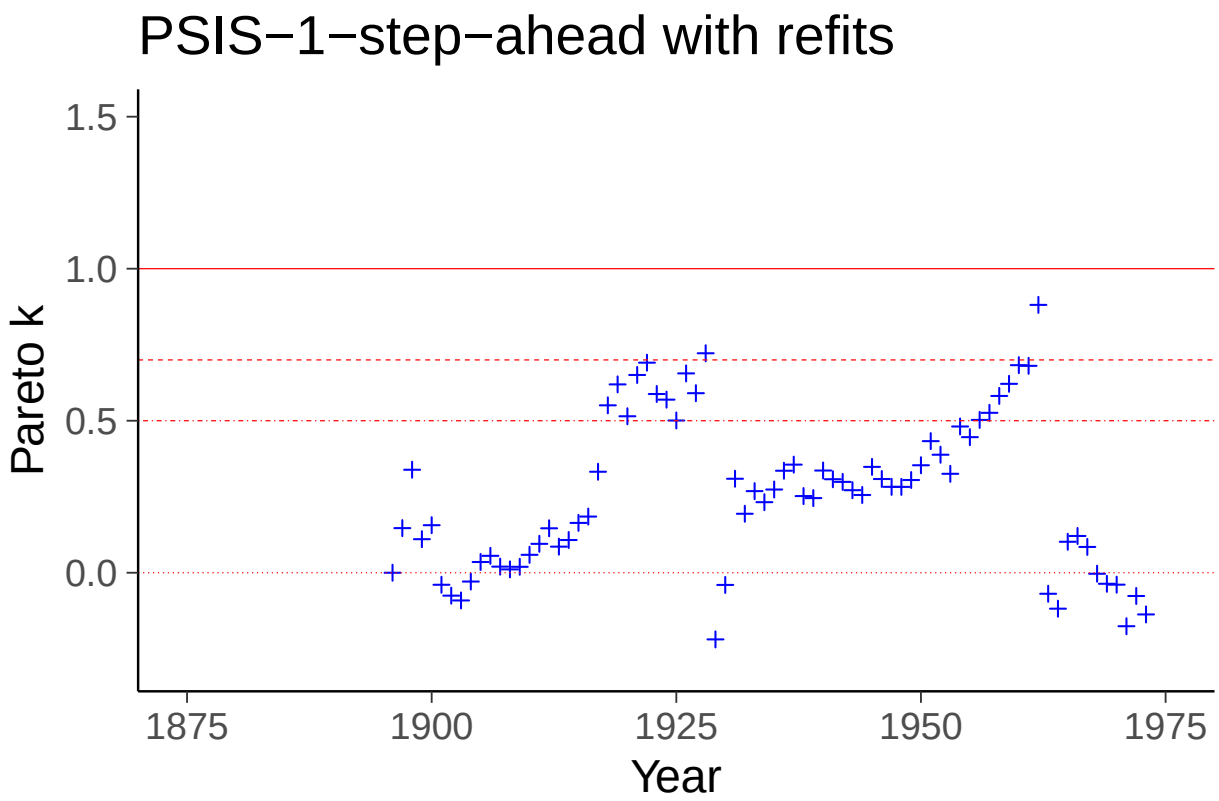




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# PSIS-LOO for time series




mc-stan.org/loo/articles/loo2-lfo.html



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# Akaike information criterion

1. Recall the gap between training lppd and LOO performance

$$\text{elpd}_{\text{loo}} = \sum_i \log p(y_i|y) - p_{\text{loo}}$$



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# Akaike information criterion

1. Recall the gap between training lppd and LOO performance

$$\text{elpd}_{\text{loo}} = \sum_i \log p(y_i|y) - p_{\text{loo}}$$

2. Akaike information criterion

$$\text{elpd}_{\text{AIC}} = \sum_i \log p(y_i|\hat{\theta}_{\text{MLE}}) - k$$

3. AIC

$$\text{AIC} = -2\text{elpd}_{\text{AIC}}$$



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# Deviance information criterion

1. Deviance information criterion

$$\text{elpd}_{\text{DIC}} = \sum_i \log p(y_i|\hat{\theta}_{\text{Bayes}}) - p_{\text{DIC}}$$

2. We estimate the effective number of parameters with

$$p_{\text{DIC1}} = 2(\log p(y|\hat{\theta}_{\text{Bayes}}) - \text{E}_{p(\theta|y)}(\log p(y|\theta)))$$

or

$$p_{\text{DIC2}} = 2V_{p(\theta|y)}(\log p(y|\theta))$$

3. DIC

$$\text{DIC} = -2\text{elpd}_{\text{DIC}}$$



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# Widely applicable information criterion (WAIC)

1. Watanabe information criterion

$$\text{elpd}_{\text{WAIC}} = \sum_i \log p(y_i|y) - p_{\text{WAIC}}$$

2. We estimate the effective number of parameters with

$$p_{\text{WAIC1}} = 2\sum_{i}^{n}(\log \mathrm{E}_{p(\theta|y)}(p(y_i|\theta)) - \mathrm{E}_{p(\theta|y)}(\log p(y_i|\theta)))$$

or

$$p_{\text{WAIC2}} = \sum_{i=1}^{n} \text{Var}_{\theta|y}[\log p(y_i \mid \theta)]$$

3. WAIC

$$\text{WAIC} = -2\text{elpd}_{\text{WAIC}}$$

4. WAIC can be seen as an approximation of the $elpd_{\text{loo}}$



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# WAIC vs PSIS-LOO



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# WAIC vs PSIS-LOO

1. WAIC and LOO estimate pointwise predictive performance; they can differ substantially with finite data.



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# WAIC vs PSIS-LOO

1. WAIC and LOO estimate pointwise predictive performance; they can differ substantially with finite data.
2. PSIS-LOO is generally preferable when its importance-sampling diagnostics are satisfactory.



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4. LOO makes the held-out prediction task explicit; another validation split may be needed for a different task.
5. ELPD: higher is better. On the $-2 \times$ ELPD scale, lower is better; the ranking contains the same information.